ESE 4125 Complete Study Pack

Safety and Environmental Aspects of Energy Projects · KUET · past papers 2020, 2021, 2022, 2023, 2024 · answers from the course slides

24-hour priority list

You answer **3 of 4 questions in each part**, so each part has its own list. Study down both lists until the green **pass line**: about 6 h for Part A and 5 h for Part B, which leaves time for sleep and a last read of the one-shot notes. Alternate between parts (A block 1, B block 1, A block 2 ...) so neither is left short. **Chance**: how often the topic came up, newer papers weigh more (2024 ×5, 2023 ×4, 2022 ×3, 2021 ×2, 2020 ×1). **Covered**: share of that part's past-paper marks you have studied once you finish the row.

Next if you have time: the three short numericals (A3 dead time, A15 AFR, B5 TWA) take 10 minutes each with the calculators and give sure marks. This course is almost all lists: read each memory trick box twice, then write the list from memory once.

Section A: Occupational safety, hazards, fire, risk, PPE, accidents and EPP

Block 1: very likely and easy · 1.7 h

Topic	Chance	Level	Covered
A14 Causes of accidents: Heinrich's domino theory, 88-10-2 rule, man-man, man-machine and man-environment relationships	Very likely 100%	Easy	8%
A1 Basic definitions: hazard, risk, accident, occupational safety, health, safety management; the three parts of OSHEM	Very likely 100%	Easy	15%
A15 Accident investigation and reporting sheet; accident statistics AFR, ASR, FAR (with numerical); HAZAN, HAZOP; safety audit	Very likely 93%	Easy	25%
A2 Sources of hazard, categories of hazard by working situation, consequences, control steps, psychosocial hazards	Very likely 93%	Easy	34%

Block 2: very likely, medium effort · 0.8 h

Topic	Chance	Level	Covered
A10 Hazard identification methods: checklist, what-if, PHA, HAZOP, FMEA, plant drawings; limitations of hazard evaluation	Very likely 87%	Medium	42%

Block 4: likely · 3.5 h

Topic	Chance	Level	Covered

A13 Personal protective equipment (PPE): types, how each protects, respiratory protective equipment	Likely 73%	Easy	48%
A7 Classes of fire and fire-fighting materials (water, foam, dry powder, CO_2, wet chemical); fire-fighting procedure	Likely 67%	Easy	55%
A16 Emergency preparedness plan (EPP), Triple Bottom Line (TBL), safety committee	Likely 67%	Easy	61%
A11 Anatomy of a process incident; event tree vs fault tree	Likely 67%	Medium	65%
A8 BLEVE (boiling liquid expanding vapour explosion): mechanism, sequence of events, effects, prevention	Likely 67%	Medium	69%
A12 Risk management, hierarchy of controls (and where PPE fits), bow-tie diagram	Likely 60%	Medium	76%

✔ Pass line: 76% of Section A marks covered, enough to choose 3 of the 4 questions (about 5.9 h)

Only if time is left · 2.4 h

Topic	Chance	Level	Covered
A6 Fire basics: fire triangle and tetrahedron, factors contributing to fire, methods of extinguishment, types of explosion	Possible 53%	Easy	81%
A4 Physical and thermal hazards: heat cramps, heat stroke, frostbite, asphyxiation, cave in, radiation, lighting, housekeeping	Possible 47%	Easy	90%
A5 Electrical hazards (severity factors, protection) and biological hazards	Possible 47%	Easy	94%
A9 Fire management in a high-rise building; PASS	Possible 47%	Easy	97%
A3 Numericals: hazard matrix rating, fractional dead time and hazard rate of a protective system	Possible 33%	Medium	100%

Section B: Occupational health, chemicals, pollution and environmental management

Block 1: very likely and easy · 4.2 h

Topic	Chance	Level	Covered
B8 MSDS (material safety data sheet): purpose, 16 headings, key sections, responsibilities of supplier, employer, worker	Very likely 100%	Easy	11%
B11 Greenhouse effect and global warming, acid rain, smog: causes, effects, control	Very likely 100%	Easy	22%

B6 Hazardous chemicals: categories and health effects; hazardous vs toxic; routes of entry; local/systemic and acute/chronic effects	Very likely 100%	Easy	31%
B1 Occupational health, OSH Act and OSHA, occupational health service (functions, how to provide)	Very likely 100%	Easy	40%
B2 Occupational diseases: definitions, vs normal disease; lung disease, blood disorders, urticaria, pigmentary changes, sore eye, stress	Very likely 100%	Easy	49%
B3 Occupational ergonomics: definition, importance, five examples of good ergonomics	Very likely 100%	Easy	54%
B12 Sustainable development: definition, environmental/economic/socio-political sustainability, pollution prevention and clean technology, challenges	Very likely 93%	Easy	60%
B4 Occupational exposure, TLV and PEL, TLV-TWA, TLV-STEL, TLV-C	Very likely 93%	Easy	65%
B15 ISO 14001: purpose and major components of an environmental management system	Very likely 93%	Easy	70%
B10 Local, regional and global environmental impacts and their causes (thermal power plants, petroleum industry)	Very likely 93%	Easy	74%

Block 2: very likely, medium effort · 1.5 h

Topic	Chance	Level	Covered
B13 IEE and EIA: why needed, differences, EIA process (flow chart), red category, EMP	Very likely 100%	Medium	84%
✔ Pass line: 84% of Section B marks covered, enough to choose 3 of the 4 questions (about 4.9 h)			
B14 Environmental clearance in Bangladesh (Green, Orange-A, Orange-B, Red), Department of Environment (DoE) and its history	Very likely 93%	Medium	90%

Block 4: likely · 0.4 h

Topic	Chance	Level	Covered
B9 Pollution and pollutants: classification, sources of air and water pollution and their control	Likely 67%	Easy	95%

Only if time is left · 0.8 h

Topic	Chance	Level	Covered
B7 Dose-response relationship; lethal, sub-lethal, acute	Possible 47%	Easy	98%
B5 Numerical: time-weighted average (TWA) exposure and comparison with TLV	Possible 33%	Easy	100%

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★ One-shot notes

One-shot notes: ESE 4125 Safety and Environmental Aspects of Energy Projects

Core words

Hazard = can hurt (not yet). **Risk** = likelihood × consequence. **Accident** (Heinrich) = unplanned, uncontrolled event giving injury/damage. **Incident** = near miss.

OSHEM = safety (sudden) + health (slow) + environment (surroundings). Safety = unintentional losses; security = intentional.

Hazard categories and sources

Physical, chemical, biological, psychosocial, human-factor ("Please Check Before People Hurt").

Sources: dangerous substances, fire/explosion, machinery, electricity, radiation, overhead work, noise/vibration, temperature/humidity, dust/fumes. Effects: human, economic, environmental.

Control steps: identify → inventory → rank → probability → rank assessment → eliminate/reduce.

Hazard matrix and FDT

Severity 20/15/10/3 × frequency 10/8/6/4/2. 100-200 unacceptable, 60-90 undesirable, 20-50 conditionally acceptable, 6-18 acceptable.

FDT = F T / 2; H = D × FDT. 2022: F 0.5/yr, T 1 week, D 1/yr → FDT 0.0048 ≈ 0.005, H = once in 200 yr.

Thermal and electrical

Heat: cramps (salt) → exhaustion (water) → stroke (thermostat fails, emergency). Cold: hypothermia (core < 35 °C), frostbite (local freezing).

Shock severity: voltage, body resistance, path, duration. 10-20 mA cannot let go; 50-100 mA fibrillation. Protection: insulation, guarding, grounding, fuses/CB/GFCI, lockout, PPE, training.

Radiation limit 5 rem/yr whole body (eyes 15, extremities 50). Time-distance-shielding.

Fire

Triangle: fuel, oxygen, heat (+ chain reaction = tetrahedron). Extinguish: starve (fuel), smother (oxygen), cool (heat), inhibit (chain).

Classes: A solids (water), B liquids (foam), C gases, D metals (special powder), E electrical (CO_2/powder), F/K cooking oil (wet chemical). Water not on B, D, E, F.

RACE: rescue, alarm, contain, extinguish. PASS: pull, aim, squeeze, sweep.

Explosions and BLEVE

Deflagration < speed of sound (~8 bar); detonation > sound (~20 bar); dust explosion; VCE (UVCE open, CVCE confined).

BLEVE: fire heats pressurised liquefied-gas tank → vapour-space wall weakens → rupture → liquid flashes → blast + fireball + fragments. Prevent: water deluge, insulation, relief valves, drainage away, spacing, early evacuation.

High-rise fire

Prevent: design, detectors and alarm, alternate power, no temporary wiring. Fight: sprinklers, hydrants, water reserve, main + diesel standby + jockey pumps, firemen lift, drills. Evacuate: fire-proof stairs with fire doors, never lift, crawl low, assembly point.

Hazard identification

Methods: checklist, what-if, PHA, HAZOP, FMEA, FTA, ETA, relative ranking, audits. Drawings: plot plan (spacing), isometric (piping), BFD, PFD (conditions), P&ID (HAZOP input).

PHA: early; hazard-cause-effect-category (I-IV)-measures. HAZOP: node, intention, parameter + guide word (NO, MORE, LESS, AS WELL AS, PART OF, REVERSE, OTHER THAN) = deviation → causes, consequences, safeguards, actions.

Limitations: completeness, reproducibility, inscrutability, relevance of experience, subjectivity.

Anatomy, FTA, ETA

Normal (optimise; process equipment, control system, procedures) → Abnormal (return to normal; shutdown, protective systems) → Emergency (safe state, minimise losses; containment, emergency response).

FTA deductive from top event (AND/OR); ETA inductive from initiating event (success/failure). Top event of FT = initiating event of ET.

Risk management and bow-tie

Steps: describe system → identify hazards → scenarios → consequence + likelihood → evaluate → reduce → monitor.

Hierarchy: elimination > substitution > engineering > administrative > PPE (last line).

Bow-tie: threats → prevention barriers → TOP EVENT → mitigation barriers → consequences; degradation factors and controls.

PPE

Head (hard hat G 2,200 V, E 20,000 V, C none), eye/face (glasses, goggles, face shield, welding helmet), hearing (plugs, muffs; 90 dBA 8 h), respiratory, hands, feet (steel toe), body, fall harness.

Respirators: APR (dust mask, cartridge, gas mask, PAPR) clean the air; SCBA/SAR supply air (O_2 < 19.5 %, IDLH).

Accidents

Heinrich dominoes: ancestry → fault of person → unsafe act/condition → accident → injury. 88 % acts, 10 % conditions, 2 % unavoidable.

Relationships: man-machine (fitness, knowledge, training, maintenance), man-man (dues, participation), man-environment (noise, heat, light, rules).

Report sheet: identification, person, injury/damage, description, causes, actions, sign-off.

Accident statistics

AFR = injuries × 10^6 / man-hours. ASR = days lost × 1000 / man-hours. FAR = deaths × 10^8 / hours.

2021: 600 × 50 × 40 = 1.2 M h × 0.95 = 1.14 M; AFR = 60 × 10^6/1.14 × 10^6 = **52.63**.

HAZAN = quantify how often and how bad. Safety audit: walk-through, comprehensive, specific, internal/external, compliance.

EPP, TBL, committee

EPP levels: 1 local, 2 whole plant, 3 neighbours + outside help, 4 national. Steps: list disasters, severity, level, plan, resources, procedures, declare/terminate, teams, leader, communication, evacuation; drills.

TBL (Elkington 1997): people, planet, profit. Committee: chairman (manager), member secretary (safety officer), doctor, personnel officer, foremen, worker reps.

Occupational health

ILO/WHO: highest physical, mental, social well-being; adapt work to man. OSH Act 1970 = law; OSHA = agency (standards, inspections). Service (ILO C161): assess, monitor, examine (pre-employment, periodic), first aid, advise, train, records, rehabilitate.

Diseases

Lung: silica → silicosis, asbestos → asbestosis/mesothelioma, coal → pneumoconiosis, cotton → byssinosis, isocyanates/flour → asthma.

Blood: benzene (aplasia, leukaemia), lead (anaemia), aniline/nitrites (methaemoglobinaemia), arsine (haemolysis).

Urticaria = itchy red welts (latex). Pigmentary = darker/lighter skin (tar, UV, plants). Sore eye: VDU, welding, dust, chlorine. Stress: workload, shifts, bullying, insecurity.

Ergonomics

"Fit the job to the worker" (ergon + nomos). Examples: adjustable chair, screen at eye level, neutral wrists, lifting aids, items within reach, breaks/rotation, good lighting.

Exposure limits

TLV (ACGIH, advice) vs PEL (OSHA, law). TWA 8 h/40 h; STEL 15 min, max 4/day, 60 min apart; Ceiling never. IDLH = escape within 30 min.

TWA = $\Sigma C_i t_i/8$. 2024: 1138/8 = **142.25 ppm** < 200, not exceeded.

Chemicals

Hazardous (health or physical danger) ⊃ toxic (poison). Categories: toxic, corrosive, oxidising, dehydrating, reactive, radioactive, flammable; target organs (hepato-, nephro-, neuro-, haemato-toxins).

Routes: inhalation, ingestion, injection, skin/eye. Local vs systemic; acute (short, high) vs chronic (long, low; benzene → leukaemia).

Dose-response: none < A (threshold) < reversible < B < permanent < C < death (D). Lethal, sub-lethal, acute.

MSDS

16 sections: 1 identification, 2 hazards, 3 composition, 4 first aid, 5 fire, 6 release, 7 handling/storage, 8 exposure/PPE, 9 properties, 10 stability, 11 toxicology, 12 ecology, 13 disposal, 14 transport, 15 regulatory, 16 other.

Supplier prepares and supplies; employer keeps current (≤ 3 yr), accessible, trains; worker follows and knows where it is.

Pollution

Disposal: biodegradable, persistent (slow), non-biodegradable. Existence: quantitative (natural, too much) vs qualitative (man-made).

Air: CO/CO_2, SO_2, NO_x, metals, SPM from fuel burning. Water: industry, sewage, farm runoff, oil spills; control: ETP, STP, IPM.

Local (city air, rivers), regional (acid rain, seas), global (greenhouse, ozone, biodiversity).

Greenhouse, acid rain, smog

Sunlight in → Earth emits IR → GHGs (H_2O, CO_2, CH_4, N_2O, CFCs) trap it → warming.

Acid rain pH < 5.6: SO_2/NO_x → H_2SO_4/HNO_3; wet/dry deposition; fish, forests, buildings. Prevent: scrubbers, low-S fuel, catalytic converters.

Smog: London (coal SO_2 + fog, 1952) vs LA photochemical (NO_x + HC + sun → ozone, PAN).

Sustainable development

Brundtland 1987: meet present needs without compromising future generations. Environmental, economic, socio-political pillars. Needs pollution prevention + clean technology. Challenges: population, urbanisation, intensive agriculture, declining resources, altered cycles, social disintegration.

IEE, EIA, EMP

IEE = quick, pre-feasibility; EIA = detailed, feasibility, quantified. Red: IEE → site clearance, EIA → environmental clearance.

EIA: screening → IEE → scoping/ToR → full EIA → review → decision → monitoring/audit. EMP = mitigation + monitoring plan for construction, operation, decommissioning.

Clearance and DoE

Categories Green, Orange-A, Orange-B, Red. General: application (Form-3, NOC) → site clearance → apply EC → ECC → renew 30 days before expiry (Green 3 yr, others 1 yr).

Red: IEE + ToR → site clearance (30 d) → EIA + ETP approval (60 d) → ECC (60 d).

DoE: 1973 water pollution project → 1977 ordinance/board → 1982 DEPC → 1989 DoE under MoEF → ECA 1995, ECR 1997.

ISO 14001

EMS standard, only certifiable one in ISO 14000. PDCA: policy, aspects + legal, objectives, implement, training, documents, emergency, monitoring/audit, corrective action, management review, continual improvement.

Abbreviations used in this pack

Every short form, written out. The answers also spell each one out the first time it appears.

Short form	Full meaning
3R	reduce, reuse, recycle
ACGIH	American Conference of Governmental Industrial Hygienists
AFR	accident frequency rate
AIT	auto-ignition temperature
ANSI	American National Standards Institute
APR	air-purifying respirator
ASR	accident severity rate
BFD	block flow diagram
BLEVE	boiling liquid expanding vapour explosion
CAS	Chemical Abstracts Service (registry number of a chemical)
CB	circuit breaker
CCl_4	carbon tetrachloride
CEO	chief executive officer
CFC	chlorofluorocarbon
CNG	compressed natural gas
CPSC	Consumer Product Safety Commission (USA)
CSR	corporate social responsibility
CVCE	confined vapour cloud explosion
dB	decibel (unit of sound level)
dBA	A-weighted decibel (sound level as the human ear hears it)
DDT	dichloro-diphenyl-trichloroethane (an insecticide)
DEPC	Department of Environmental Pollution Control
DIFE	Department of Inspection for Factories and Establishments (Bangladesh)
DoE	Department of Environment (Bangladesh)
DOT	Department of Transportation (USA)
ECA	Environment Conservation Act (Bangladesh, 1995)
ECC	Environmental Clearance Certificate
ECR	Environment Conservation Rules (Bangladesh, 1997; new rules 2023)
EIA	environmental impact assessment
EMP	environmental management plan
EMS	environmental management system
EPA	Environmental Protection Agency (USA)
EPP	emergency preparedness plan
EPS	earnings per share
ESP	electrostatic precipitator
ETA	event tree analysis

ETP	effluent treatment plant
FAR	fatal accident rate
FDT	fractional dead time
FMEA	failure modes and effects analysis
FTA	fault tree analysis
GFCI	ground-fault circuit interrupter
GHG	greenhouse gas
GHS	Globally Harmonised System of classification and labelling of chemicals
HAZAN	hazard analysis
HAZOP	hazard and operability study
HFC	hydrofluorocarbon
HI	hazard identification
HIRA	hazard identification and risk assessment
HR	human resources
ICC	International Chamber of Commerce
ICRP	International Commission on Radiological Protection
IDLH	immediately dangerous to life or health
IEE	initial environmental examination
ILO	International Labour Organization
IR	infrared
ISO	International Organization for Standardization
LC50	lethal concentration for 50 % of a test population
LD50	lethal dose for 50 % of a test population
LEL	lower explosive limit
LNG	liquefied natural gas
LOTO	lockout/tagout
LPG	liquefied petroleum gas
mA	milliampere = 1/1000 ampere
MIC	methyl isocyanate
MoEF	Ministry of Environment and Forest
MoEFCC	Ministry of Environment, Forest and Climate Change
MSDS	material safety data sheet
mSv	millisievert (radiation dose unit)
NDT	non-destructive testing
NEP	National Environment Policy
NFPA	National Fire Protection Association (USA)
NIOSH	National Institute for Occupational Safety and Health (USA)

NO_x	nitrogen oxides
NOC	no objection certificate
NoC	no objection certificate
NRR	noise reduction rating
NSC	National Safety Council (USA)
OEL	occupational exposure limit
OHS	occupational health and safety
OHSAS	Occupational Health and Safety Assessment Series
OSH	occupational safety and health
OSHA	Occupational Safety and Health Administration (US government agency)
OSHEM	Occupational Safety, Health and Environment Management
P&ID	piping and instrumentation diagram
PAN	peroxyacetyl nitrate
PAPR	powered air-purifying respirator
PASS	Pull, Aim, Squeeze, Sweep (using an extinguisher)
PCB	polychlorinated biphenyl
PDCA	Plan-Do-Check-Act
PEL	permissible exposure limit
PFD	process flow diagram
PHA	preliminary hazard analysis
PM	particulate matter ($PM_{2.5}$ = particles below 2.5 micrometres)
PPE	personal protective equipment
ppm	parts per million
QRA	quantitative risk assessment
RACE	Rescue, Alarm, Contain, Extinguish/Evacuate
REL	recommended exposure limit
rem	roentgen equivalent man (unit of radiation dose; 1 rem = 10 mSv)
RFP	request for proposal
SAR	supplied-air respirator
SCBA	self-contained breathing apparatus
SDS	safety data sheet
SO_x	sulphur oxides
SPM	suspended particulate matter
STEL	short-term exposure limit
TBL	triple bottom line
TDI	toluene diisocyanate

TEDE	total effective dose equivalent
TLV	threshold limit value
ToR	terms of reference
TWA	time-weighted average
UEL	upper explosive limit
UV	ultraviolet
UVCE	unconfined vapour cloud explosion
VCE	vapour cloud explosion
VDU	visual display unit (computer screen)
VOC	volatile organic compound
WHO	World Health Organization

Section A: Occupational safety, hazards, fire, risk, PPE, accidents and EPP

A14 Causes of accidents: Heinrich's domino theory, 88-10-2 rule, man-man, man-machine and man-environment relationships Must study

Very likely · 100% Easy

PAST-PAPER QUESTIONS

- **2024 Q4(a)** [10] Briefly explain Heinrich's domino theory of accident. Also explain the man to machine relationship of an accident.
- **2023 Q4(a)** [7] How does Heinrich's domino theory explain the causation of accidents in the workplace?
- **2022 Q3(b)** [13] What are the major causes of workplace accidents? Explain the accident that occurs due to imbalance of the man and machine relationship. Suggest the remedies.
- **2021 Q2(c)** [10] Explain the three basic relationships between man, machine and environment. What are the remedial measures to ensure safety?
- **2020 Q3(d)** [12] What are the major causes of workplace accidents? Explain the man and machine relationship.

★ REMEMBER IT

- Heinrich's 5 dominoes: **Ancestry/social environment → fault of person → unsafe act or condition → accident → injury**. Remove the middle domino (unsafe act/condition) and the chain stops.
- 88-10-2 rule: 88 % of accidents from unsafe acts, 10 % unsafe conditions, 2 % unavoidable (acts of God).

Before you start: why accidents happen

Heinrich (an American insurance engineer, 1931) studied 75,000 accident reports and concluded: accidents do not just happen, they are **caused**, mostly by people's **unsafe acts** (88 %), some by **unsafe conditions** (10 %), and very few are truly unavoidable (2 %).

He pictured the causes as five dominoes standing in a row: if the first falls, all fall and someone is injured. Take out one domino (usually the unsafe act/condition) and the chain stops.

Heinrich's domino theory (2024 4a, 2023 4a)

1. **Ancestry and social environment**: inherited traits and upbringing (stubbornness, recklessness, lack of education).
2. **Fault of the person**: carelessness, bad temper, ignorance, nervousness - caused by domino 1.
3. **Unsafe act and/or mechanical or physical hazard (unsafe condition)**: standing under a load, removing a guard, no guard, poor lighting.
4. **Accident**: fall, being struck, caught in a machine.
5. **Injury**: cut, fracture, death - the result.

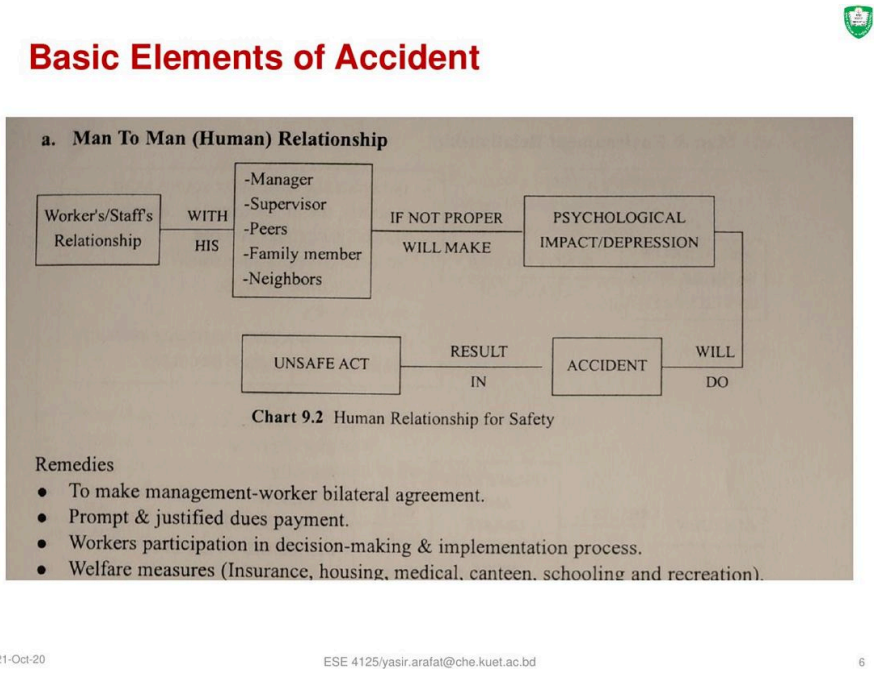
Each domino knocks the next. **Removing the central domino (3)** - by correcting unsafe acts and conditions through training, supervision, guards and inspection - breaks the chain, so the injury does not occur even if the earlier factors exist. Heinrich's data: 88 % unsafe acts, 10 % unsafe conditions, 2 % beyond control (acts of God). (NSC: 18 % mechanical, 19 % personal, 63 % both.)

Major causes of workplace accidents (2022 3b, 2020 3d)

An accident arises from four elements: human activities, equipment/machines, materials/substances and the process/system. Causes:

- **Unsafe acts** (personal factors): working without authority, removing guards, not using PPE (personal protective equipment), horseplay, working fast or under the influence, wrong lifting, using defective tools, ignoring procedures.
- **Unsafe conditions** (environmental/mechanical): unguarded machines, defective equipment, poor housekeeping, poor lighting/ventilation, noise, heat, slippery floors, hazardous materials, no safe procedures.
- **Management factors**: poor training, poor supervision, wrong selection of workers, no maintenance, no safety rules or culture.
- **Human factors**: fatigue, stress, boredom, lack of knowledge, physical defects (eyesight, hearing), accident proneness.

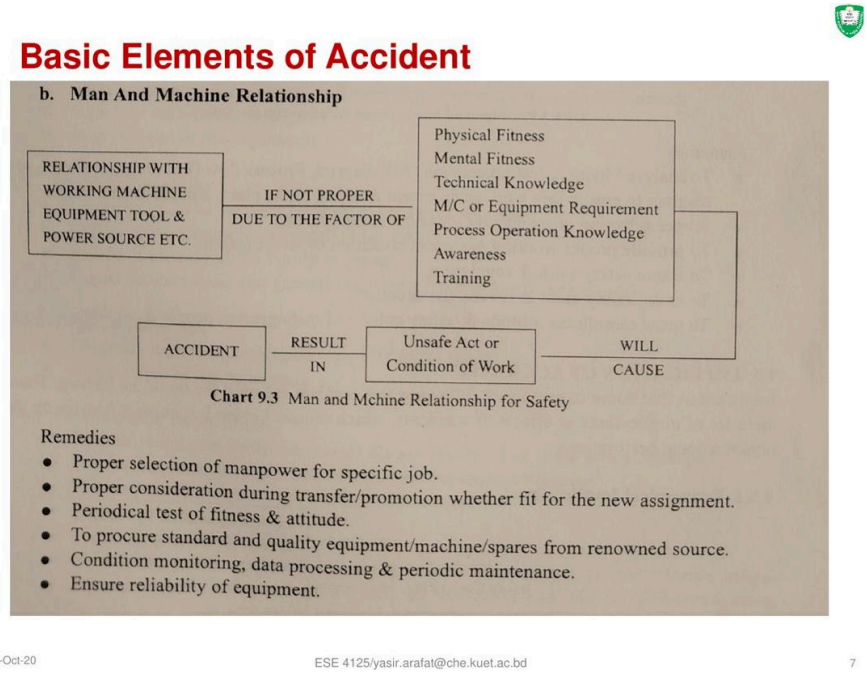
Three basic relationships (2021 2c)



Man-machine relationship: factors leading to unsafe acts/conditions and accidents

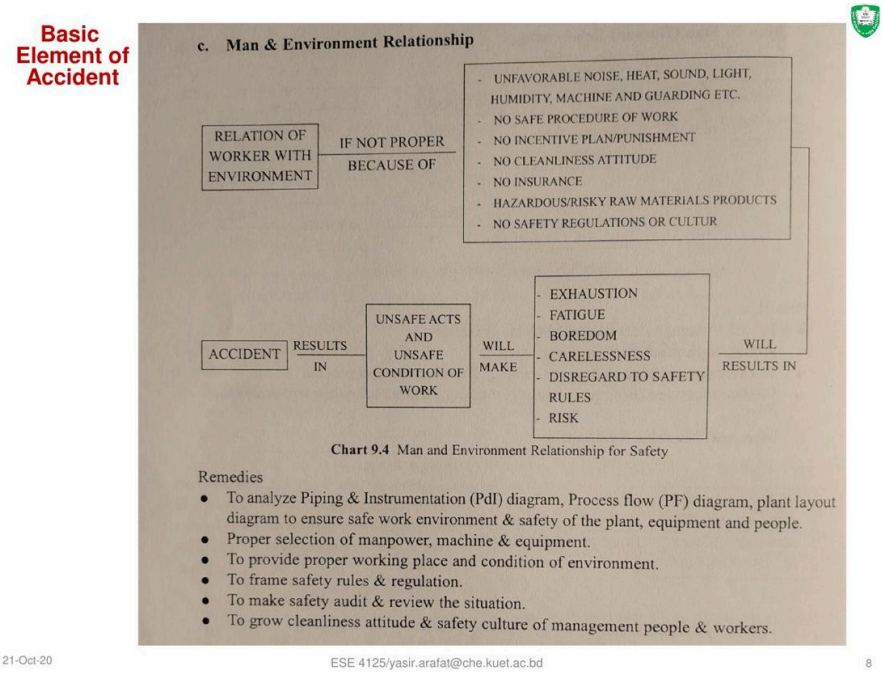
1. Man-machine relationship. The worker must be fit for the machine and the machine fit for the worker. Factors: physical fitness, mental fitness, technical knowledge, machine requirements, process knowledge, awareness, training. If any is lacking (e.g. an untrained operator on a high-speed press, a poorly maintained machine), it causes an unsafe act or unsafe condition and then an accident.

Remedies: proper selection of operators, fitness check on transfer/promotion, periodic fitness tests, training, standard equipment with guards, condition monitoring and maintenance, reliable machines, ergonomic design.



Man-man relationship

2. Man-man relationship. Relations between workers, and between workers and management (supervisor, union). Conflict, mistrust, delayed wages, poor communication lead to frustration, carelessness and unsafe acts. **Remedies:** bilateral agreements, prompt payment of dues, workers' participation, welfare, good communication.



Man-environment relationship

3. Man-environment relationship. Unfavourable noise, heat, light, humidity; no safe procedures; no incentive; no cleanliness; no insurance; hazardous materials; no regulations → exhaustion, fatigue, boredom, carelessness → unsafe acts. **Remedies:** analyse P&ID (piping and instrumentation diagram)/PFD/layout, proper workplace design, safety rules, safety audits, safety culture.

of musculoskeletal system. Some movements not only produce a local mechanical stress on the muscles & joints, but also require an expenditure of energy on the parts of muscle, heads & lungs. Physical & chemical environmental factors, such as noise, vibration, lighting, humidity and presence of chemical substance can also affect workers health and comfort. Ergonomics actually deals with the probable personal hazard associated with the work and prescribe remedies of them.

g. **Mental peace at work**

Workers mental condition at the working place is a vital issue related to safety. A mentally depressed worker at the working place will not be able to seriously concentrate to his job and has the probability to do mistakes resulting accident. It is one of the prime duties of the manger to keep their subordinates' moral high and mind peaceful. Job satisfaction not only reduces mental stress but also helps in promoting safety and in increasing productivity. So managers should try to have good working relation and shall control and manage other event which will promote a congenial work environment. Manager must appreciate workers for their good performance, shall promptly make receivable payments, timely promotions, logical compensations etc. Workers peace can also be hampered by act of outside people such as--his neighboures, friends, associates etc. Present day, companies provide residence for the workers, so they leave in a standard/controlled society. Company also provides certain entertainment facilities for recreation in the evening, schooling for their children, transport for shopping, organize sports, picnic etc. All these efforts promote better understanding among workers; grow teambuilding mind, better attitude to work and loyalty for the company.

h. **Human errors-slips, lapses, mistakes and violations**

Understanding human failures we need to understand human errors and violations.

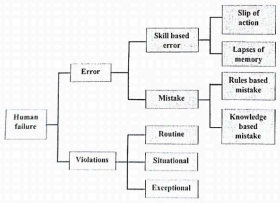


Chart 8.1 Human Errors and Violations

CamScanner

Human failure: errors (skill-based slips and lapses, rule- and knowledge-based mistakes) and violations

Memory trick: Dominoes: **"A Fault Under Any Injury"** = Ancestry, Fault of person, Unsafe act/condition, Accident, Injury. Heinrich ratio **88-10-2**.

A1 Basic definitions: hazard, risk, accident, occupational safety, health, safety management; the three parts of OSHEM

Must study · Very likely · 100% · Easy

PAST-PAPER QUESTIONS

- **2024 Q1(a)** [12] Define and explain: (i) Hazard, (ii) Occupational risk, (iii) Occupational accident, (iv) Safety management.
- **2023 Q1(a)** [12] Define and explain: (i) Hazard, (ii) Occupational accident, (iii) Occupational safety, (iv) Safety management.
- **2023 Q1(b)** [7] How do occupational safety, health and environmental management practices intersect to ensure comprehensive industrial workplace safety? Explain.
- **2022 Q5(d)** [7] Explain the necessity of study of the subject "Safety and Environmental Aspects of Energy Projects".
- **2021 Q1(a)** [5] What are the components of OSHEM (Occupational Safety, Health and Environment Management)? Which areas do they address?
- **2020 Q1(a)** [6] Define: hazard, occupational accident and occupational safety.

★ REMEMBER IT

- **Hazard** = something that *can* hurt (a wet floor). **Risk** = chance × how bad (how likely someone slips and how badly). **Accident** = it actually happened. Hazard is the dog, risk is the bite chance, accident is the bite.
- OSHEM = Occupational Safety (stop injuries) + Health (stop diseases) + Environment Management (stop harm outside the fence).

Before you start: what this subject is about

Every workplace (a power plant, a refinery, a factory) can hurt people in two ways: **suddenly** (a fall, a fire, an explosion: this is **safety**) and **slowly** (breathing dust for 20 years, noise for 10 years: this is **health**). It can also hurt people far away and future generations by polluting air, water and soil (this is **environment**).

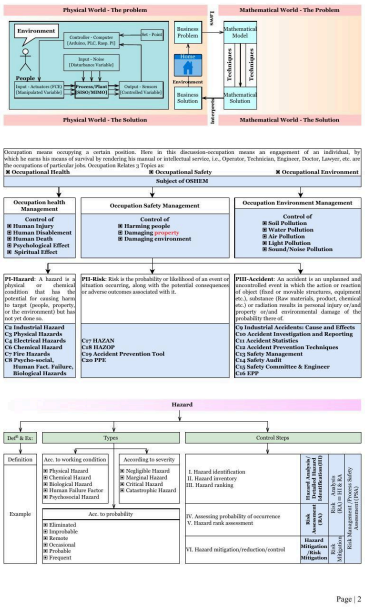
So the subject OSHEM = **O**ccupational **S**afety, **H**ealth and **E**nvironment **M**anagement has exactly these three parts. "Occupational" just means "related to a job".

Keep these three words apart in your head: **hazard** = the thing that can hurt (a wet floor), **risk** = how likely and how badly it will hurt (high in a busy corridor), **accident** = it actually hurt someone (a worker slipped and broke a leg).

Definitions (2024 1a, 2023 1a, 2020 1a)

Term	Definition (write this)	Example
Hazard	Any real or potential condition or act that could cause injury, illness or death to people, damage to property or the environment, or disrupt a process - **but has not yet done so**. (Webster: "a chance of being harmed or injured".)	LPG cylinder near a flame; unguarded rotating shaft; noise of 100 dB
Occupational risk	The probability (likelihood) that a workplace hazard will actually cause harm, combined with the severity of that harm. Risk = likelihood × consequence.	A worker near the noisy turbine 8 h every day has high risk of hearing loss
Occupational accident	An unplanned and uncontrolled event at work in which the action or reaction of an object, substance or radiation results in personal injury, property damage or environmental damage (Heinrich).	A ladder slips and the electrician falls 4 m
Occupational safety	Safety at the working place: workers know the hazards of their job and have the knowledge, training and protection to eliminate or handle them, so they stay unharmed. It protects people, property and environment from **sudden** harm.	Machine guarding, lockout before repair, safety training
Occupational health	Keeping the working conditions such that the worker does not develop a work-related (legally compensable) illness and stays free from discomfort - physically, mentally and socially.	Dust control to prevent silicosis; hearing protection
Safety management	The systematic process of ensuring a safe working environment by identifying, assessing and controlling risks to prevent accidents, injuries and illnesses. It keeps people unharmed, property undamaged and the environment unchanged. (Safety = management of *unintentional* losses; security = management of *intentional* losses.)	Safety policy, risk assessment, permit system, audits, safety committee, EPP (emergency preparedness plan)

Side note: Why does hazard come before accident? Because accidents are the **end result** of hazards that were not controlled. Chain: hazard → accident → loss of life and wealth → extra cost and interruption → possibly out of business.



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Teacher's summary: OSHEM = health management + safety management + environment management, and the definitions of hazard, risk and accident (Mostafizur Rahaman notes)

Components of OSHEM and what each addresses (2021 1a)

Component	Controls / addresses
Occupational **health** management	Health deterioration and disablement: human injury, disablement, death, psychological and spiritual effects of long exposure (dust, noise, chemicals, stress)
Occupational **safety** management	Sudden harm: harming people, damaging property, damaging environment by accidents, fire, explosion, toxic release
Environment management	Environmental degradation: air, water, soil, noise, light pollution; protects the community and future generations

How safety, health and environment management intersect (2023 1b)

Industrial activities have two kinds of effects: **slow and gradual** (from normal operation: slow damage to workers' health and slow environmental degradation) and **sudden** (from accidents: instantly harming people, property and environment). One event usually touches all three areas, so they cannot be managed separately:

- **Same root causes:** a gas leak is at once a safety hazard (fire/explosion), a health hazard (workers inhale it) and an environmental hazard (air pollution). One hazard identification and one risk assessment (HAZOP, bow-tie) covers all three.
- **Same control tools:** the hierarchy of controls (elimination, substitution, engineering, administrative, PPE (personal protective equipment)), training, permits, audits, emergency preparedness plans serve all three.
- **Same management system:** integrated management systems combine ISO (International Organization for Standardization) 45001 (OHS), ISO 14001 (environment) and ISO 9001 (quality) with one policy, one audit and one management review.
- **Failure in one shows up in another:** poor housekeeping causes slips (safety), dust exposure (health) and spills (environment).
- **Common goal (Triple Bottom Line):** people (safety + health), planet (environment) and profit are protected together.

Necessity of studying "Safety and Environmental Aspects of Energy Projects" (2022 5d)

1. Energy projects (power plants, refineries, gas fields, LPG plants) handle huge amounts of energy, flammable fuels, high pressure, high voltage and toxic chemicals - a single failure can kill many people (e.g. Bhopal 1984, CO_2 stripper explosion at UFFL Ghorashal 1991).
2. Engineers must identify hazards and design them out at the design stage, which is far cheaper than redesign after an accident.
3. Energy production is the biggest source of air pollution and greenhouse gases; EIA (environmental impact assessment) and environmental clearance (ECA 1995, ECR (Environment Conservation Rules (Bangladesh, 1997; new rules 2023)) 1997) are legally required for power projects (Red category).
4. Laws (Bangladesh Labour Act 2006, OSHA (Occupational Safety and Health Administration (US government agency)), ISO 14001) demand it; accidents bring compensation, litigation, insurance cost and loss of reputation.
5. Safe and clean plants are efficient plants: fewer shutdowns, higher productivity, better worker morale and public acceptance (sustainable development).

> **Memory trick: H-R-A**: **H**azard = can hurt, **R**isk = how likely × how bad, **A**ccident = did hurt. OSHEM = **S**udden (safety) + **S**low (health) + **S**urroundings (environment).

A15 Accident investigation and reporting sheet; accident statistics AFR, ASR, FAR (with numerical); HAZAN, HAZOP; safety audit Must study

Very likely · 93% Easy

PAST-PAPER QUESTIONS

- **2024 Q4(b)** [10] Prepare an accident reporting sheet for accidental investigation.
- **2024 Q4(c)** [5] Define: (i) accident frequency rate, (ii) accident severity rate, (iii) fatal accident rate, (iv) HAZAN (hazard analysis), (v) Triple Bottom Line (TBL).
- **2023 Q4(b)** [7] What are the purposes of accident report analysis? Explain AFR (accident frequency rate), ASR (accident severity rate) and FAR (fatal accident rate) with their significance.
- **2022 Q3(c)** [12] Briefly explain accident investigation and reporting. Prepare a simple accident reporting sheet.
- **2022 Q4(c)** [12] Define and explain: (i) AFR, (ii) ASR, (iii) FAR, (iv) HAZAN, (v) HAZOP (hazard and operability study), (vi) EPP (emergency preparedness plan).
- **2021 Q4(b)** [9] What are the different types of safety audits? Explain the important aspects to be kept during execution of a safety audit.
- **2021 Q4(c)** [6] Define: (i) accident frequency rate, (ii) accident severity rate, (iii) fatal accident rate.
- **2021 Q4(d)** [10] An undertaking with 600 workers, working 50 weeks a year with 40 working hours a week, had 60 accidents; the workers were absent during 5 % of the aggregate working time. Find the accident frequency rate.

UNDERSTAND **Read this to understand; you do not need to write it**

Accident rates are normalised by hours of exposure so big and small factories can be compared: per **million man-hours** for AFR and ASR, per **10^8 hours** for FAR. Absent workers are not exposed, so remove absence from the hours first.

CALCULATOR **Accident frequency, severity and fatal accident rates**

Change any number; every step is recalculated and shown with the numbers put in.

1. Total man-hours: **600** × **50** × **40** = **1200000** h
2. Hours actually worked: **1200000** × (1 - **5** / 100) = **1140000** h
3. Accident frequency rate: AFR = accidents × 10^6 / man-hours = **60** × 10^6 / **1140000** = **52.63** per 10^6 h
4. Accident severity rate: ASR = days lost × 10^6 / man-hours = **0** × 10^6 / **1140000** = **0.00** days per 10^6 h
5. Fatal accident rate: FAR = fatalities × 10^8 / man-hours = **0** × 10^8 / **1140000** = **0.00** per 10^8 h

Before you start: measuring safety with numbers

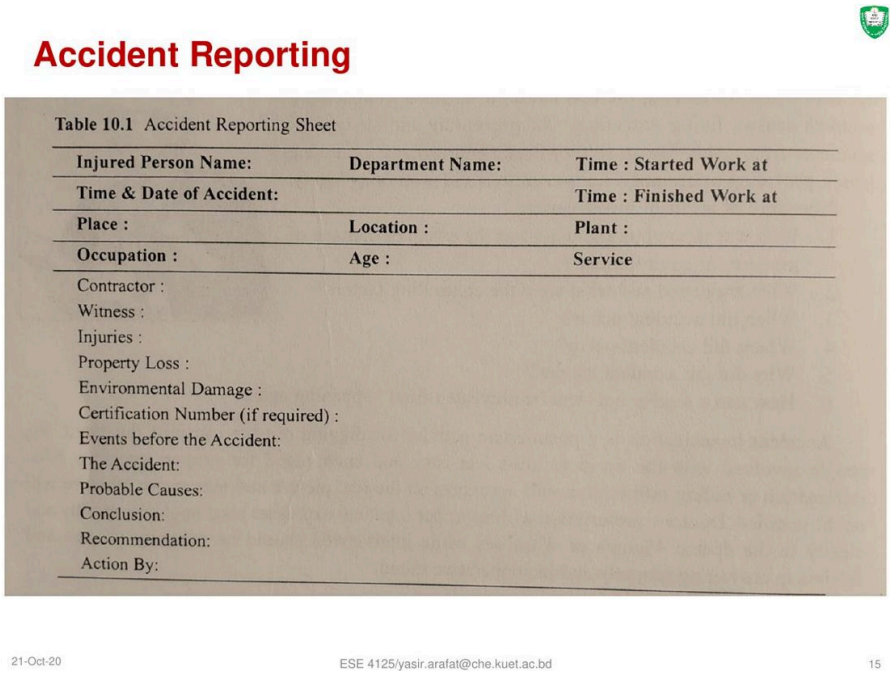
To compare the safety of two factories of different size, we divide the number of accidents by the total hours worked. Hours worked = number of workers × weeks × hours per week (minus absence). Because the answer is tiny, we multiply by 1 million (10^6) for AFR, by 1000 for ASR and by 100 million (10^8) for FAR.

AFR counts **how often** accidents happen, ASR counts **how serious** they are (time lost), FAR counts **deaths**.

Accident investigation and reporting (2022 3c)

Accident investigation is a systematic procedure to find the real (root) causes of an accident, not to blame people, so that similar accidents can be prevented. It answers: **who** was involved, **what** happened, **when**, **where**, **why** it happened, and **how** to prevent it. Steps: secure the scene and give first aid, collect evidence (photos, sketches, samples), interview witnesses, establish the sequence of events, find unsafe acts/conditions and root causes, recommend corrective actions, write the report, follow up. **Reporting** is the formal written record sent to management (and to the inspector of factories when required by the Labour Act 2006) for statistics, compensation and prevention.

Accident reporting sheet (2024 4b, 2022 3c)



Accident Reporting

Table 10.1 Accident Reporting Sheet

Injured Person Name:	**Department Name:**	**Time : Started Work at**
Time & Date of Accident:		**Time : Finished Work at**
Place :	**Location :**	**Plant :**
Occupation :	**Age :**	**Service**

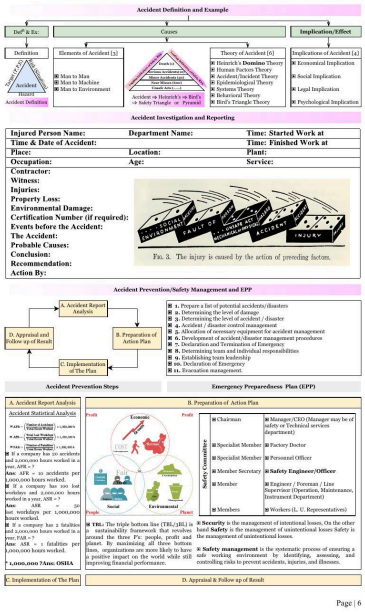
Contractor :
Witness :
Injuries :
Property Loss :
Environmental Damage :
Certification Number (if required) :
Events before the Accident:
The Accident:
Probable Causes:
Conclusion:
Recommendation:
Action By:

21-Oct-20 ESE 4125/yasir.arafat@che.kuet.ac.bd 15

Accident investigation report sheet (Yasir Arafat slides)

Section	Information to record
1. Identification	Organisation/department, report no., date and time of accident, date of report, exact location
2. Injured person	Name, ID, age, sex, designation/job, experience, shift, supervisor
3. Injury / damage	Nature of injury, part of body, severity (first aid, lost time, fatal), treatment given, hospital; property/equipment damaged, estimated cost; environmental release
4. Description	What the person was doing; sequence of events; equipment, materials and tools involved; witnesses' names and statements; sketch/photo
5. Causes	Immediate causes (unsafe act, unsafe condition); root causes (training, procedure, maintenance, supervision)
6. Corrective actions	Recommendations, responsible person, target date, follow-up status
7. Sign-off	Investigated by, reviewed by safety officer, approved by manager, date

Purposes of accident report analysis (2023 4b)



Teacher's one-page summary: accident theories, investigation, reporting sheet, AFR/ASR/FAR, EPP, TBL (triple bottom line), safety committee

- Find the causes and trends of accidents (which department, job, time, type).
- Plan and prioritise preventive actions and training.
- Compare safety performance between periods, plants and industries.
- Measure the effectiveness of the safety programme.
- Provide data for compensation, insurance and legal requirements.
- Motivate management and workers (accident prevention cycle: report analysis → action plan → implementation → appraisal and follow-up).

AFR, ASR, FAR (2024 4c, 2023 4b, 2022 4c, 2021 4c)

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AFR = number of lost-time injuries × 10^6 / total man-hours worked
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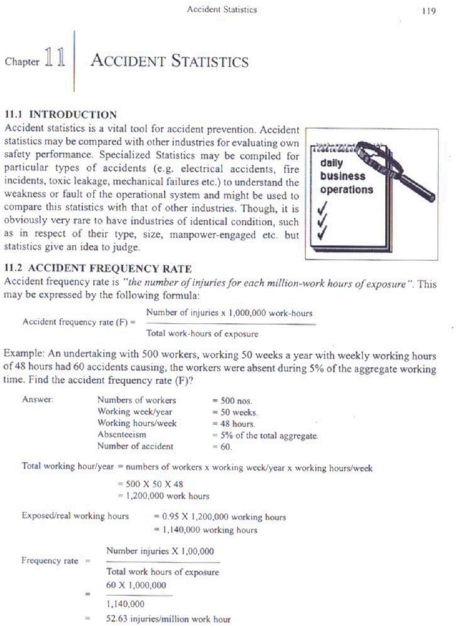
```
ASR = man-days (or hours) lost × 1000 / total man-hours worked
```

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FAR = number of fatalities × 10^8 / total man-hours of exposure
```

Rate	Meaning	Significance
AFR (accident frequency rate)	Lost-time injuries per million man-hours	How **often** accidents happen; compares plants of different size
ASR (accident severity rate)	Days/hours lost per 1000 man-hours (death and permanent disablement counted by standard equivalent days)	How **serious** the accidents are
FAR (fatal accident rate)	Deaths per 10^8 hours of exposure (about 1000 workers' working lifetimes)	Compares the **risk of death** between jobs and activities

> **Side note:** The teacher's notes also use the OSHA (Occupational Safety and Health Administration (US government agency)) style with 10^6 for all three: 2,000,000 h, 20 accidents, 100 lost days, 2 deaths → AFR 10, ASR 50, FAR 1. Write the formula you use and keep the multiplier stated.

Numerical (2021 4d)



Accident Statistics 119

Chapter 11 ACCIDENT STATISTICS

11.1 INTRODUCTION

Accident statistics is a vital tool for accident prevention. Accident statistics may be compared with other industries for evaluating own safety performance. Specialized Statistics may be compiled for particular types of accidents (e.g. electrical accidents, fire incidents, toxic leakage, mechanical failures etc.) to understand the weakness or fault of the operational system and might be used to compare this statistics with that of other industries. Though, it is obviously very rare to have industries of identical condition, such as in respect of their type, size, manpower-engaged etc. but statistics give an idea to judge.

11.2 ACCIDENT FREQUENCY RATE

Accident frequency rate is *"the number of injuries for each million-work hours of exposure"*. This may be expressed by the following formula:

$$\text{Accident frequency rate (F)} = \frac{\text{Number of injuries x 1,000,000 work-hours}}{\text{Total work-hours of exposure}}$$

Example: An undertaking with 500 workers, working 50 weeks a year with weekly working hours of 48 hours had 60 accidents causing, the workers were absent during 5% of the aggregate working time. Find the accident frequency rate (F)?

Answer:

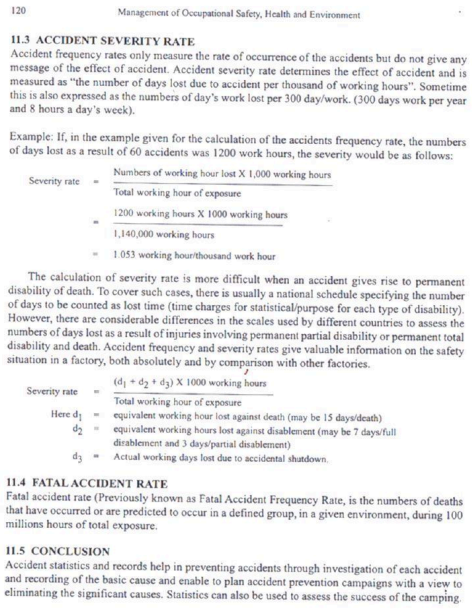
Numbers of workers = 500 nos.
Working week/year = 50 weeks.
Working hours/week = 48 hours.
Absenteeism = 5% of the total aggregate.
Number of accident = 60.

Total working hour/year = numbers of workers x working week/year x working hours/week
= 500 X 50 X 48
= 1,200,000 work hours

Exposed/real working hours = 0.95 X 1,200,000 working hours
= 1,140,000 working hours

$$\text{Frequency rate} = \frac{\text{Number injuries X 1,00,000}}{\text{Total work hours of exposure}} = \frac{60 \text{ X } 1,000,000}{1,140,000} = 52.63 \text{ injuries/million work hour}$$

Textbook chapter 11: accident frequency rate with the same worked example (52.63 per million hours)



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11.3 ACCIDENT SEVERITY RATE

Accident frequency rates only measure the rate of occurrence of the accidents but do not give any message of the effect of accident. Accident severity rate determines the effect of accident and is measured as "the number of days lost due to accident per thousand of working hours". Sometime this is also expressed as the numbers of day's work lost per 300 day/work. (300 days work per year and 8 hours a day's week).

Example: If, in the example given for the calculation of the accidents frequency rate, the numbers of days lost as a result of 60 accidents was 1200 work hours, the severity would be as follows:

$$\text{Severity rate} = \frac{\text{Numbers of working hour lost X 1,000 working hours}}{\text{Total working hour of exposure}}$$

$$= \frac{\text{1200 working hours X 1000 working hours}}{\text{1,140,000 working hours}}$$

$$= \text{1.053 working hour/thousand work hour}$$

The calculation of severity rate is more difficult when an accident gives rise to permanent disability of death. To cover such cases, there is usually a national schedule specifying the number of days to be counted as lost time (time charges for statistical/purpose for each type of disability). However, there are considerable differences in the scales used by different countries to assess the numbers of days lost as a result of injuries involving permanent partial disability or permanent total disability and death. Accident frequency and severity rates give valuable information on the safety situation in a factory, both absolutely and by comparison with other factories.

$$\text{Severity rate} = \frac{(d_1 + d_2 + d_3) \text{ X 1000 working hours}}{\text{Total working hour of exposure}}$$

Here d_1 = equivalent working hour lost against death (may be 15 days/death)

d_2 = equivalent working hours lost against disablement (may be 7 days/full disablement and 3 days/partial disablement)

d_3 = Actual working days lost due to accidental shutdown.

11.4 FATAL ACCIDENT RATE

Fatal accident rate (Previously known as Fatal Accident Frequency Rate, is the numbers of deaths that have occurred or are predicted to occur in a defined group, in a given environment, during 100 millions hours of total exposure.

11.5 CONCLUSION

Accident statistics and records help in preventing accidents through investigation of each accident and recording of the basic cause and enable to plan accident prevention campaigns with a view to eliminating the significant causes. Statistics can also be used to assess the success of the camping.

Textbook: accident severity rate and fatal accident rate

Step (write this)	Why / what it means
`Workers = 600, 50 weeks/yr, 40 h/week, 60 accidents, absence 5 %`	Write the data.
`Total man-hours = 600 × 50 × 40 = 1,200,000 h`	Every worker's hours added together.
`Actual hours worked = 1,200,000 × (1 - 0.05) = 1,140,000 h`	Absent workers are not exposed, so remove 5 %.
`AFR = 60 × 10^6 / 1,140,000`	Accidents per million hours actually worked.
`AFR = 52.63 accidents per million man-hours`	Final answer.

Other definitions (2024 4c, 2022 4c)

- **HAZAN (hazard analysis)**: the quantitative estimation of how often a hazard can occur and how severe its consequences are, compared with a target, to decide whether the risk is acceptable. HAZOP finds *what* can go wrong; HAZAN calculates *how likely and how bad*.
- **HAZOP (hazard and operability study)**: a systematic team study of a process, node by node, using guide words to find deviations, their causes, consequences and safeguards (A10).
- **EPP (emergency preparedness plan)**: a written plan for responding to emergencies (A16).
- **TBL (triple bottom line)**: judging a business on people, planet and profit (A16).

Types of safety audit and aspects during execution (2021 4b)

A **safety audit** is a systematic, independent examination of a plant's safety management, hazards and practices against standards, to find weaknesses and recommend improvements.

- **Types:** (1) **Walk-through** (preliminary) audit - quick visual inspection; (2) **Comprehensive (detailed)** audit - complete review of all systems and records; (3) **Specific/topical** audit - one area (fire, electrical, process safety, PPE (personal protective equipment)); (4) **Internal** audit by the company's own team vs **external/third-party** audit by independent experts or regulators; (5) **compliance** audit against laws; (6) pre-start-up audit of a new plant.
- **Aspects to keep during execution:** clear objective and scope; competent, independent auditors; checklist prepared beforehand; management support and cooperation of workers; inspect the actual site, not just papers; examine records (accidents, permits, maintenance, training, MSDS

(material safety data sheet)); interview operators; check emergency equipment and drills; rate findings by risk; write a clear report with recommendations, responsibilities and target dates; follow up implementation.

Memory trick: Multipliers: **AFR million, ASR thousand, FAR hundred million**. Man-hours = **workers × weeks × hours × (1 − absence)**.

WRITE IN EXAM **Write this in the exam (this much is enough for full marks)**

2021 4(d) Given: 600 workers, 50 weeks, 40 h/week, 60 accidents, absence 5 %

Total man-hours: workers × weeks × hours

= 600 × 50 × 40

= **1,200,000 h**

Actual hours worked: 1,200,000 × (1 − 0.05)

= 1,140,000

= **1,140,000 h**

AFR: accidents × 10^6 / man-hours

= 60 × 10^6 / 1,140,000

= **52.63 per million man-hours**

★ REMEMBER IT

- **F**requency counts accidents, **S**everity counts days lost, **F**atal counts deaths (per 10^8 h, roughly a working lifetime of 1000 people).

A2 Sources of hazard, categories of hazard by working situation, consequences, control steps, psychosocial hazards

Must study · Very likely · 93% · Easy

PAST-PAPER QUESTIONS

- **2024 Q1(b)** [12] Explain the various categories of hazards according to working situation. What are the major sources of hazards and their effects?
- **2023 Q1(d)** [6] What are the psychosocial hazards and how do they affect workplace safety?
- **2022 Q1(a)** [10] Define hazard. What are the typical sources of hazards and consequences of them?
- **2022 Q1(b)** [15] Explain different categories of hazards according to the working situation. (numerical part in A3)
- **2022 Q1(c)** [10] Discuss the measures to control the hazards.
- **2021 Q1(b)** [10] Define hazard. What are the typical sources of hazard? What are the consequences of hazard?

★ REMEMBER IT

- Hazard categories, **"Please Check Every Bio-Psych Ergo"**: Physical, Chemical, Electrical/mechanical, Biological, Psychosocial, Ergonomic.
- Control steps: identify → assess → control (hierarchy) → monitor and review.

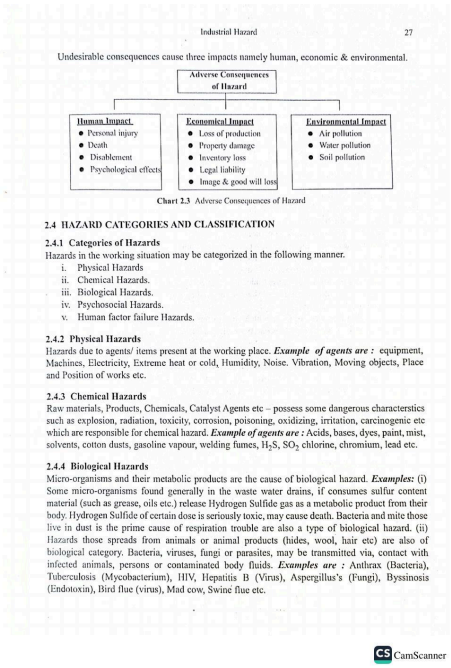
Before you start: the idea of hazard categories

There are thousands of hazards, so we sort them by **what kind of agent** causes the harm. Ask: is it energy or a physical thing (physical)? a substance (chemical)? a living thing (biological)? the mind (psychosocial)? a human mistake (human-factor)? That gives the five categories.

Then for each hazard we look at **where it comes from** (source) and **what it can do** (consequence).

Categories of hazards according to the working situation (2024 1b, 2022 1b)

Category	Meaning	Examples of agents
1. Physical	Agents/items present at the workplace that transfer energy to the body	Machines, moving objects, electricity, extreme heat or cold, humidity, noise, vibration, radiation, working at height, position of work
2. Chemical	Raw materials, products, catalysts with dangerous properties: explosive, toxic, corrosive, oxidizing, irritant, carcinogenic	Acids, bases, solvents, paint, cotton dust, gasoline vapour, welding fumes, H_2S, SO_2, chlorine, chromium, lead
3. Biological	Micro-organisms and their metabolic products, animals and animal products	Bacteria (anthrax, TB), viruses (HIV, hepatitis B, bird flu), fungi (aspergillus), endotoxin (byssinosis), H_2S from bacteria in drains, mites in dust
4. Psychosocial	From working relationships or situations that put mental pressure on the worker	Stress, monotony, fatigue, brain fag, irritation, insecurity, poor relations, bullying
5. Human-factor failure	From human faults and lapses	Unsafe work practice, ergonomic factors (forceful movement, lifting heavy weight), poor communication, poor training, improper manning, accident proneness, negligence of management



Industrial Hazard 27

Undesirable consequences cause three impacts namely human, economic & environmental.

Chart 2.3 Adverse Consequences of Hazard

2.4 HAZARD CATEGORIES AND CLASSIFICATION

2.4.1 Categories of Hazards

Hazards in the working situation may be categorized in the following manner.

i. Physical Hazards
ii. Chemical Hazards.
iii. Biological Hazards.
iv. Psychosocial Hazards.
v. Human factor failure Hazards.

2.4.2 Physical Hazards

Hazards due to agents/ items present at the working place. ***Example of agents are :*** equipment, Machines, Electricity, Extreme heat or cold, Humidity, Noise. Vibration, Moving objects, Place and Position of works etc.

2.4.3 Chemical Hazards

Raw materials, Products, Chemicals, Catalyst Agents etc – possess some dangerous characterstics such as explosion, radiation, toxicity, corrosion, poisoning, oxidizing, irritation, carcinogenic etc which are responsible for chemical hazard. ***Example of agents are :*** Acids, bases, dyes, paint, mist, solvents, cotton dusts, gasoline vapour, welding fumes, H_2S, SO_2 chlorine, chromium, lead etc.

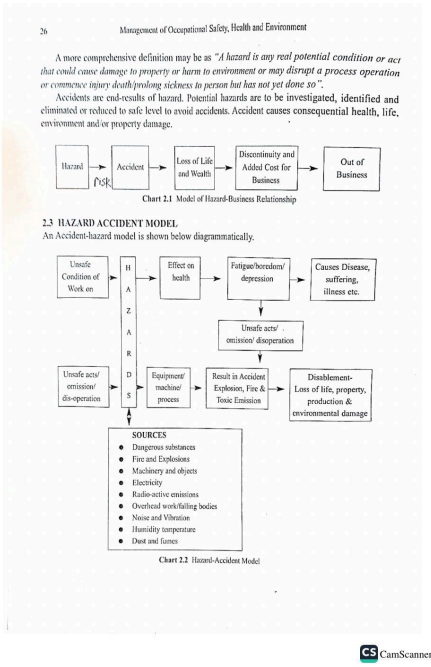
2.4.4 Biological Hazards

Micro-organisms and their metabolic products are the cause of biological hazard. ***Examples:*** (i) Some micro-organisms found generally in the waste water drains, if consumes sulfur content material (such as grease, oils etc.) release Hydrogen Sulfide gas as a metabolic product from their body. Hydrogen Sulfide of certain dose is seriously toxic, may cause death. Bacteria and mite those live in dust is the prime cause of respiration trouble are also a type of biological hazard. (ii) Hazards those spreads from animals or animal products (hides, wool, hair etc) are also of biological category. Bacteria, viruses, fungi or parasites, may be transmitted via, contact with infected animals, persons or contaminated body fluids. ***Examples are :*** Anthrax (Bacteria), Tuberculosis (Mycobacterium), HIV, Hepatitis B (Virus), Aspergillus's (Fungi), Byssinosis (Endotoxin), Bird flue (virus), Mad cow, Swine flue etc.

CamScanner

Adverse consequences of hazard (human, economic, environmental) and the five hazard categories (textbook ch. 2)

Major sources of hazards (2024 1b, 2022 1a, 2021 1b)



Hazard-accident model: unsafe conditions and unsafe acts turn sources of hazard into disease or accidents (textbook ch. 2)

- Dangerous substances (toxic, flammable, corrosive chemicals)
- Fire and explosions
- Machinery and moving objects
- Electricity
- Radio-active emissions
- Overhead work and falling bodies
- Noise and vibration
- Humidity and temperature extremes
- Dust and fumes

These sources become dangerous through **unsafe conditions of work** (no guard, poor ventilation) and **unsafe acts** (omission, wrong operation). They lead either to **slow effects on health** (fatigue → disease, suffering) or to **sudden accidents** (explosion, fire, toxic emission → loss of life, property, production, environment).

Consequences (effects) of hazards

Human impact	Economic impact	Environmental impact
Personal injury, death, disablement, psychological effects	Loss of production, property damage, inventory loss, legal liability, loss of image and goodwill	Air pollution, water pollution, soil pollution

Severity and probability classes (used later in the hazard matrix)

- **Catastrophic**: death, widespread illness, loss of entire production. **Critical**: death or serious injury (loss of eye/limb), major damage. **Marginal**: minor injury (cuts), minor illness, slight damage. **Negligible**: no serious injury or damage.
- Probability: **frequent** (almost every time), **probable** (most of the time), **occasional** (sometimes), **remote** (very unlikely), **improbable** (no chance).

Measures to control hazards (2022 1c)

Hazard control is a continuous process from the **design phase**, through **operation**, to **dismantling**. The six basic steps:

1. **Hazard identification**: experts audit the workplace and list every hazard (checklist, what-if, HAZOP (hazard and operability study)).
2. **Hazard inventory**: record each hazard, where it is and how much material is involved.
3. **Hazard ranking**: rank in descending order of severity.
4. **Assessing probability of occurrence**: how often it may happen.
5. **Hazard rank assessment**: severity × probability rating (hazard matrix, see A3) to decide acceptable or not.
6. **Elimination / reduction / control**: remove the major hazard or replace it by a less severe one; if impossible, prepare to handle it.

Practical control measures follow the **hierarchy of controls** (details in A12): **elimination** → **substitution** → **engineering controls** (guards, ventilation, interlocks, relief valves) → **administrative controls** (procedures, permits, training, signs, job rotation, housekeeping) → **PPE** (personal protective equipment). Plus monitoring, inspection, maintenance and an emergency preparedness plan.

Psychosocial hazards and workplace safety (2023 1d)

Psychosocial hazards arise when a worker fails to adapt to the psychological and social environment of work: frustration, lack of job satisfaction, insecurity, poor human relationships, emotion, tension, monotony, fatigue, excessive workload, long hours and shift work, bullying or harassment.

Effect on safety: a stressed, tired or bored worker becomes less attentive, slower in action and careless about precautions, so **unsafe acts** (and accidents) increase; mental fatigue also lowers productivity and causes absenteeism, depression and illness. **Control:** congenial work environment, fair pay and timely promotion, job rotation, rest breaks, participation, counselling, good supervisor relations.

> **Memory trick:** Five categories = **"Please Check Before People Hurt"**: **P**hysical, **C**hemical, **B**iological, **P**sychosocial, **H**uman-factor.

A10 Hazard identification methods: checklist, what-if, PHA, HAZOP, FMEA, plant drawings; limitations of hazard evaluation Must study

Very likely · 87% Medium

PAST-PAPER QUESTIONS

- **2024 Q2(c)** [10] How do plot plans, isometric drawings, BFDs, PFDs and P&IDs provide the essential system information needed for effective hazard identification?
- **2023 Q3(d)** [5] Differentiate between Preliminary Hazard Analysis (PHA) and HAZOP (hazard and operability study) analysis.
- **2022 Q4(a)** [12] Make a list of various kinds of hazard identification methodologies. (anatomy part in A11)
- **2022 Q4(b)** [11] A design feeds hydrogen sulfide (H_2S) from a cylinder to a processing unit. The analyst knows only that H_2S is toxic and flammable and identifies its potential release as hazardous. Do a preliminary hazard analysis (PHA).
- **2020 Q1(b)** [12] What are the methods of hazard identification? Describe the HAZOP procedure.
- **2020 Q3(a)** [5] What are the limitations of a hazard evaluation study?

LEARN IT FROM ZERO

Hazard identification methods, from zero

Before building or running a plant, engineers ask "what could go wrong?". The methods differ in how deep and formal the asking is:

- **Checklist**: tick a ready list of known problems. Fast, but only finds what is on the list.
- **What-if**: a team asks "what if the pump stops? what if the operator opens the wrong valve?".
- **PHA** (preliminary hazard analysis) (preliminary hazard analysis): early design stage, rough list of hazards and their severity, before details exist.
- **HAZOP** (hazard and operability study): detailed design, takes the P&ID (piping and instrumentation diagram) line by line ("node"), and applies **guide words** (NO, MORE, LESS, REVERSE, OTHER THAN, AS WELL AS, PART OF) to each parameter (flow, temperature, pressure). "NO flow" → causes → consequences → safeguards → actions.
- **FMEA** (failure modes and effects analysis) (failure mode and effects analysis): takes each piece of equipment and asks how it can fail and what happens.

PHA vs HAZOP: PHA is early, quick and rough; HAZOP is later, systematic and needs the full drawings.

★ REMEMBER IT

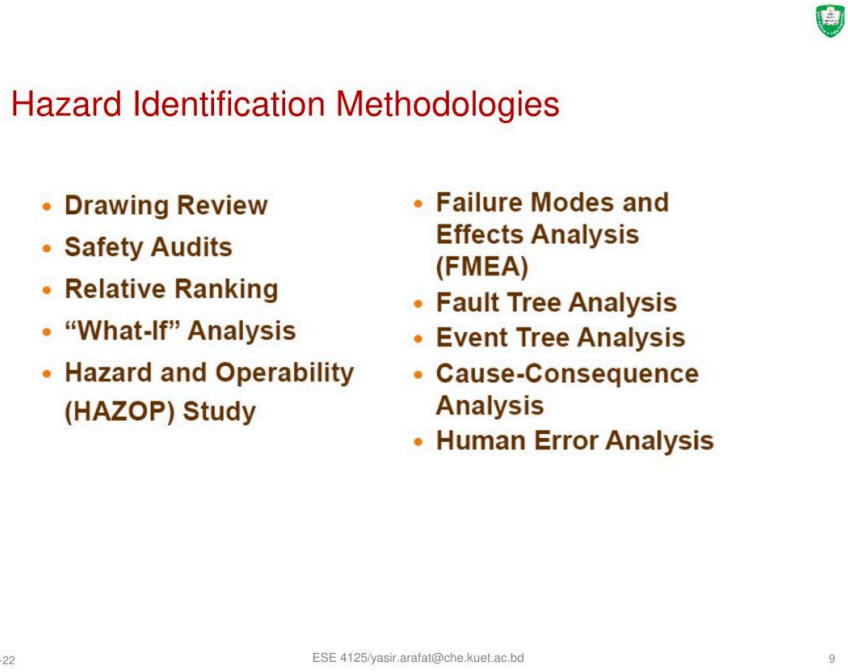
- HAZOP guide words: **No, More, Less, Reverse, Other than, As well as, Part of** ("No More Less, Reverse Others' Parts").
- Drawings for hazard ID: plot plan (where things are), BFD (blocks), PFD (process flow with conditions), P&ID (every pipe, valve, instrument), isometric (3-D piping).

Before you start: finding hazards before they hurt

Hazard identification (HI) answers **"what can go wrong?"** on paper, before it happens in the plant. You cannot look at a whole refinery at once, so you study its **drawings** and go through it piece by piece with a method (a checklist, a brainstorm, or a strict word-by-word study like HAZOP).

Then risk assessment asks **"how likely?"** and **"how bad?"** (A3, A12).

Hazard identification methodologies (2022 4a, 2020 1b)



Common hazard identification methods and the hazard evaluation methodologies

Method	What it is	When used
Analysing material properties and process conditions	Check flammability, toxicity, reactivity (MSDS), pressure, temperature	Always, first step
Reviewing organisational and industry experience	Past incidents in this and similar plants	Always
Interaction matrix	Grid of chemical × chemical (or material) to find dangerous combinations	Reactive chemicals
Checklist	List of standard questions/requirements ticked yes/no	Any stage; quick
Safety audit / drawing review	Inspection of plant or drawings by experts	Existing plant
Relative ranking (Dow, Mond indices)	Score units to rank their hazard	Choosing where to study first
PHA (preliminary hazard analysis)	Rough early list of major hazards	Concept/early design
What-if	Team brainstorm of "what if ...?" questions	Any stage
HAZOP (hazard and operability study)	Systematic node-by-node study with guide words on P&ID	Detailed design
FMEA (failure modes and effects analysis)	Each component: how can it fail, what happens	Equipment level
FTA (fault tree analysis) / **ETA** (event tree analysis), cause-consequence, human error analysis	Logic trees for causes and outcomes (A11)	Quantifying risk

How plant drawings provide system information for HI (2024 2c)

You can only identify hazards in a system you can see clearly. Each drawing shows a different layer:

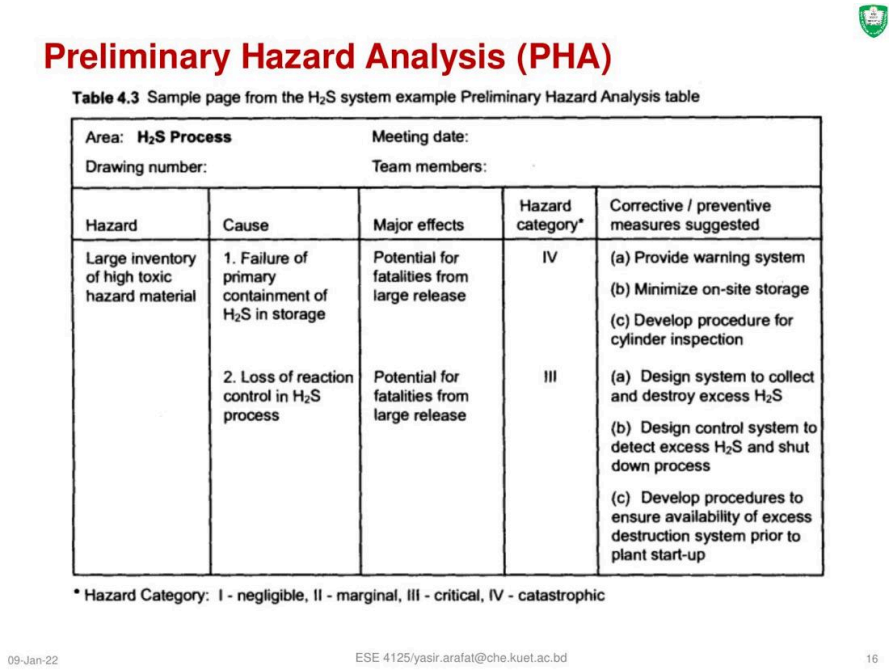
Drawing	Shows	Use for hazard identification
Plot plan	Top view of the whole site: location of units, tanks, buildings, roads, distances	Spacing between hazardous units, fire/explosion spread, escape routes, location of control room, emergency access, wind direction
Isometric drawing	3-D single-line drawing of piping with dimensions, elevations, fittings	Pipe stress, supports, low points (liquid traps), flanges and leak points, access for maintenance
BFD (block flow diagram) (block flow diagram)	Simple blocks for each process step with main streams	Overall picture: where hazardous materials go, which sections to study first
PFD (process flow diagram) (process flow diagram)	Main equipment, flows, temperatures, pressures, mass and energy balance	Inventories, operating conditions, what happens if flow/temperature/pressure change
P&ID (piping and instrumentation diagram)	Every line, valve, instrument, control loop, alarm, trip, relief valve	The main input to HAZOP: nodes, deviations, existing safeguards

Together with the single-line electrical diagram and wiring drawings, they let the team divide the plant into nodes, know the intention of each part, see existing safeguards and find the weak points.

PHA (preliminary hazard analysis)

PHA is done in the **early (concept) phase** of a new plant, when little detail exists, to list the major hazards and decide what to design out. It covers: raw materials, products and their reactivity; equipment; interfaces between parts; operating environment; operations (testing, maintenance); facility; safety equipment. Hazard categories: **I negligible, II marginal, III critical, IV catastrophic**. Worksheet columns: **hazard | cause | major effects | hazard category | corrective/preventive measures**.

PHA for an H_2S cylinder feed (2022 4b)



Preliminary Hazard Analysis (PHA)

Table 4.3 Sample page from the H_2S system example Preliminary Hazard Analysis table

Area: **H_2S Process**		Meeting date:		
Drawing number:		Team members:		
Hazard	Cause	Major effects	Hazard category*	Corrective / preventive measures suggested
Large inventory of high toxic hazard material	1. Failure of primary containment of H_2S in storage	Potential for fatalities from large release	IV	(a) Provide warning system (b) Minimize on-site storage (c) Develop procedure for cylinder inspection
	2. Loss of reaction control in H_2S process	Potential for fatalities from large release	III	(a) Design system to collect and destroy excess H_2S (b) Design control system to detect excess H_2S and shut down process (c) Develop procedures to ensure availability of excess destruction system prior to plant start-up

* Hazard Category: I - negligible, II - marginal, III - critical, IV - catastrophic

09-Jan-22 ESE 4125/yasir.arafat@che.kuet.ac.bd 16

Textbook PHA worksheet for the H_2S system

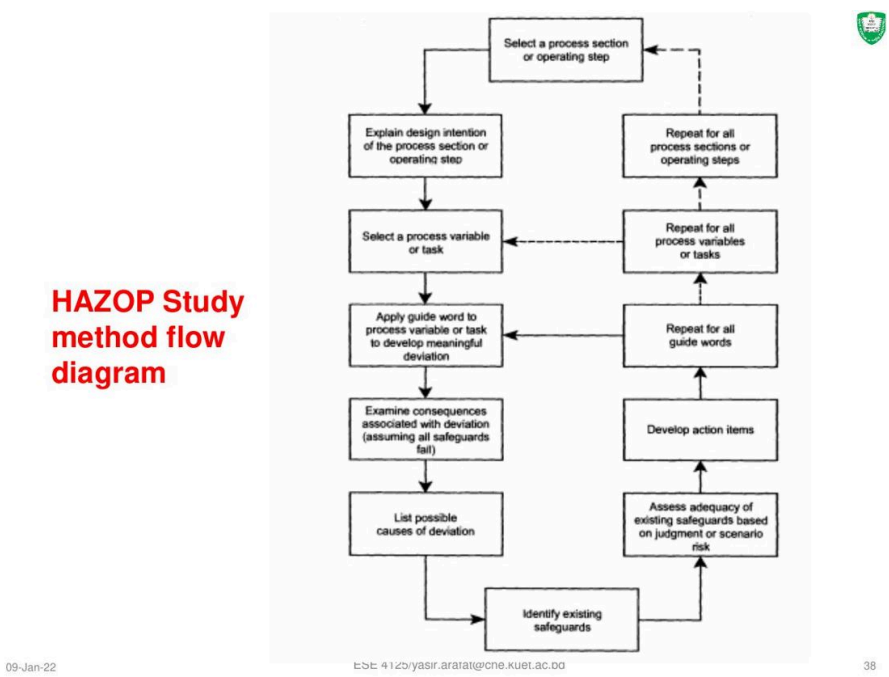
Hazard	Cause	Major effects	Category	Corrective / preventive measures
Toxic release (large inventory of a highly toxic material)	Failure of primary containment of H_2S in storage (cylinder, valve, connection leaks)	Potential for fatalities from a large release	IV	(a) Provide a warning (H_2S detection and alarm) system; (b) minimise on-site storage; (c) develop a procedure for cylinder inspection
Toxic release	Loss of reaction control in the H_2S process (excess H_2S not consumed)	Potential for fatalities from a large release	III	(a) Design a system to collect and destroy excess H_2S (scrubber/flare); (b) control system to detect excess H_2S and shut down the process; (c) procedures so the destruction system is available before start-up
Fire / explosion (H_2S is flammable, LEL (lower explosive limit) 4 %)	Leak meets an ignition source	Burns, fire spreading, SO_2 formed	III	Ventilated cylinder enclosure, no ignition sources, flameproof equipment, excess-flow valve, emergency shut-off

PHA vs HAZOP (2023 3d)

	PHA	HAZOP
Stage	Early concept / preliminary design	Detailed design (P&ID ready) or existing plant
Detail	Rough, broad; whole system	Very detailed, node by node, line by line
Method	Experience-based list: hazard → cause → effect → category → measure	Structured guide words (NO, MORE, LESS, REVERSE ...) + parameters give deviations
Team / time	Small team, short (hours to days)	Multidisciplinary team, long (weeks)
Output	List of major hazards and categories I-IV	Deviations, causes, consequences, safeguards, actions; also operability problems

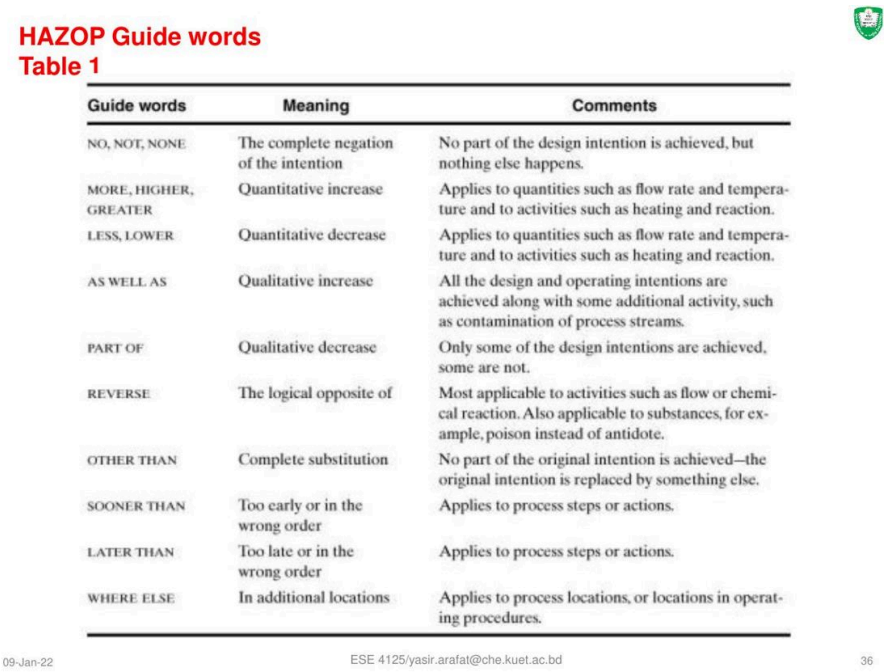
HAZOP procedure (2020 1b)

HAZOP (hazard and operability study) is a formal procedure that lets the team's imagination run free in a controlled way to find every way a process can deviate from its design intention. It needs PFD, P&ID, equipment specifications, materials data and mass and energy balances.



HAZOP procedure flow chart

1. Begin with a detailed flow sheet (P&ID); break it into process **units**.
2. Choose a **study node** (a vessel, a line, an operating instruction).
3. Describe the design **intention** of the node (e.g. transfer 10 m^3/h of feed at 5 bar).
4. Pick a **process parameter** (flow, pressure, temperature, level, composition).
5. Apply a **guide word** to the parameter to make a **deviation** (NO + flow = no flow).
6. If the deviation is possible, list its **causes** and the existing **safeguards** (protective systems).
7. Evaluate the **consequences** of the deviation.
8. Recommend **action** (what, by whom, by when).
9. **Record** everything on the HAZOP worksheet.
10. Repeat steps 5-9 for all guide words, then step 4 for all parameters, then step 2 for all nodes, then step 1 for all units.



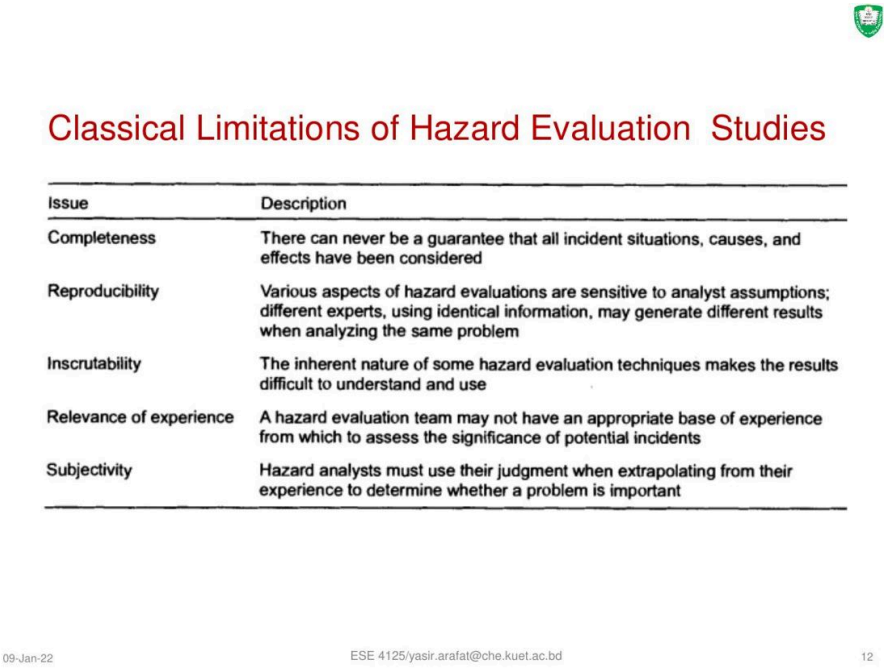
HAZOP Guide words
Table 1

Guide words	Meaning	Comments
NO, NOT, NONE	The complete negation of the intention	No part of the design intention is achieved, but nothing else happens.
MORE, HIGHER, GREATER	Quantitative increase	Applies to quantities such as flow rate and temperature and to activities such as heating and reaction.
LESS, LOWER	Quantitative decrease	Applies to quantities such as flow rate and temperature and to activities such as heating and reaction.
AS WELL AS	Qualitative increase	All the design and operating intentions are achieved along with some additional activity, such as contamination of process streams.
PART OF	Qualitative decrease	Only some of the design intentions are achieved, some are not.
REVERSE	The logical opposite of	Most applicable to activities such as flow or chemical reaction. Also applicable to substances, for example, poison instead of antidote.
OTHER THAN	Complete substitution	No part of the original intention is achieved—the original intention is replaced by something else.
SOONER THAN	Too early or in the wrong order	Applies to process steps or actions.
LATER THAN	Too late or in the wrong order	Applies to process steps or actions.
WHERE ELSE	In additional locations	Applies to process locations, or locations in operating procedures.

HAZOP guide words and their meaning

Guide words: NO/NOT/NONE (complete negation), MORE/HIGHER (quantitative increase), LESS/LOWER (decrease), AS WELL AS (extra material/activity), PART OF (only some), REVERSE (opposite), OTHER THAN (complete substitution), SOONER/LATER (timing), WHERE ELSE (location).

Limitations of a hazard evaluation study (2020 3a)



Classical Limitations of Hazard Evaluation Studies

Issue	Description
Completeness	There can never be a guarantee that all incident situations, causes, and effects have been considered
Reproducibility	Various aspects of hazard evaluations are sensitive to analyst assumptions; different experts, using identical information, may generate different results when analyzing the same problem
Inscrutability	The inherent nature of some hazard evaluation techniques makes the results difficult to understand and use
Relevance of experience	A hazard evaluation team may not have an appropriate base of experience from which to assess the significance of potential incidents
Subjectivity	Hazard analysts must use their judgment when extrapolating from their experience to determine whether a problem is important

Limitations of hazard evaluation

1. **Completeness**: no method can guarantee that every hazard, cause and effect has been considered.
2. **Reproducibility**: results depend on the analysts' assumptions; another team may get different answers.
3. **Inscrutability**: results (trees, numbers) can be hard for managers and operators to understand and use.
4. **Relevance of experience**: the team may lack experience with this process, so hazards are missed.

5. **Subjectivity**: judgement is used to decide what is credible and acceptable.

> **Memory trick:** HAZOP in one line: **"Node, Intention, Parameter + Guide word = Deviation → Cause, Consequence, Safeguard, Action"**. Limitations = **"CRIRS"**: Completeness, Reproducibility, Inscrutability, Relevance, Subjectivity.

A13 Personal protective equipment (PPE): types, how each protects, respiratory protective equipment

Must study · Likely · 73% · Easy

PAST-PAPER QUESTIONS

- **2024 Q3(a)** [10] What are the various PPEs used in industry and how do they protect humans against various hazards associated with industry?
- **2022 Q3(a)** [10] How is PPE (personal protective equipment) used in general industry to protect against various hazards associated with humans?
- **2021 Q4(a)** [10] What is meant by PPE? Briefly write about respiratory protective equipment.
- **2020 Q3(b)** [8] What are the common PPEs used in industry? Mention their advantages.

★ REMEMBER IT

- PPE head to toe: helmet → goggles/face shield → ear plugs/muffs → respirator → gloves → apron/coverall → safety harness → safety shoes. Respirators: air-purifying (filter, cartridge) vs air-supplying (SCBA) for low O_2.

Before you start: PPE in one idea

PPE (personal protective equipment) is anything a worker **wears** to stop a hazard from reaching the body: helmet on the head, goggles on the eyes, plugs in the ears, mask over the nose and mouth, gloves, shoes, suits. Each part of the body has its own PPE.

Remember: PPE is the last level of control (A12). It is used when the hazard cannot be removed or engineered away.

What is PPE (2021 4a)

Personal protective equipment (PPE) is equipment worn by a worker to minimise exposure to hazards that cause serious workplace injuries and illnesses - from chemical, radiological, physical, electrical, mechanical or other workplace hazards. The employer must assess the hazards, first eliminate or reduce them with engineering and administrative controls, then provide suitable PPE free of charge, train the workers and maintain/replace it; the worker must wear it properly, attend training, take care of it and report damage.



Examples of PPE for head, eyes, face, ears, hands, feet and body

Types of PPE and how they protect (2024 3a, 2022 3a, 2020 3b)

Body part / hazard	PPE	How it protects / advantage
Head: falling objects, bumps, electric shock	Hard hats: class G (general, 2,200 V), class E (electrical, 20,000 V), class C (conductive, no electrical protection); type I (top), type II (top and sides)	Hard shell spreads the impact, suspension absorbs the shock; insulates from electricity
Eye and face: flying particles, chemical splashes, dust, molten metal, intense light	Safety glasses with side shields, goggles (direct/indirect-vented, non-vented), face shields (used **with** glasses/goggles), welding helmets with filter lens, laser goggles	Impact-resistant lenses stop particles; sealed goggles stop splashes; filter lenses absorb UV/IR from welding
Hearing: noise above 85 dB (required at 90 dBA (A-weighted decibel (sound level as the human ear hears it)) for 8 h)	Foam ear plugs, moulded plugs, ear muffs, noise-cancelling muffs	Reduce sound reaching the ear by the noise reduction rating (NRR); prevent hearing loss
Respiratory: dust, fumes, gases, oxygen deficiency	Air-purifying respirators (dust mask, cartridge, gas mask, PAPR (powered air-purifying respirator)) or atmosphere-supplying (SCBA, supplied-air)	Filter/absorb contaminants or supply clean air (see below)
Hands: cuts, punctures, chemical absorption, burns, vibration	Leather-palm, cut-resistant, chemical-resistant (rubber, nitrile, neoprene), heat-resistant, anti-vibration, insulating rubber gloves	Barrier between skin and the hazard; chosen to match the hazard
Feet: falling/rolling objects, crushing, sharp objects, hot metal, slippery or electrical hazards	Safety shoes/boots with steel or composite toe, metatarsal guards, heat-resistant and slip-resistant soles, chemical-resistant boots, conductive or insulating footwear	Toe cap resists impact and compression; soles resist puncture, slip, heat, electricity
Body: heat, chemicals, cuts, radiation, dust	Lab coats, coveralls, aprons, vests, full suits: paper fibre (dust), treated wool/cotton (fire), duck (cuts), leather (heat/flames), rubber/neoprene (chemicals); hazmat suits EPA (Environmental Protection Agency (USA)) level A-D	Covers skin from splashes, sparks, heat and contamination
Falls from height	Full-body harness, lanyard, lifeline	Arrests the fall and spreads the load over the body



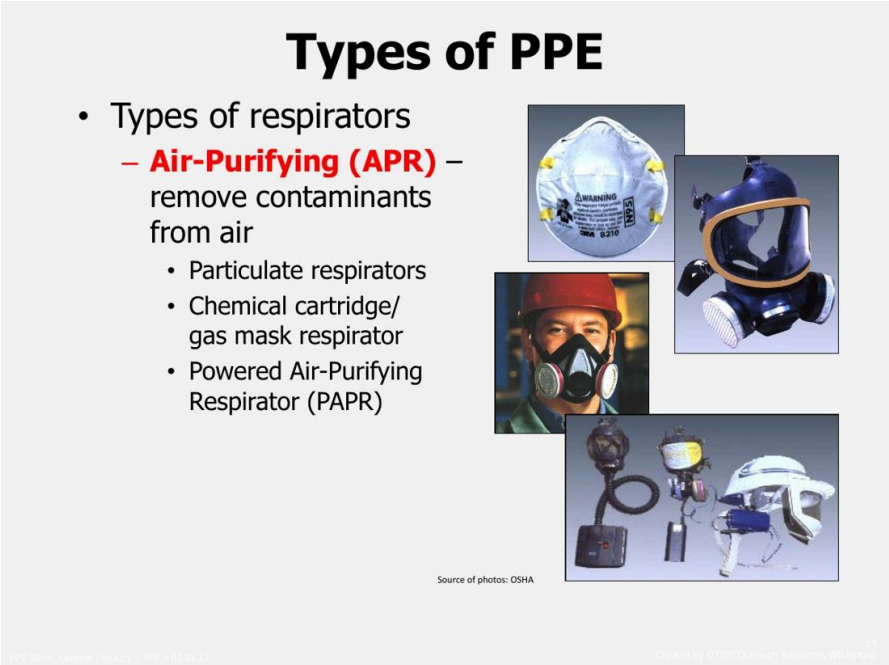
Eye and face protection: safety glasses, goggles, face shields, welding shields

Advantages of PPE: immediate, simple and relatively cheap protection; works where other controls are not feasible (maintenance, emergencies, short jobs); portable; protects against several hazards together; reduces injury severity and compensation cost.

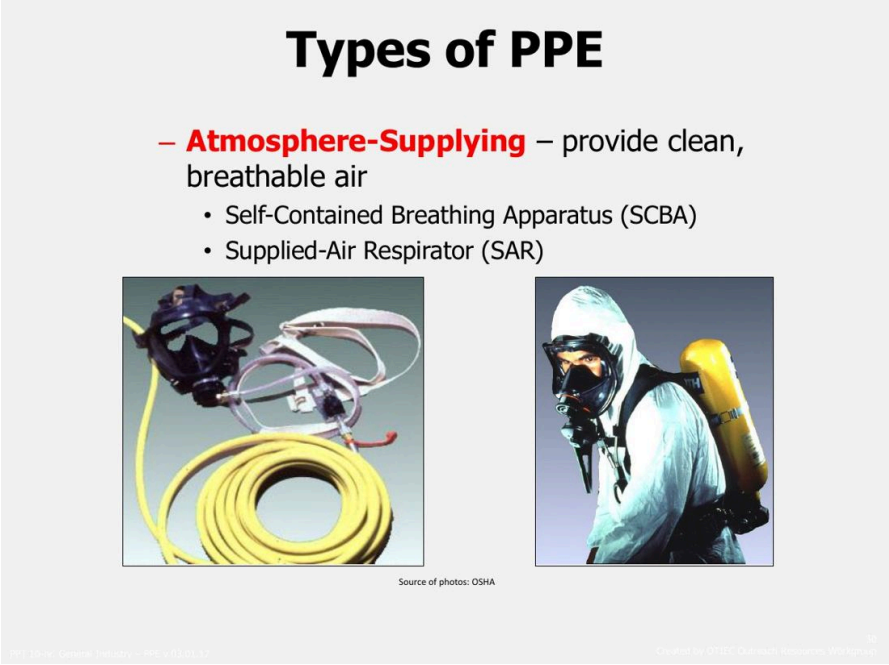
Respiratory protective equipment (2021 4a)

Respirators are used only when engineering controls (ventilation, enclosure) are not feasible or while they are being installed. A medical evaluation and fit test is needed before use; respirators must be inspected, cleaned, sanitised and stored properly.

Type	How it works	Examples	Use / limitation
Air-purifying respirator (APR)	Filters or chemically absorbs contaminants from the surrounding air	Particulate respirator (dust mask, N95), chemical cartridge respirator, gas mask (canister), **PAPR** (powered air-purifying respirator: blower pushes air through the filter)	Only when oxygen is enough (≥ 19.5 %) and the contaminant and its concentration are known and below IDLH (immediately dangerous to life or health)
Atmosphere-supplying respirator	Supplies clean breathing air from a source independent of the work air	**SCBA** (self-contained breathing apparatus) (self-contained breathing apparatus: air cylinder on the back), **SAR** (supplied-air respirator) (supplied-air respirator: air line from a compressor)	Oxygen-deficient or IDLH atmospheres, confined spaces, fire fighting, unknown contaminants; SCBA limited by cylinder time, SAR by hose length



Air-purifying respirators



Atmosphere-supplying respirators: SCBA and supplied-air

Memory trick: PPE top to toe: **"Hat, Eyes, Ears, Breath, Hands, Feet, Body"**. Respirators: **APR (air-purifying respirator) cleans the air you have; SCBA/SAR brings new air**.

A7 Classes of fire and fire-fighting materials (water, foam, dry powder, CO_2, wet chemical); fire-fighting procedure

Must study · Likely · 67% · Easy

PAST-PAPER QUESTIONS

- **2023 Q2(b)** [13] Define fire and flame. Briefly explain the fire-fighting procedure and fire-fighting materials for each class of fire.
- **2022 Q2(b)** [13] Briefly explain the fire classes based on the types of fuel involved, and give an overview of commonly used fire-fighting materials for each class.
- **2021 Q3(b)** [10] What are the fire-fighting materials used in industry? Can water be used to extinguish all types of fire? Explain.
- **2020 Q2(c)** [14] Discuss the applications of fire-fighting materials for different classes of fire.

★ REMEMBER IT

- Classes (by fuel): **A** = Ash (wood, paper, cloth) → water. **B** = Barrel (flammable liquids) → foam, CO_2, powder. **C** = Cylinder (gases) → shut off gas, powder. **D** = Dense metals (Mg, Na) → special dry powder. **E**/electrical → CO_2, powder (never water). **F/K** = Fat/Kitchen oil → wet chemical.
- Water cannot be used on oil (spreads it), live electricity (conducts) or reactive metals (reacts, can explode).

Before you start: why fires are put into classes

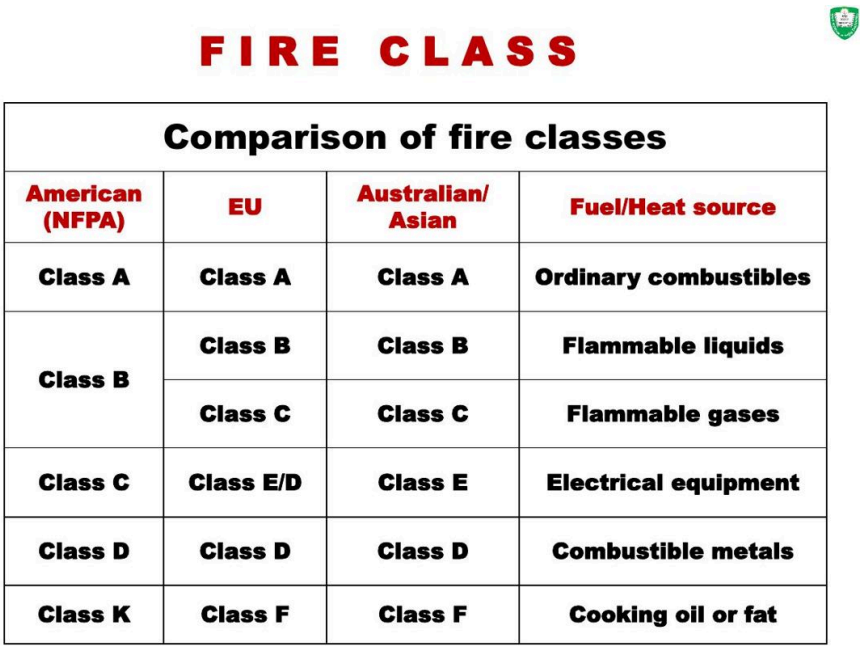
Different fuels need different extinguishers. Water cools wood very well, but on burning oil it spreads the fire, on live electrics it conducts current to you, and on burning metals (sodium, magnesium) it reacts violently. So fires are labelled by fuel (class A, B, C, D, E/K) and each extinguisher is marked with the classes it can fight.

Fire and flame (2023 2b)

Fire: rapid exothermic oxidation of an ignited fuel producing heat and light. **Flame**: the visible, luminous zone of burning gases/vapour in a fire.

Classes of fire (2022 2b)

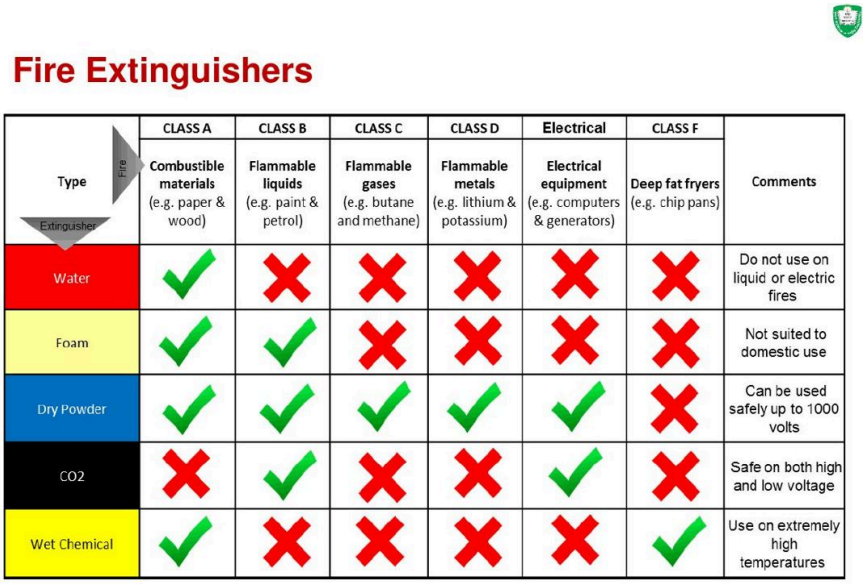
Class (EU/Asia)	Fuel	Principle needed
A	Ordinary combustible solids: wood, paper, cloth, coal, sawdust, textiles, fibres	Cooling by water is essential
B	Flammable/combustible liquids: petrol, oil, grease, solvents, paint, varnish	Blanketing (smothering) is essential
C	Flammable gases under pressure: natural gas, LPG, propane, butane	Stop the gas and dilute with inert gas or dry powder
D	Combustible metals: magnesium, aluminium, zinc, sodium, potassium	Special dry powder (metal reacts with water)
E	Energised electrical equipment	Non-conductive agent (CO_2, dry powder); isolate power
F / K	Cooking oils and fats	Wet chemical; never water



Comparison of fire classes: in the American (NFPA) system class C = electrical and class K = cooking oil

Fire-fighting materials (2021 3b, 2020 2c)

Material	How it works	Use on	Do not use on
Water (jet, spray, fog, sprinkler)	Cools; absorbs huge heat and turns to steam which also smothers	A	B (spreads burning oil), D (reacts), E (conducts)
Foam	Blanket on the surface cuts air and cools	A and B (petrol, diesel)	Live electrical, metals
Dry powder (ABC, BC, special D powder)	Smothers and breaks chain reaction; "universal" extinguisher	A, B, C, electrical, D (special powder)	Leaves residue on equipment
CO_2 and inert gases	Displaces oxygen, non-conductive, no residue	E (electrical), B	Confined rooms with people (asphyxiation); weak on A
Wet chemical	Forms a soapy layer on hot oil and cools	F/K (cooking oil), also A	Electrical
Steam, sand, fire blanket	Smother	Small fires, B, D (dry sand)	-



Which extinguisher on which class: water only A; foam A and B; dry powder almost all; CO_2 B and electrical; wet chemical A and F

Can water be used on all types of fire? (2021 3b)

No. Water is the cheapest and best cooling agent for class A fires, but: (1) on **oil/petrol (B)** fires water is heavier, sinks, boils violently and splashes burning oil around, spreading the fire; (2) on **live electrical (E)** fires water conducts current and electrocutes the fire-fighter; (3) on **metal (D)** fires (Na, K, Mg) water reacts to give hydrogen and explosions; (4) on **cooking oil (F)** it causes a fireball. Only fine water mist/fog used by experts is an exception for some B fires.

Fire-fighting procedure (2023 2b)



Fire-fighting procedure RACE (Rescue, Alarm, Contain, Extinguish/Evacuate) and how to use an extinguisher, PASS (Pull, Aim, Squeeze, Sweep (using an extinguisher))

1. **R - Rescue** anyone in immediate danger.
2. **A - Alarm**: activate the fire alarm and call the fire service/control room.
3. **C - Contain**: close doors and windows to confine the fire and smoke; shut off fuel and power.
4. **E - Extinguish** a small fire with the right extinguisher, or **evacuate** if it is spreading.

Use an extinguisher only if the fire is small (waste-bin size), confined to one item, everyone else is out, you have a clear exit and you are trained. Use **PASS**: **P**ull the pin, **A**im the nozzle at the base of the fire, **S**queeze the handle, **S**weep side to side. Large fires: trained team connects hoses to hydrants, attacks from upwind, cools adjacent tanks, crawls low under smoke, rescues trapped people.

> **Memory trick:** Classes: **A** = **A**sh (solids leave ash), **B** = **B**arrel (liquids), **C** = **C**ylinder (gas), **D** = **D**ense metal, **E** = **E**lectric, **F** = **F**rying pan.

A16 Emergency preparedness plan (EPP), Triple Bottom Line (TBL), safety committee

Must study · Likely · 67% · Easy

PAST-PAPER QUESTIONS

- **2024 Q4(d)** [10] Briefly explain the steps of preparing an Emergency Preparedness Plan (EPP).
- **2023 Q4(c)** [7] What is the Triple Bottom Line (TBL) framework? Who are the typical members of a safety committee and what roles do they play?
- **2023 Q4(d)** [14] What are the steps for an emergency preparedness plan? Briefly explain them.
- **2020 Q3(c)** [5] Classify the EPP (emergency preparedness plan).

★ REMEMBER IT

- EPP (emergency preparedness plan) steps: **identify emergencies → assess resources → assign roles (chain of command) → set alarms and communication → evacuation routes and assembly point → train and drill → review**.
- TBL (triple bottom line) = People, Planet, Profit.

Before you start: planning for the worst day

An emergency (fire, explosion, toxic leak, flood, earthquake) gives no time to think. An **emergency preparedness plan (EPP)** decides in advance who does what, with which equipment, how people are warned and evacuated, and when outside help is called. Example the teacher uses: FR Tower fire, Banani, 28 March 2019.

The **Triple Bottom Line** says a company should report not just profit but also its effect on people and the planet. A **safety committee** is the group of managers and workers that runs safety day to day.

Classification of EPP: four disaster levels (2020 3c)

Level	Extent	Who handles it
Level 1	Local: one unit/area	People on the spot with local equipment (like storm signal 1-4)
Level 2	Whole plant affected	Plant's own emergency team and resources are enough (signal 5-7)
Level 3	Spreads to neighbouring plants/community	Outside help: fire service, ambulance, hospitals, neighbouring plants (signal 8-10)
Level 4	National-level disaster; plant has lost control	Government, police, army, national agencies (signal 11)

EPPs can also be classified as **on-site** (inside the plant, by the management) and **off-site** (outside the boundary, by the district authority).

Steps of preparing an EPP (2024 4d, 2023 4d)

1. **List possible accidents/disasters** from operations and nature (fire, explosion, toxic release, flood, earthquake), using history, similar plants, what-if, HAZOP (hazard and operability study), fault and event trees and experience.
2. **Determine the severity** of each possible event (people, property, environment affected).

3. **Determine the disaster level** (1-4) of each event.
4. **Make a control/management plan** for each: causes, prevention, response (rescue, evacuation, first aid, hospital).
5. **Arrange resources and equipment**: walkie-talkies, torches, extinguishers, fire water network, hydrants, foam, pumps, siren, ambulance, helmets, masks, blankets, gloves, first-aid kits, cutters; emergency phone numbers.
6. **Develop written procedures**: simple, clear, concise, complete.
7. **Emergency declaration and termination**: who declares the emergency and its level, and who declares it over.
8. **Team and individual responsibilities**: fire-fighting, rescue, first aid, evacuation, communication, security teams.
9. **Team leadership**: incident controller and chain of command (coloured helmets for identification).
10. **Emergency announcement and communication**: alarms, PA system, telephones, walkie-talkies tested regularly; informing authorities and the public.
11. **Evacuation management**: escape routes, assembly (meeting) point, head count, transport.

Then train everyone and hold regular **mock drills**, and update the plan after each drill or change.

Triple Bottom Line (TBL) (2023 4c, 2024 4c)

TBL is a sustainability framework (John Elkington, 1997, "Cannibals with Forks") that measures a company's performance on three bottom lines instead of profit alone: **People** (social), **Planet** (environmental) and **Profit** (economic) - the "3 Ps" or three capitals.

Bottom line	Covers	Indicators / standards
People (social capital)	Fair wages, safe workplace, freedom, community, stakeholders	Employment, wages, turnover, accident rates (AFR, ASR (accident severity rate), FAR (fatal accident rate)), CSR (corporate social responsibility), women employed; OHS (occupational health and safety) standard (ISO 45001, earlier OHSAS (Occupational Health and Safety Assessment Series) 18001)
Planet (environmental capital)	Air, water, raw materials, waste, ecosystems	Environmental category (Green/Orange/Red), pollution levels, ETP (effluent treatment plant), 3R; ISO (International Organization for Standardization) 14001
Profit (economic capital)	Profitability and economic value	Profit, turnover, EPS, dividend, growth; quality ISO 9001

A business is sustainable only where environmental protection, social responsibility and economic success overlap (lessons from Rana Plaza, Tazreen and Bhopal).



TBL Venn diagram: environmental protection, social responsibility and economic success overlap in the sustainable principle (teacher notes, in Bangla)

Safety committee: members and roles (2023 4c)

Member	Role
Chairman: factory manager / CEO	Presides over meetings, approves agenda and decisions, provides resources
Member secretary: safety engineer / safety officer	Prepares agenda and minutes, follows up actions, keeps accident statistics, advises
Specialist members: factory doctor, personnel (HR) officer	Records occupational diseases and health data; welfare, compensation, training records
Engineers, foremen, line supervisors	Bring shop-floor problems, carry out decisions in their sections
Worker representatives (elected/nominated by the union, rotating)	Voice of workers, report hazards, promote cooperation

Functions: formulate safety rules and policy; promote cooperation and obedience to rules; counteract indifference; conduct periodic inspections; investigate accidents and advise remedies; record accidents and sickness; ensure workers use safety gear and procure it; arrange training, seminars and posters; publish safety manuals and bulletins.

> **Memory trick:** EPP 11 steps: **"List, Severity, Level, Plan, Resources, Procedures, Declare, Teams, Leader, Communicate, Evacuate"**. TBL = **3 Ps: People, Planet, Profit**.

A11 Anatomy of a process incident; event tree vs fault tree Must study

Likely · 67% Medium

PAST-PAPER QUESTIONS

- **2024 Q3(b)** [13] Briefly explain the anatomy of a process incident with their modes, states, operational goals and critical systems.
- **2022 Q4(a)** [12] Describe the anatomy of a process incident mentioning their different modes, states and goals.
- **2021 Q3(c)** [6] "Event tree analysis is the logical counterpart of the fault tree" - explain.

UNDERSTAND **Read this to understand; you do not need to write it**

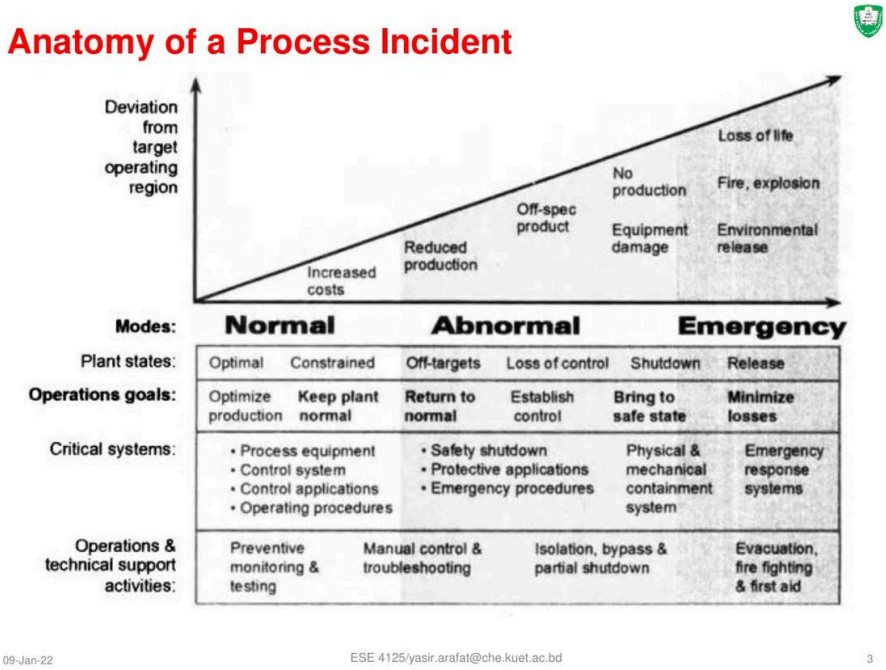
A process incident does not jump from normal to disaster. The plant moves through **states**: normal operation → abnormal (a deviation, e.g. pressure rising) → emergency (loss of control) → incident (release, fire). At each step a **layer of protection** (control system, alarm and operator, safety valve, containment, emergency response) tries to bring it back. An incident happens only when every layer fails.

Fault tree works backwards from the top event to its causes (AND/OR gates). Event tree works forwards from an initiating event through each safeguard working or failing to the possible outcomes. One is the mirror of the other.

Before you start: how a normal plant ends in a disaster

A plant does not jump straight from fine to explosion. It **drifts**: first a small deviation (off-target), then loss of control, then shutdown or release. At each stage the operators have a different goal and different systems to rely on. This step-by-step picture is the "anatomy" (structure) of a process incident.

Anatomy of a process incident (2024 3b, 2022 4a)



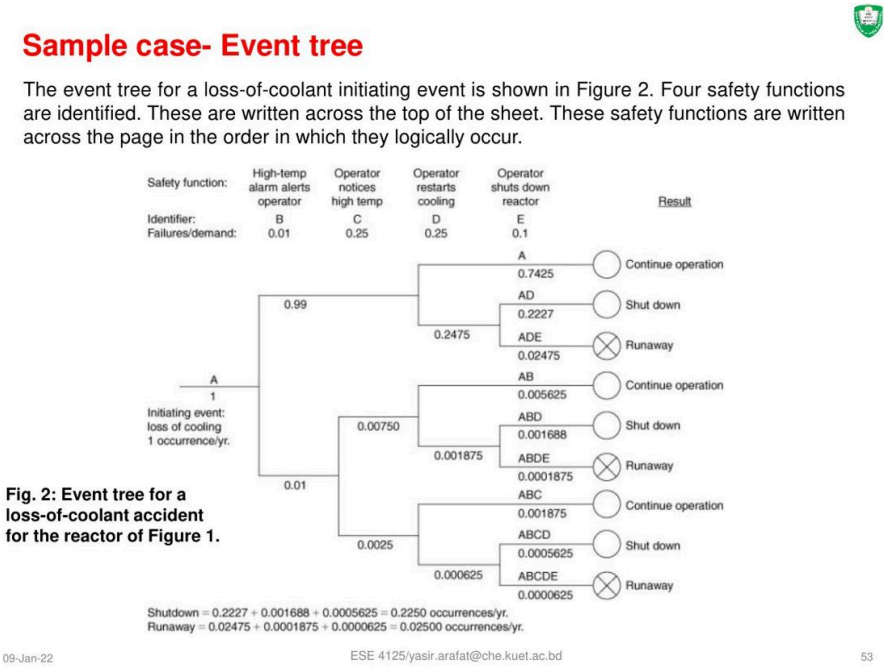
Anatomy of a process incident: modes, plant states, operational goals, critical systems and support activities

Mode of operation	Plant state	Operational goal	Critical systems	Support activities
Normal	Optimal → constrained	Optimise production; keep the plant normal	Process equipment, control system and applications, operating procedures	Preventive monitoring, testing and maintenance
Abnormal	Off-targets → loss of control	Return to normal; establish control	Safety shutdown, protective applications (alarms, interlocks, relief), emergency procedures	Manual control and troubleshooting; isolation, bypass and partial shutdown
Emergency	Shutdown → release	Bring the plant to a safe state; minimise losses	Physical and mechanical containment (dikes, relief, flare), emergency response systems	Evacuation, fire fighting and first aid

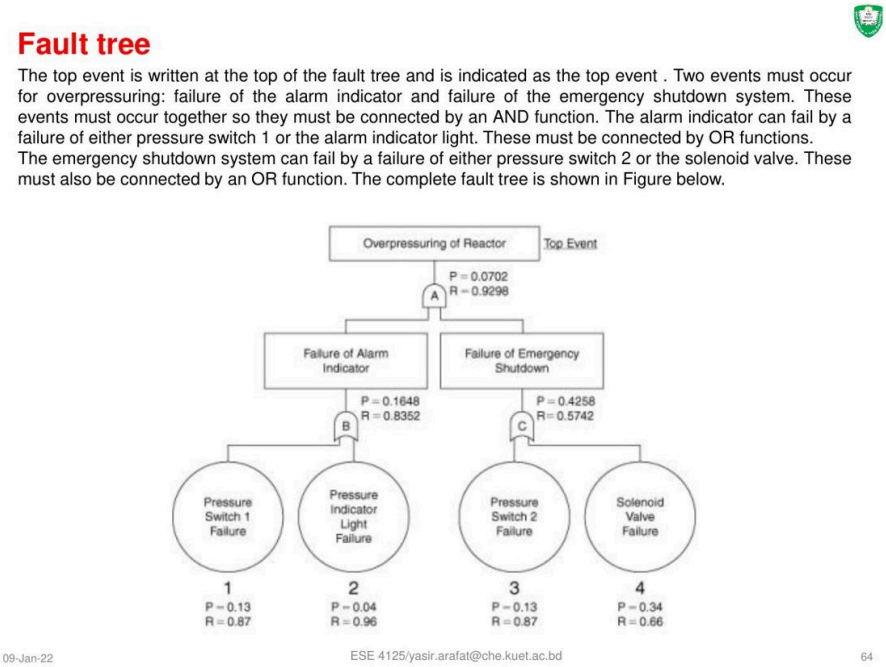
Consequences grow as the plant moves down the states: increased costs → reduced production → off-spec product → no production / equipment damage → loss of life, fire, explosion, environmental release. Each layer of protection (control, alarms, shutdown, containment, emergency response) is meant to stop the drift at that stage.

"Event tree analysis is the logical counterpart of the fault tree" (2021 3c)

	Fault tree analysis (FTA)	Event tree analysis (ETA)
Logic	**Deductive** (backward): from effect to causes	**Inductive** (forward): from cause to effects
Starts from	A **top event** (undesired, e.g. reactor over-pressure)	An **initiating event** (e.g. loss of cooling)
Branches	AND / OR gates down to basic events (component failures)	Success (up) / failure (down) of each safety function in time order
Answers	How can this accident happen, and how often?	What can happen after this failure, and how often?
Result	Frequency of the top event	Frequency of each outcome (safe shutdown, runaway ...)



Event tree for loss of cooling in a reactor: safety functions succeed or fail, runaway 0.025/yr (once in 40 years)



Fault tree for reactor over-pressure: alarm fails AND shutdown fails

Fault tree

The circles denote basic events and the rectangles denote intermediate events. The fishlike symbol represents the OR logic function. It means that either of the input events will cause the output state to occur.

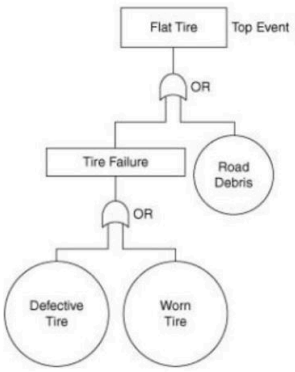


Fig: A fault tree describing the various events contributing to a flat tire.

09-Jan-22 ESE 4125/yasir.arafat@che.kuet.ac.bd 57

Simple fault tree: flat tyre from tyre failure OR road debris; circles = basic events, rectangles = intermediate events

Why counterparts: they look at the same accident from opposite sides. The **top event of the fault tree is the initiating event (or a failure branch) of the event tree**: FTA (fault tree analysis) calculates how often the initiating event happens, and ETA (event tree analysis) then shows what follows from it. Used together (bow-tie / cause-consequence analysis) they give the complete chain: causes → event → consequences.

Side note: Reactor example: loss of coolant 1/yr. Alarm, operator and shutdown each work with some probability. Multiply along each path: runaway = 0.025/yr (once in 40 years) - too frequent, so an extra automatic shutdown is added.

Memory trick: Fault tree = Find causes (looks back); Event tree = Expect outcomes (looks ahead). Anatomy = **"Normal-Abnormal-Emergency: Optimise, Control, Safe"**.

A8 BLEVE (boiling liquid expanding vapour explosion): mechanism, sequence of events, effects, prevention

Must study · Likely · 67% · Medium

PAST-PAPER QUESTIONS

- **2024 Q2(b)** [10] Explain the mechanism of BLEVE (boiling liquid expanding vapour explosion) with the sequence of events leading to the BLEVE.
- **2022 Q2(c)** [10] What is BLEVE? "All combustible materials do not create explosion" - explain.
- **2021 Q2(d)** [8] Define BLEVE. Why is prevention of BLEVE necessary?

LEARN IT FROM ZERO

BLEVE, from zero

Think of a pressure cooker full of liquefied gas (LPG). Inside, the liquid is far above its normal boiling point, held liquid only by the pressure. Now a fire heats the tank. Where the liquid touches the wall, the liquid keeps the steel cool. Above the liquid line only vapour touches the wall, so that steel **overheats, weakens and splits**.

The instant the tank opens, the pressure drops, and the whole superheated liquid **boils at once**, expanding about 250 times. That blast throws tank pieces hundreds of metres. If the liquid is flammable, the cloud ignites into a huge **fireball**.

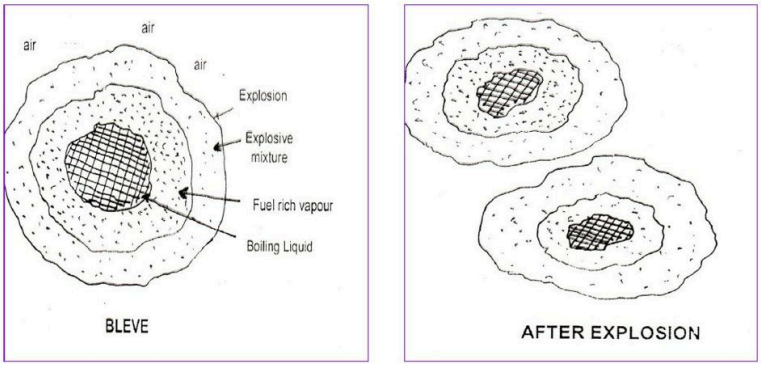
Sequence to write: **fire around tank → pressure rises, relief valve opens → liquid level falls → unwetted wall overheats and weakens → tank ruptures → sudden boiling → vapour explosion (blast + fragments) → fireball**. Prevention: water spray cooling, insulation/fireproofing, mounding, relief valves, keep fires away.

Before you start: boiling and pressure

LPG, propane, butane and ammonia are gases at room conditions but are kept **liquid** in tanks by keeping them under pressure. Inside the tank the liquid is already above its normal boiling point; only the pressure stops it boiling.

If the tank suddenly splits, the pressure drops to atmospheric in a moment, and the whole liquid boils at once ("flash"), turning into about 200-250 times its volume of vapour. That sudden expansion is the explosion; if the vapour is flammable, it ignites into a huge **fireball**.

Boiling Liquid Expanding Vapor Explosion (BLEVE)



13-Oct-20 ESE 4125/yasir.arafat@che.kuet.ac.bd 33

BLEVE: the tank heated by fire bursts, the boiling liquid flashes to vapour and forms an explosive mixture and fireball

What is BLEVE (2022 2c, 2021 2d)

BLEVE (Boiling Liquid Expanding Vapour Explosion), also called a fireball, is the explosion of a vessel containing a liquid stored above its atmospheric boiling point (pressurised liquefied gas) when the vessel ruptures, usually after being heated by an external fire. It is a combination of fire and explosion with an intense radiant heat emission within a short time.

Mechanism / sequence of events (2024 2b)

1. A pressurised tank of liquefied flammable gas (e.g. LPG) is exposed to an external fire (pool fire from a leak, or a jet fire).
2. Heat enters the tank: the liquid absorbs heat, its vapour pressure rises; the pressure relief valve opens and vents vapour (which may also burn).
3. As vapour escapes, the liquid level falls. The tank wall in contact with **liquid** stays cool (liquid absorbs heat), but the wall above the liquid (vapour space) is not cooled, overheats and loses strength.
4. The weakened metal in the vapour space cracks or tears even though the pressure is below the design pressure.
5. The pressure drops suddenly to atmospheric; the superheated liquid **flashes** (boils explosively) and expands ~250 times, producing a pressure blast that rips the tank apart.
6. The released vapour-liquid mixture is ignited by the surrounding fire, giving a huge rising **fireball**; tank fragments are thrown hundreds of metres.

Effects: huge fire and fireball, intense radiant heat, rapid expansion of explosive mixture, blast/shock waves, metal splinters at bullet velocity, flying vessel parts (rocketing), spread of fire to other tanks (domino effect).

Why prevention of BLEVE is necessary (2021 2d) and how

A BLEVE kills and injures people over a wide area within seconds (radiant heat can cause fatal burns hundreds of metres away), destroys the plant and nearby tanks, and cannot be fought once it starts - fire-fighters are among the most frequent victims. Hence prevention is the only real protection:

- Water deluge/spray to cool tanks exposed to fire; fire-proof insulation of tank walls.
- Correctly sized pressure relief valves; emergency depressurisation.
- Slope the ground and drain spills away from tanks so no pool fire forms under them; bund walls.
- Mounded or buried storage; adequate spacing between tanks and from public areas.
- Leak detection, emergency shut-off valves, ignition source control.
- Evacuate people early (at least 1 km) if a pressurised tank is engulfed in fire.

"All combustible materials do not create explosion" (2022 2c)

An explosion needs a combustible **mixed with air within its flammability (explosive) limits**, an ignition source and, usually, confinement so that pressure builds up rapidly. Most combustible solids (a log of wood, a sheet of paper, cloth) burn slowly from the surface only, because air can reach only the surface, so they give a **fire** but not an explosion. The same material can explode only when finely divided and suspended as a dust cloud, or when it gives vapour/gas that mixes with air between the LEL (lower explosive limit) and UEL (upper explosive limit). Liquids below their flash point do not give enough vapour. So whether a combustible explodes depends on its physical state, particle size, concentration in air, confinement and ignition energy.

> **Side note: LEL/UEL** = lower/upper explosive limit: e.g. methane burns only between 5 % and 15 % in air. Below LEL the mixture is too lean, above UEL too rich.

> **Memory trick:** BLEVE in five words: **"Fire, Heat, Weak wall, Split, Fireball"**.

A12 Risk management, hierarchy of controls (and where PPE fits), bow-tie diagram

Must study · Likely · 60% · Medium

PAST-PAPER QUESTIONS

- **2024 Q2(d)** [7] Briefly explain the correct order of the risk control hierarchy from most effective to least effective.
- **2024 Q3(c)** [12] Describe the structure of a bow-tie diagram with the identification of key elements.
- **2023 Q3(a)** [10] Define risk management. Briefly explain the key steps in the hazard identification process.
- **2023 Q3(b)** [10] What is a bow-tie diagram and how is it used in risk management? Explain.
- **2023 Q3(c)** [10] What is the hierarchy of mitigation controls for managing workplace hazards and how do different types of PPE (personal protective equipment) fit into it?

LEARN IT FROM ZERO

Bow-tie and hierarchy of controls, from zero

Draw a bow tie. The knot in the middle is the **top event** (loss of control, e.g. gas leak). The left wing lists the **threats** (causes) with **preventive barriers** between them and the knot. The right wing lists the **consequences** with **mitigating (recovery) barriers**. Escalation factors can weaken any barrier. The picture shows the whole risk on one page: left side stops it happening, right side limits the damage.

Hierarchy of controls (most to least effective): **Elimination** (remove the hazard) → **Substitution** (use something safer) → **Engineering controls** (guards, ventilation) → **Administrative controls** (training, rules, rotation) → **PPE**. PPE is the **last** line because it protects only the wearer and fails if worn wrongly.

★ REMEMBER IT

- Hierarchy: **"Every Smart Engineer Avoids PPE"** = Eliminate, Substitute, Engineer, Administer, PPE.

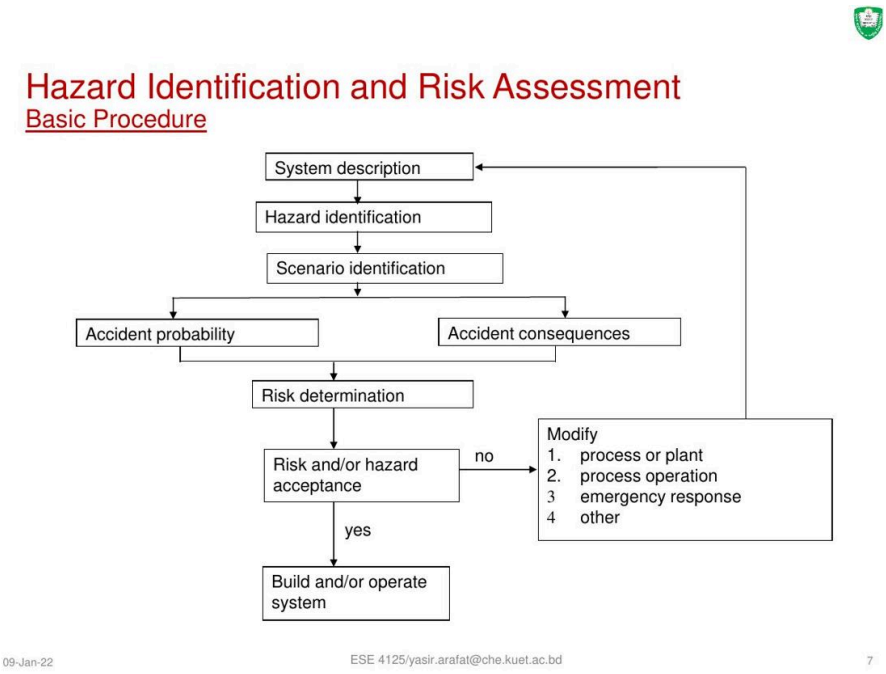
Before you start: what risk management means

Risk = likelihood × consequence. Risk management simply means: find the hazards, estimate how likely and how bad, decide if it is acceptable, reduce it if not, and keep checking.

Reducing risk has a clear order of preference: first try to **remove** the hazard; only if that is impossible, protect people from it. PPE (gloves, masks) is the last choice because it protects only the wearer, and only if worn correctly.

Risk management and the key steps (2023 3a)

Risk management is the systematic process of identifying hazards, analysing and evaluating the risks they pose, and controlling those risks (eliminating or reducing them to an acceptable level), followed by monitoring and review, so that accidents, injuries, illness and losses are prevented.

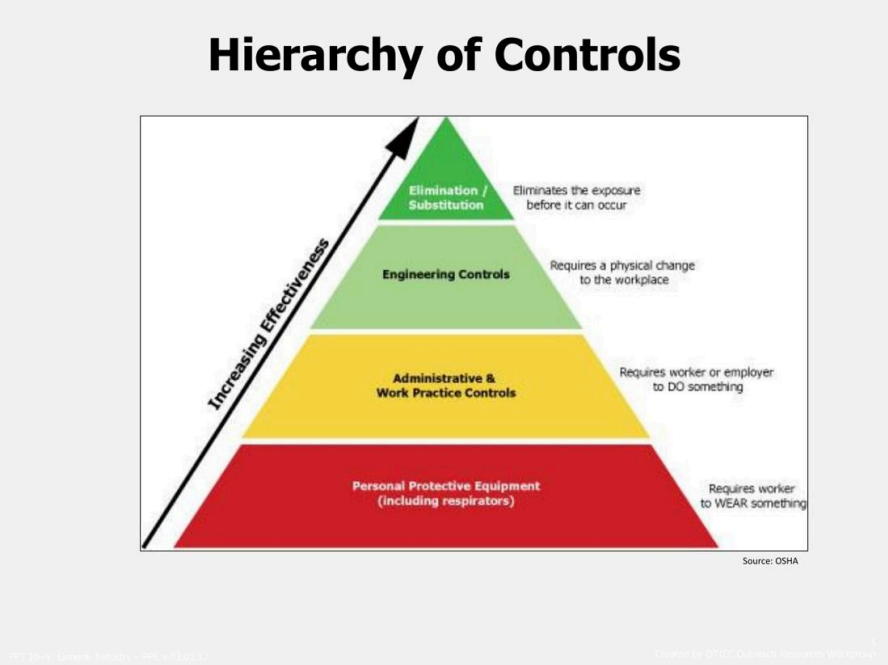


Hazard identification and risk assessment (HIRA) procedure

Key steps in the hazard identification and risk assessment process:

1. **System description**: collect plot plans, isometrics, BFD (block flow diagram), PFD (process flow diagram), P&ID (piping and instrumentation diagram), wiring diagrams (A10).
2. **Hazard identification**: analyse material properties and process conditions, review experience, interaction matrices, apply methods (checklist, PHA (preliminary hazard analysis), what-if, HAZOP (hazard and operability study), FMEA (failure modes and effects analysis)).
3. **Scenario identification**: what incidents can happen.
4. **Estimate consequences and likelihood** (FTA, ETA (event tree analysis), consequence models).
5. **Determine and evaluate the risk** (hazard matrix, A3): is it acceptable?
6. **If not acceptable**: modify the process, operation or emergency response and re-assess; **if acceptable**: build/operate.
7. **Implement controls and monitor/review** regularly.

Hierarchy of controls (2024 2d, 2023 3c)

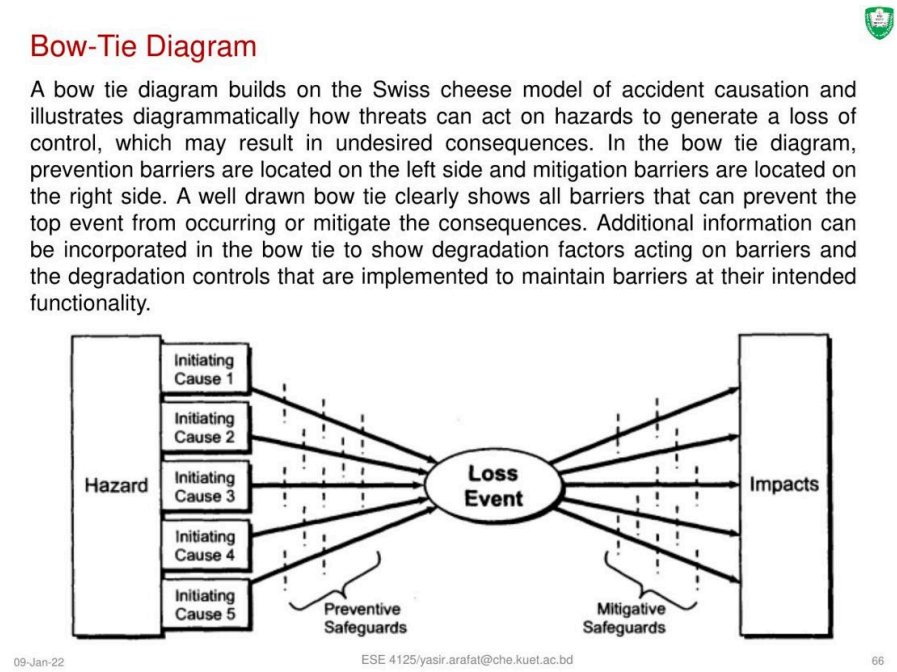


Hierarchy of controls: most effective at the top, least effective at the bottom

Order	Control	Meaning	Examples
1 (most effective)	**Elimination**	Physically remove the hazard	Do the work at ground level instead of at height; remove a hazardous step
2	**Substitution**	Replace with something less hazardous	Water-based paint instead of solvent paint; less toxic chemical
3	**Engineering controls**	Isolate people from the hazard	Machine guards, enclosures, local exhaust ventilation, interlocks, relief valves, sensors, barriers
4	**Administrative controls** (work practice)	Change the way people work	Procedures, permits to work, training, warning signs, job rotation, limited exposure time, housekeeping, supervision
5 (least effective)	**PPE**	Protect the worker with worn equipment	Helmets, goggles, ear plugs, respirators, gloves, safety shoes

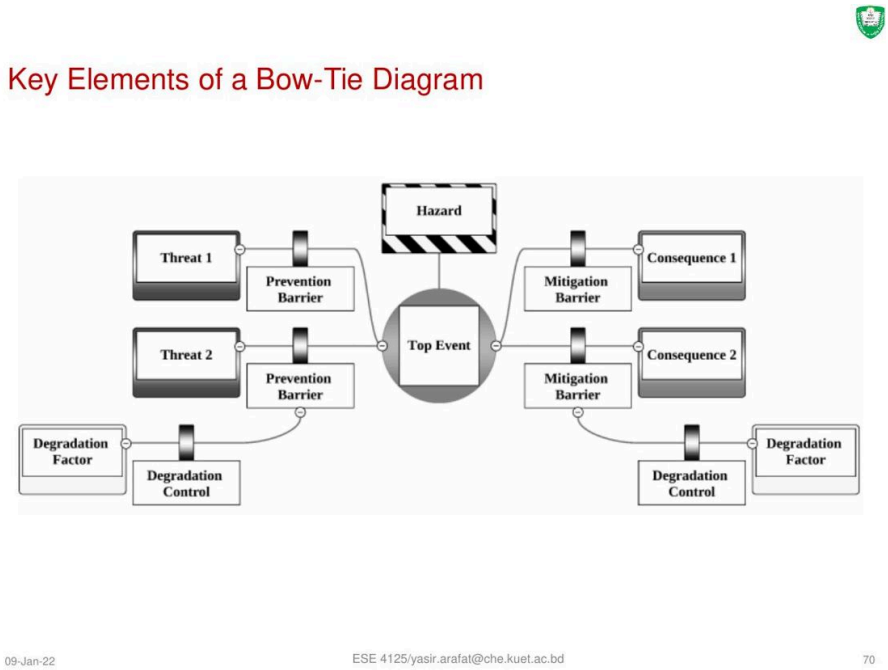
Where PPE fits: PPE is the **last line of defence**. It is used only when the hazard cannot be eliminated or adequately controlled by engineering and administrative measures, while those are being installed, or in emergencies. It depends on correct selection, fit, training and the worker wearing it every time, and it fails without warning. Different PPE match different hazards: head (hard hat), eye/face (goggles, face shield), hearing (plugs, muffs), respiratory (air-purifying or air-supplying), hands, feet, body (A13).

Bow-tie diagram: structure and key elements (2024 3c, 2023 3b)



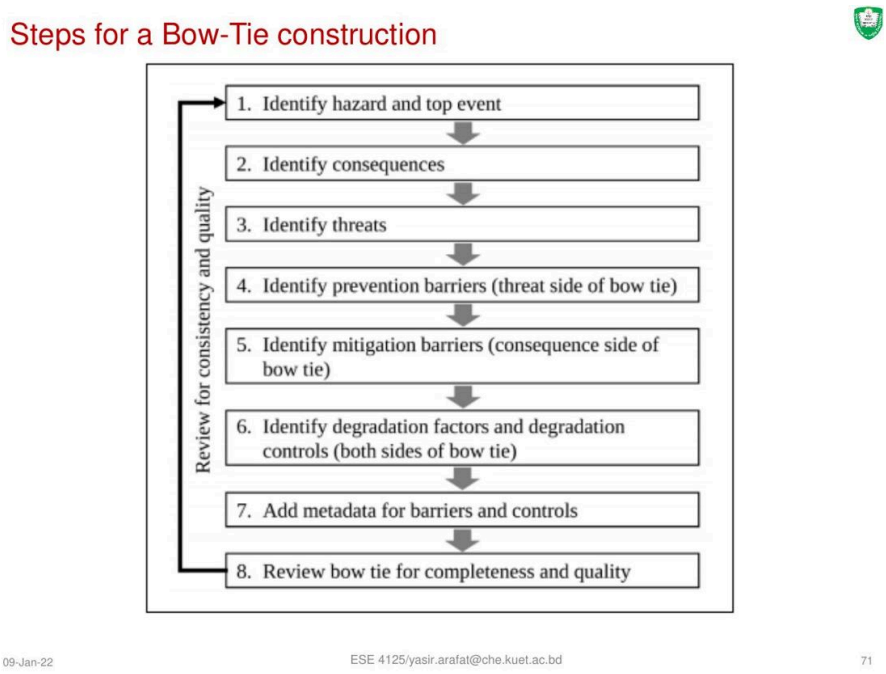
Bow-tie diagram: threats on the left, top event in the centre, consequences on the right, with prevention and mitigation barriers

A **bow-tie diagram** is a risk-assessment picture shaped like a bow tie. It joins a fault tree (left side, causes) and an event tree (right side, outcomes) around one central event, and shows the barriers that stop each threat or consequence. It is based on the "Swiss cheese" idea: each barrier has holes, and an accident happens only when the holes line up.



Key elements of a bow-tie

Element	Meaning	Pipeline example
Hazard	Something with potential to cause harm	Gas under pressure in a pipeline
Top event (centre knot)	The moment control over the hazard is lost	Loss of containment (gas leak)
Threats (left)	Causes that can release the top event	Corrosion, excavation damage, over-pressure
Prevention barriers (left)	Stop a threat from causing the top event	Coating and cathodic protection, marker posts, pressure trip
Consequences (right)	Results of the top event	Fire, explosion, injury, environmental damage
Mitigation (recovery) barriers (right)	Reduce the consequences after the top event	Gas detection, emergency shutdown valve, fire fighting, evacuation
Degradation (escalation) factors	Conditions that make a barrier fail	Lack of maintenance, untrained operator
Degradation controls	Measures that keep barriers working	Inspection schedule, training, audits



Eight steps for a bow-tie construction

Steps to build it: (1) identify the hazard and top event, (2) list consequences, (3) list threats, (4) add prevention barriers, (5) add mitigation barriers, (6) add degradation factors and controls, (7) add metadata (barrier owner, effectiveness), (8) review with the team.

Use in risk management: one clear picture of causes, consequences and barriers; shows which barriers are weak or missing; assigns responsibility for each barrier; easy to explain to managers and workers; links hazard identification to emergency planning.

> **Memory trick:** Hierarchy: **"Every Smart Engineer Avoids PPE"** = Elimination, Substitution, Engineering, Administrative, PPE. Bow-tie: **Threats → Prevent → TOP EVENT → Mitigate → Consequences**.

A6 Fire basics: fire triangle and tetrahedron, factors contributing to fire, methods of extinguishment, types of explosion Must study Possible · 53% Easy

PAST-PAPER QUESTIONS

- **2024 Q1(c)** [11] What are the different types of explosion? Explain the different methods of fire extinguishment.
- **2021 Q3(a)** [7] "All combustible materials are not fuel" - explain.
- **2020 Q2(a)** [6] What are the methods of fire extinguishment? Define each.
- **2020 Q2(b)** [10] Explain the fire triangle. What factors contribute to a fire incident?

LEARN IT FROM ZERO

Fire, from zero

Fire needs three things together: **fuel** (something that burns), **oxygen** (from air) and **heat** (to reach ignition temperature). That is the **fire triangle**. Add the **chain reaction** of free radicals that keeps the flame going and you get the **fire tetrahedron**.

Every way to put out a fire removes one side: **starving** (take the fuel away, close the gas valve), **smothering** (cut off oxygen: blanket, foam, CO_2), **cooling** (remove heat: water), and **breaking the chain** (dry chemical powder, halons).

"All combustible materials are not fuel": a log of wood will not burn from a match until it is heated enough to give off flammable vapour, and a vapour only burns between its lower and upper flammable limits. It must be in the right state, mixed with the right amount of air, at the right temperature.

★ REMEMBER IT

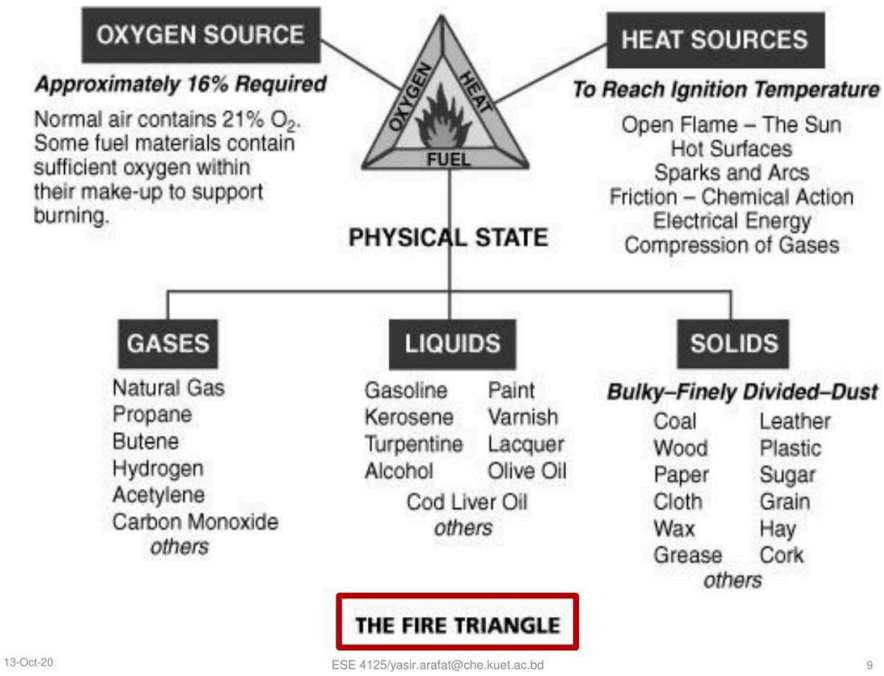
- Extinguishing = **Starve, Smother, Cool, Break the chain**.

Before you start: what fire needs

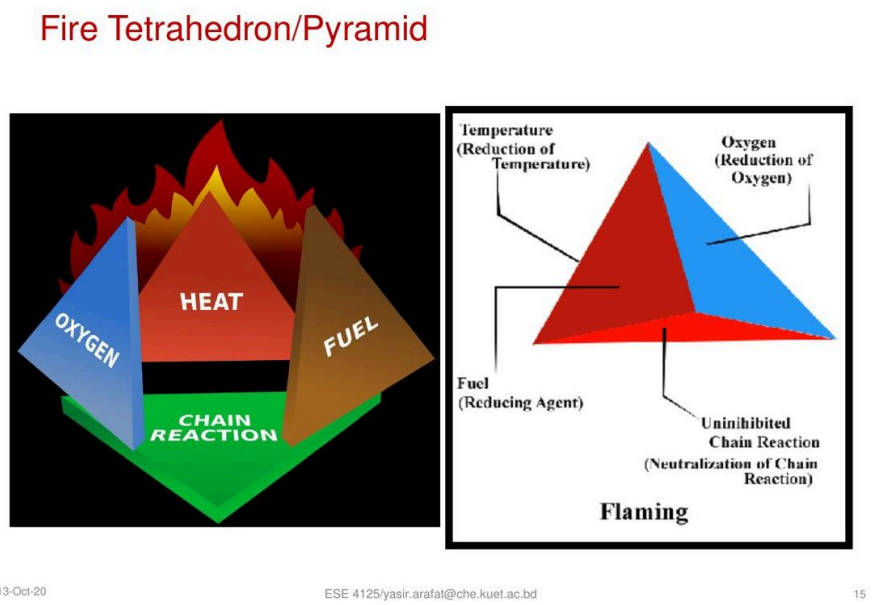
Fire is a fast chemical reaction of a **fuel** with **oxygen** that gives heat and light. It needs three things together: fuel, oxygen (about 16 % or more in air) and enough heat to reach the ignition temperature. That is the **fire triangle**. Once burning, free radicals keep a **chain reaction** going: add that and you get the **fire tetrahedron**.

Every way of putting out a fire simply removes one side: take away fuel, take away oxygen, take away heat, or break the chain reaction.

An **explosion** is burning (or expansion) so fast that gas expands suddenly and makes a pressure (shock) wave.



The fire triangle: oxygen source, heat sources and fuels (gases, liquids, solids)



Fire tetrahedron: heat, fuel, oxygen and the chain reaction

Fire triangle and factors contributing to fire (2020 2b)

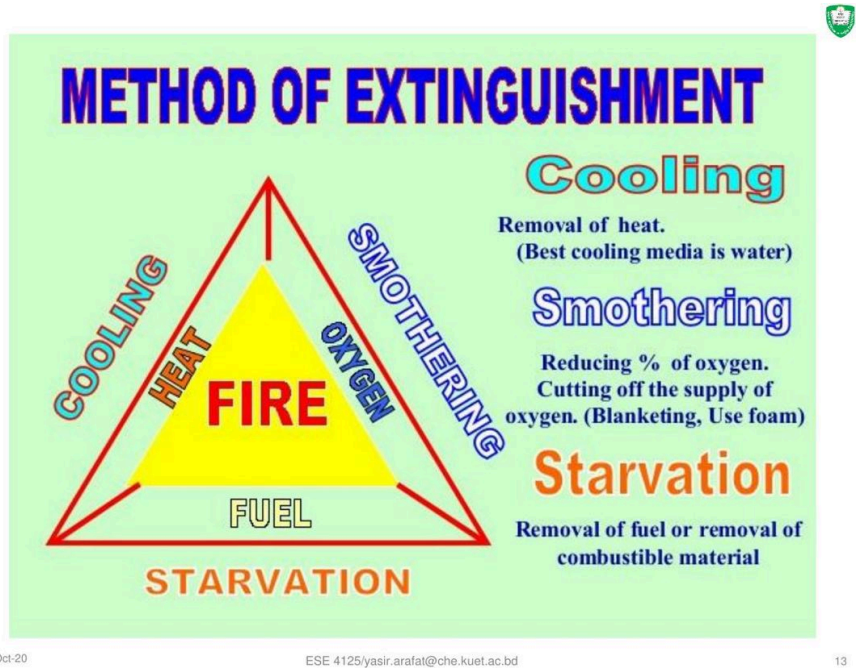
Fire (combustion) is a rapid exothermic oxidation of an ignited fuel, producing heat and usually visible light. **Flame** is the visible, glowing gas-phase part of a fire where vapour burns. Combustion always occurs in the vapour phase: liquids evaporate and solids decompose to vapour first.

- **Fuel**: any combustible material - gases (natural gas, LPG, hydrogen), liquids (petrol, kerosene, paint), solids (wood, paper, cloth, coal, dust).
- **Oxygen**: normally from air (21 %); about 16 % is enough for burning; oxidisers can also supply it.
- **Heat**: to raise the fuel to its ignition temperature - flames, hot surfaces, sparks and arcs, friction, static, chemical reaction, compression.
- **Chain reaction** (4th side of the tetrahedron): free radicals that keep the fire going.

Easily detectable contributing factors: (1) combustible rubbish, solvents, paper, wood lying around; (2) easy availability of air/oxidisers; (3) ignition sources: sparks, static, hot surfaces, friction; (4) open handling of flammables; (5) machinery run without lubrication/maintenance; (6) non-flameproof electrical fittings in flammable areas; (7) smoking in flammable areas; (8) no fire detectors; (9) no extinguishers; (10) not following fire rules. **Hidden factors:** runaway chemical reaction, sudden loss of cooling, metal parts trapped in rollers giving sparks, no inert gas blanket, lightning.

Spread of fire increases with larger exposed combustible area, more heat released, more heat conduction, low humidity and high wind.

Methods of fire extinguishment (2024 1c, 2020 2a)



Method of extinguishment: cooling removes heat, smothering removes oxygen, starvation removes fuel

Method	Side removed	How	Examples
Starving (starvation)	Fuel	Remove or isolate the fuel from the fire	Close the gas/oil valve, remove nearby combustibles, pump oil out of a burning tank, cut fire breaks in forests
Smothering (blanketing)	Oxygen	Cut off air or dilute oxygen below ~15 %	Foam blanket on oil, CO_2, fire blanket, sand, lid on a burning pan, steam
Cooling (quenching)	Heat	Absorb heat to bring fuel below ignition temperature	Water jets and sprays (best cooling medium), wet chemical
Inhibition (breaking chain reaction)	Chain reaction	Chemicals capture free radicals	Dry chemical powder, halon substitutes

"All combustible materials are not fuel" (2021 3a)

A **combustible material** is anything that ignites easily and releases heat on burning (paper, cloth, plastic, rubbish). A **fuel** is a substance used on purpose as a reducing agent that gives up electrons to an oxidiser in a controlled combustion to release **useful** energy economically (coal, natural gas, petrol, LPG, wood). So every fuel is combustible, but many combustibles (clothes, furniture, plastic

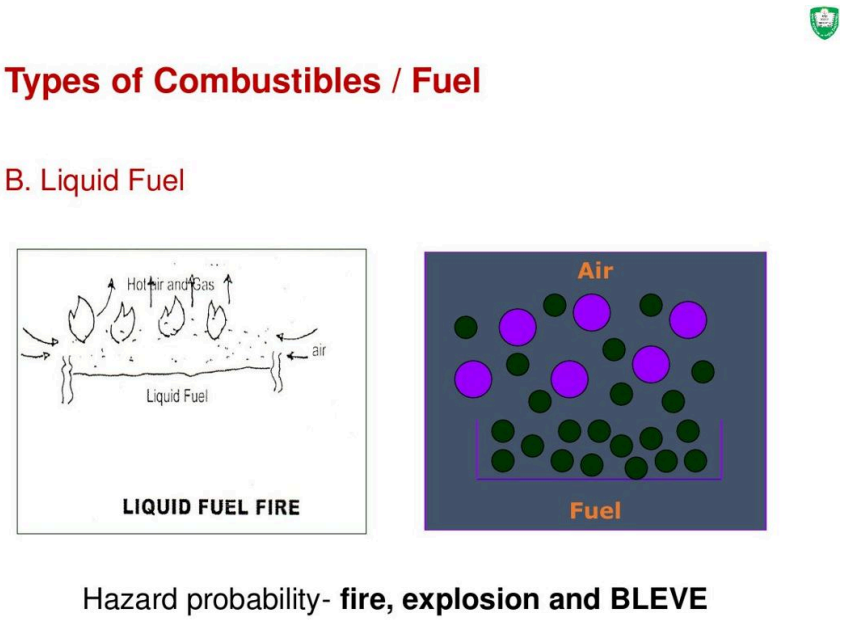
bags, paint) are not fuels because they are not used or suitable for producing useful heat: they burn badly, give toxic smoke, are expensive or have low heating value. **All fuels are combustible but all combustible materials are not fuel.**

Types of explosion (2024 1c)

Explosion = rapid combustion (or expansion) with a sudden, violent change of pressure, liberating and expanding a large volume of gas; fire gives burning damage, explosion gives both **blast and burning** damage.

Type	Meaning
Dust explosion	Rapid combustion of fine combustible particles suspended in air (coal, flour, sugar, wood, plastics, aluminium, magnesium). Prevent: avoid dust clouds, wet process, remove ignition sources.
Deflagration	Flame front/shock wave moves **slower than the speed of sound**; peak pressure ~8 bar in a closed vessel.
Detonation	Shock wave moves **faster than sound**; ~20 bar or more; much more destructive.
Vapour cloud explosion (VCE)	Large release of flammable vapour forms a cloud that drifts to an ignition source. Unconfined (UVCE) in open air, threatens a large area (Beirut 2020 is a famous blast); confined (CVCE) inside vessels/buildings gives higher pressure.
BLEVE (boiling liquid expanding vapour explosion)	Boiling liquid expanding vapour explosion of a pressurised liquefied-gas tank heated by fire (see A8).
Physical / pressure explosion	Bursting of a boiler or vessel by over-pressure without burning.

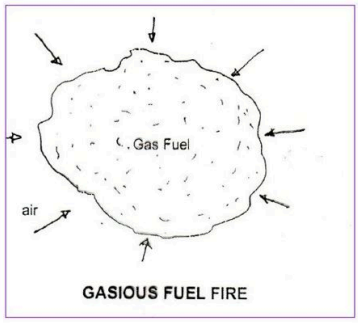
> **Side note:** Solid fuels mainly give fire (but dust can explode); liquid fuels give fire, explosion and BLEVE; gaseous fuels give fire, explosion, detonation and UVCE (unconfined vapour cloud explosion).

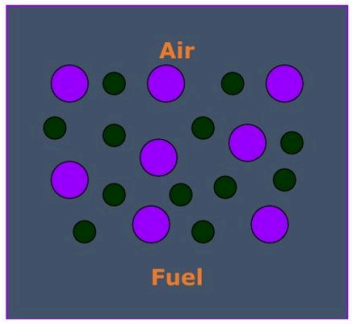


Liquid fuel: hazard of fire, explosion and BLEVE

Types of Combustibles / Fuel

C. Gaseous fuel





Hazard probability- **fire, explosion, detonation, UVCE**

13-Oct-20 ESE 4125/yasir.arafat@che.kuet.ac.bd 7

Gaseous fuel: hazard of fire, explosion, detonation, UVCE

Memory trick: Extinguishing = **"Starve, Smother, Spray (cool), Stop the chain"**. Deflagration = **D**elayed (sub-sonic), Detonation = **D**angerous (super-sonic).

A4 Physical and thermal hazards: heat cramps, heat stroke, frostbite, asphyxiation, cave in, radiation, lighting, housekeeping Must study

Possible · 47% Easy

PAST-PAPER QUESTIONS

- **2023 Q1(c)** [10] Briefly explain the thermal hazards that workers face in extreme temperature conditions. Also discuss their effects.
- **2023 Q2(c)** [10] Briefly explain: (i) heat cramps, (ii) cave in, (iii) asphyxiation, (iv) frostbite, (v) heat stroke.
- **2021 Q1(d)** [12] Write down the types of hazards imposed by radiation. Are there any limits for radiation exposure? What are the preventive measures?
- **2021 Q2(a)** [10] Short notes: (i) emergency lighting, (ii) heat cramps, (iii) cave in, (iv) asphyxiation.
- **2021 Q2(b)** [7] What do you understand by improper housekeeping? What types of accidents can occur due to improper housekeeping?
- **2020 Q1(c)** [12] Explain: frostbite, heat stroke, asphyxiation and ionizing radiation.

★ REMEMBER IT

- Heat illness ladder: **cramps** (salt loss, muscle pain) → **exhaustion** (sweaty, weak) → **stroke** (sweating stops, body above 40 °C, emergency). Cold side: frostnip → **frostbite** (tissue freezes).
- Asphyxiation = not enough O_2 (confined spaces, gas displaces air). Cave in = trench/excavation walls collapse on workers.

Before you start: how the body handles heat, cold and air

Your body keeps its core at about 37 °C. In heat it sweats (losing water **and salt**); in cold it shivers and cuts blood flow to fingers, toes, ears. Problems start when these defences are overwhelmed: too much salt lost (cramps), too much water lost (exhaustion), the thermostat itself fails (stroke), the core cools (hypothermia), or the skin freezes (frostbite).

You also need about 21 % oxygen in air. Below about 19.5 % you need breathing apparatus; at 6 % death comes in minutes.

Thermal hazards in extreme temperature and their effects (2023 1c)

Temperature extremes have **generalised** effects (whole body: hypothermia, heat exhaustion, heat stroke) and **local** effects (one part: frostbite, heat cramps, burns).

Hazard	Cause / what happens	Signs and effects	First aid / prevention
Heat cramps	Painful involuntary muscle spasms during heavy work in heat; heavy sweating loses salt, and the worker drinks water without salt	Spasms in calves, thighs, abdomen, shoulders; may start hours later; common in first days of hot work	Drink saline water or take half a spoon of salt; rest in shade; acclimatise
Heat exhaustion	Loss of water and salt, blood pools in skin	Fatigue, nausea, headache, dizziness, low pulse and blood pressure, cool clammy skin	Move to cool place, give cool liquids
Heat stroke	Excessive heat and exhaustion; the body's internal temperature-regulating mechanism **shuts down**, sweating stops	Very high body temperature, hot dry skin, confusion, collapse; **medical emergency**, may cause death	Cool the body immediately, call medical help; work-rest cycles, water, shade
Heat burns	Steam splash, contact with un-insulated hot lines, flames, molten material	Skin burns	Insulation, PPE (personal protective equipment), cold water on burn
Hypothermia	Body loses heat faster than it produces; core falls below 35 °C	Shivering (begins below 36 °C), confusion, rigidity; below 28 °C heart fibrillation; death	Warm clothing, heated shelters, limit exposure
Frostbite	Local effect of cold (even without hypothermia): fluid around cells freezes because circulation is poor	Pain then numbness in nose, fingers, toes, cheeks, ears; tissue death	Do not expose skin below 25 °F (−4 °C); gloves, insulated boots, rewarming

Short notes (2023 2c, 2021 2a, 2020 1c)

- **Asphyxiation**: interference with the normal supply of oxygen to body tissues. *Simple asphyxiants* (nitrogen, CO_2, hydrogen, methane, ethane) dilute oxygen in air below what the lungs need. *Chemical asphyxiants* (CO, H_2S, HCN, chlorine) react with haemoglobin or cells and stop oxygen use, causing permanent damage. Common in confined spaces, tanks, pits, after inert gas purging or welding. Prevent: test oxygen before entry (≥ 19.5 %), ventilate, use air-supplied respirator, standby person.
- **Cave in (engulfment)**: finely divided solids (sawdust, grain, sugar, sand, coal) trap and bury a person like quicksand. A crust over a hollow in a bin or silo collapses under the worker's weight, or material drawn from the bottom forms a funnel that sucks the worker in; death by suffocation can occur in under a minute. Prevent: never walk on stored material, lock out discharge, use harness and lifeline, permit system, shoring of trenches.
- **Emergency lighting**: lighting for important stairways, exits and control points that switches on **automatically** with minimum delay when the normal lighting fails; it must be powered from a **different source** (battery or generator) so people can evacuate safely and critical operations can be shut down.
- **Ionizing radiation**: X-rays, gamma rays and particles from radioactive isotopes (used in radiography, level gauges, medicine) that ionise body cells. Effects: acute burns, dermatitis, blood disorders; long-term bone-marrow damage, cancer, leukaemia, genetic effects.

Radiation: types, exposure limits, preventive measures (2021 1d)

Type	Source	Effect
Ionizing (X-ray, γ, α, β)	Radiography (weld testing), level gauges, isotopes, nuclear plants, medical X-ray	Burns, dermatitis, blood dyscrasias; chronic: cancer, leukaemia, genetic damage
Ultraviolet	Arc welding	Conjunctivitis and keratitis (red, painful eyes), skin burn; usually heals in a few days
Infrared / intense light	Furnaces, welding, glass	Eye and skin burns, cataract

Limits: ICRP (International Commission on Radiological Protection) maximum occupational dose = **5 rem per year** (50 mSv) whole body, measured as total effective dose equivalent (TEDE); **15 rem/yr** to the eye lens and **50 rem/yr** to extremities (hands, feet) and skin. Exposure is measured with a personal dosimeter (film badge).

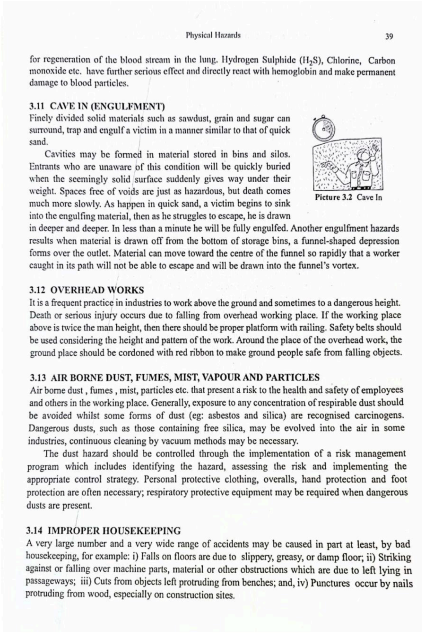
Preventive measures: (1) shielding (lead, concrete) thick enough to bring exposure below the limit; (2) **time-distance-shielding**: minimise time, maximise distance; (3) periodic medical examination (every 2 months in the textbook) and transfer of a worker who exceeds the limit; (4) pregnant women not engaged; (5) training on hazards and safe procedures; (6) warning signs and barriers around radiation areas; (7) personal dosimetry.

Improper housekeeping (2021 2b)

Improper housekeeping = an untidy, dirty, disorganised workplace: materials, tools and waste lying in passages, spills not cleaned, poor stacking, blocked exits. Accidents it causes:

1. Falls on slippery, greasy or wet floors.
2. Striking against or tripping over machine parts or objects left in passageways.
3. Cuts from sharp objects protruding from benches.
4. Punctures by nails sticking out of wood (construction sites).
5. Badly stacked materials falling on people; internal transport accidents in blocked aisles.
6. Fire from combustible waste and excess flammable material.
7. Health damage from dust and chemicals on dirty surfaces.

Good housekeeping (a management responsibility): defined storage areas, racks for tools, waste bins, non-slip floors, regular cleaning, clear aisles and exits (5S).



Physical Hazards 39

for regeneration of the blood stream in the lung. Hydrogen Sulphide (H_2S), Chlorine, Carbon monoxide etc. have further serious effect and directly react with hemoglobin and make permanent damage to blood particles.

3.11 CAVE IN (ENGULFMENT)

Finely divided solid materials such as sawdust, grain and sugar can surround, trap and engulf a victim in a manner similar to that of quick sand.

Cavities may be formed in material stored in bins and silos. Entrants who are unaware of this condition will be quickly buried when the seemingly solid surface suddenly gives way under their weight. Spaces free of voids are just as hazardous, but death comes much more slowly. As happen in quick sand, a victim begins to sink into the engulfing material, then as he struggles to escape, he is drawn in deeper and deeper. In less than a minute he will be fully engulfed. Another engulfment hazards results when material is drawn off from the bottom of storage bins, a funnel-shaped depression forms over the outlet. Material can move toward the centre of the funnel so rapidly that a worker caught in its path will not be able to escape and will be drawn into the funnel's vortex.

Picture 3.2 Cave In

3.12 OVERHEAD WORKS

It is a frequent practice in industries to work above the ground and sometimes to a dangerous height. Death or serious injury occurs due to falling from overhead working place. If the working place above is twice the man height, then there should be proper platform with railing. Safety belts should be used considering the height and pattern of the work. Around the place of the overhead work, the ground place should be cordoned with red ribbon to make ground people safe from falling objects.

3.13 AIR BORNE DUST, FUMES, MIST, VAPOUR AND PARTICLES

Air borne dust , fumes , mist, particles etc. that present a risk to the health and safety of employees and others in the working place. Generally, exposure to any concentration of respirable dust should be avoided whilst some forms of dust (eg: asbestos and silica) are recognised carcinogens. Dangerous dusts, such as those containing free silica, may be evolved into the air in some industries, continuous cleaning by vacuum methods may be necessary.

The dust hazard should be controlled through the implementation of a risk management program which includes identifying the hazard, assessing the risk and implementing the appropriate control strategy. Personal protective clothing, overalls, hand protection and foot protection are often necessary; respiratory protective equipment may be required when dangerous dusts are present.

3.14 IMPROPER HOUSEKEEPING

A very large number and a very wide range of accidents may be caused in part at least, by bad housekeeping, for example: i) Falls on floors are due to slippery, greasy, or damp floor; ii) Striking against or falling over machine parts, material or other obstructions which are due to left lying in passageways; iii) Cuts from objects left protruding from benches; and, iv) Punctures occur by nails protruding from wood, especially on construction sites.

CamScanner

Textbook: asphyxiation, cave in (engulfment), overhead work, airborne dust and improper housekeeping

Memory trick: Heat ladder **C-E-S**: **C**ramps (salt) → **E**xhaustion (water) → **S**troke (thermostat broken = emergency). Cold: hypo**thermia** = whole body, frost**bite** = a bite on one spot.

A5 Electrical hazards (severity factors, protection) and biological hazards

Good to have | Possible · 47% | Easy

PAST-PAPER QUESTIONS

- **2023 Q2(a)** [12] Briefly explain the main methods of protection against electrical hazards in industry.
- **2022 Q2(a)** [12] What are the key factors that determine the severity and consequences of electrical injuries? What are the risks posed by biological hazards in different occupational settings?

★ REMEMBER IT

- Electric shock severity: **current** (not voltage) through the body, its **path** (hand-to-hand across the heart is worst), **duration**, frequency and body resistance (wet skin). Protection: insulation, earthing, ELCB/RCD, isolation and lock-out, safe distance, PPE (personal protective equipment).

Before you start: electricity and the body

Ohm's law: **current I = voltage V / resistance R**. It is the **current** through the body that hurts. Dry skin has about 100 000 Ω, so 220 V gives only 2.2 mA (a tingle). Sweaty skin (10 000 Ω) gives 22 mA (cannot let go), wet skin (1000 Ω or less) gives 220 mA (heart stops).

mA = milliampere = 1/1000 ampere. About 10-20 mA is the "let-go" limit; 50-100 mA can make the heart quiver (ventricular fibrillation).

Factors that determine the severity of electrical injuries (2022 2a)

1. **Voltage of the circuit**: higher voltage drives more current; low-voltage deaths are usually by heart fibrillation, high-voltage deaths by deep electro-thermal burns. (Below 600 V is called low voltage.)
2. **Resistance of the body**: dry skin ~100 000 Ω, sweaty 10 000 Ω, wet ~400-1000 Ω; current I = V/R rises sharply when skin is wet or burnt.
3. **Path of current through the body**: hand-to-hand or hand-to-foot passes through heart and lungs (most dangerous); elbow-to-finger of one hand is less serious.
4. **Duration of contact**: 40-100 mA for 10 s can stop breathing; the longer the flow, the more energy and damage.
5. Also: type of current (DC freezes muscles more, AC causes fibrillation more), frequency, the person's health.

Current (mA)	Effect
< 1	Not felt
3	Painful shock
10-15	Muscles contract, many people cannot let go
30	Breathing difficult, unconsciousness
50-100	Ventricular fibrillation possible
> 200	Severe burns, heart stops

Consequences: burns (inside and outside), involuntary muscle contraction ("frozen on the circuit"), ventricular fibrillation, cardiac arrest, respiratory arrest (death in 4-6 min without oxygen),

secondary injuries (falls from ladders, fractures from jolts), and fires from sparks, short circuits, lightning and static electricity.

Main methods of protection against electrical hazards (2023 2a)

Method	How it protects
Insulation	High-resistance cover (rubber, PVC, mica, glass) on conductors prevents contact; colour code: green/green-yellow = earth, white/grey = neutral, black/red = live
Guarding	Live parts ≥ 50 V enclosed in rooms, vaults, cabinets, or placed ≥ 8 ft (2.44 m) high; warning signs; locked rooms for > 600 V
Grounding (earthing)	Low-resistance path to earth: system (neutral) ground protects equipment; **equipment ground** on metal frames makes a fault current trip the breaker instead of passing through the worker
Circuit protection devices	Fuse (melts), circuit breaker (trips) on overload/short circuit; **GFCI** (ground-fault circuit interrupter) (ground-fault circuit interrupter) compares outgoing and returning current and cuts power within 1/40 s if the leak exceeds ~5-6 mA - used in wet places
Safe work practices	De-energise before work, **lockout/tagout** (one lock, one key per worker), test before touch, safe distances (OSHA table), work in pairs, no wet hands
PPE and insulated tools	Rubber gloves, sleeves, mats, blankets, insulated boots, Class E helmets, insulated hooks and tools rated for the voltage
Design and selection	Hazardous-area classification and explosion-proof (flameproof) equipment with proper temperature class in flammable areas; lightning arrestors; earthing of tanks against static
Training and good judgment	Qualified electricians, first-aid for shock displayed, awareness

Risks posed by biological hazards in different occupational settings (2022 2a)

Biological hazard = any living agent (bacteria, viruses, fungi, parasites) or its toxin that causes infection, allergy or poisoning. They spread by contact with infected people, animals, body fluids, contaminated water, dust or food.

Setting	Risk
Hospitals, dental, laboratories, paramedics	Blood-borne diseases (hepatitis B and C, HIV) from needle-stick; TB; infections
Sewage, drains, plumbing, waste/rubbish handling	Bacteria, leptospirosis; **H_2S** produced by bacteria in drains (toxic, can kill)
Farms, livestock, meat, wool, hides	Anthrax, bird flu, swine flu, mad cow disease, brucellosis
Textile (cotton) mills, grain, hay	Byssinosis (endotoxin), farmer's lung (fungal spores), allergies from mites in dust
Child care, schools, police/prison	Measles, mumps, gastric viruses, TB
Food processing, cooling towers	Food-borne bacteria, Legionella

Control: identify the hazard, assess and control the risk (vaccination, hygiene, gloves and masks, safe disposal of sharps and waste, ventilation), and monitor.

> **Memory trick:** Shock severity = **"V-R-P-T"**: **V**oltage, **R**esistance, **P**ath, **T**ime. Protection = **"I Got Good Circuit Safety"**: **I**nsulation, **G**uarding, **G**rounding, **C**ircuit devices, **S**afe practice.

A9 Fire management in a high-rise building; PASS Good to have Possible · 47% Easy

PAST-PAPER QUESTIONS

- **2024 Q2(a)** [8] Properly explain the fire management procedure in a high rise building.
- **2021 Q3(d)** [12] What are meant by PASS (Pull, Aim, Squeeze, Sweep (using an extinguisher)) and fire triangle? Explain the fire management in high-rise buildings.

★ REMEMBER IT

- PASS (using an extinguisher): **P**ull the pin, **A**im at the base, **S**queeze the handle, **S**weep side to side.
- High-rise fire: detect & alarm → compartment (fire doors) → protected stairs (never lifts) → sprinklers and risers → smoke control → evacuation drill → fire warden.

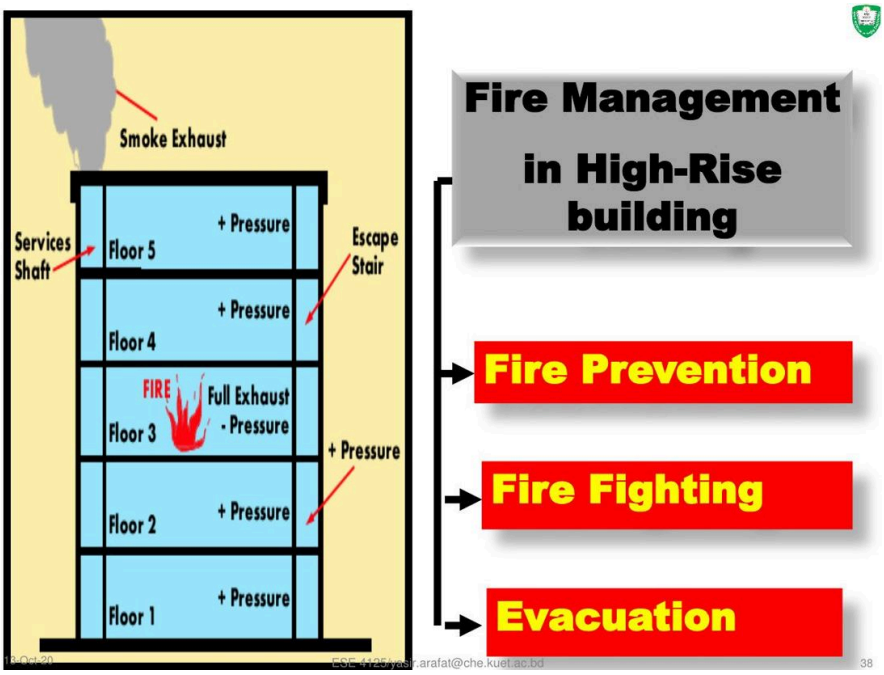
Before you start: fire in a tall building

In a high-rise (tall) building, fire and smoke climb up through stairs and shafts, people are far from the ground, fire-service ladders cannot reach the upper floors, and lifts become traps. So the building must be designed to **prevent** fire, **fight** it by itself (its own water and pumps), and **evacuate** people safely by protected stairs.

PASS and fire triangle (2021 3d)

Fire triangle: fire needs fuel + oxygen + heat together; remove any one and the fire goes out (see A6). **PASS** is how to use a portable fire extinguisher: **P**ull the pin, **A**im the nozzle at the base of the fire, **S**queeze the handle, **S**weep from side to side until the fire is out.

Fire management in a high-rise building (2024 2a, 2021 3d)



Fire management in high-rise buildings: A prevention, B fire fighting, C evacuation

A. Fire prevention

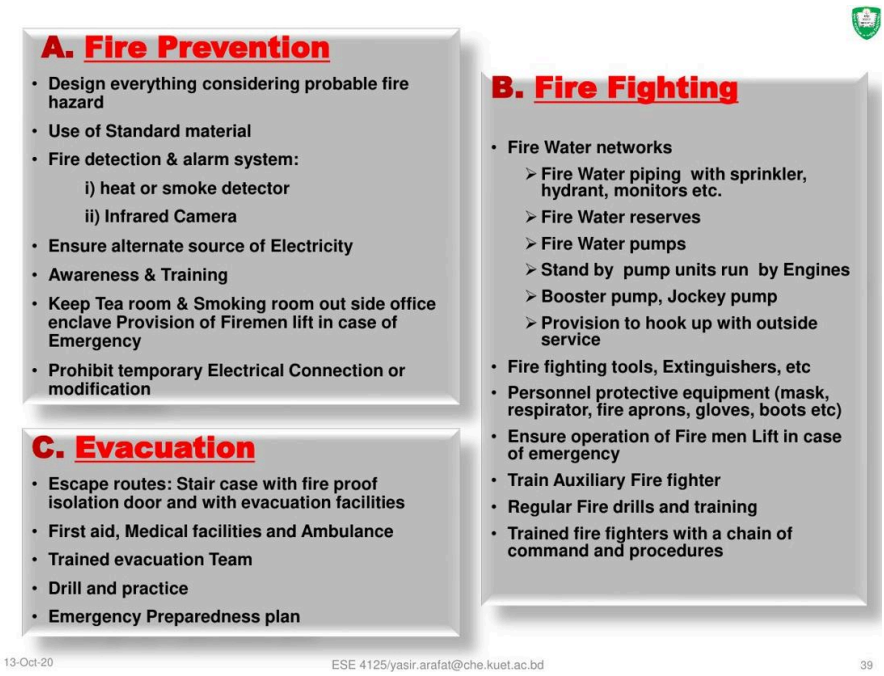
- Design the building for its fire hazard: fire-resistant (standard) materials, compartments, fire doors, fire-rated shafts.
- Automatic **fire detection and alarm** on every floor: heat and smoke detectors, manual call points, infrared cameras.
- Reliable **alternate electricity** (generator) for pumps, emergency lights, fire lifts.
- Prohibit temporary/overloaded electrical connections; maintain wiring; tea/smoking room outside the main area.
- Fire awareness and training for all occupants, regular fire audits and housekeeping.

B. Fire fighting

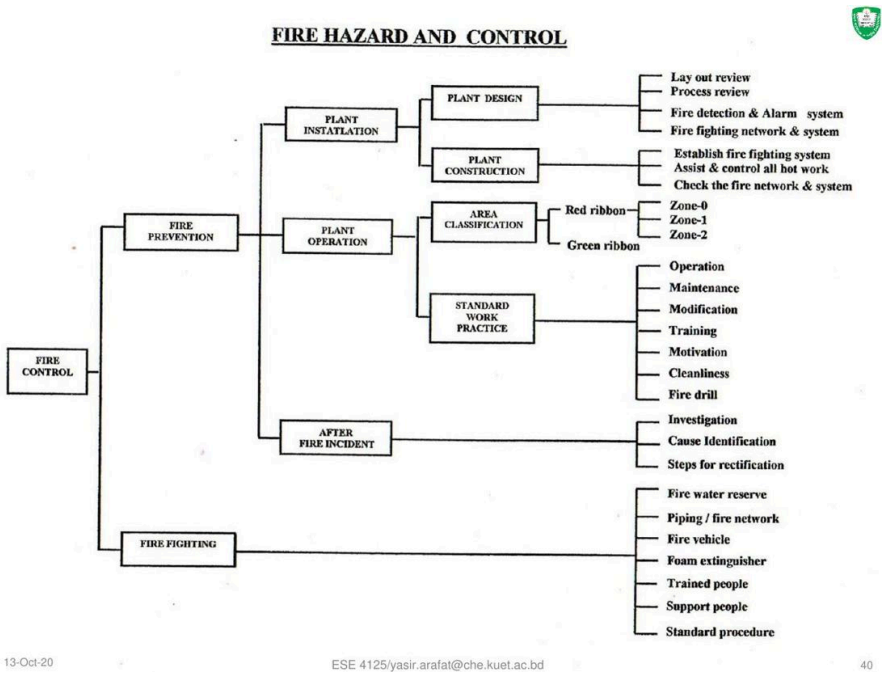
- **Fire water network**: sprinkler piping in every floor, wet risers, hydrants and hose reels, monitors.
- **Water reserve** (underground and roof tanks) sufficient for the required duration.
- **Fire pumps**: main electric pump, **standby diesel engine pump** and a small **jockey pump** that keeps the network pressurised; booster pumps for upper floors.
- Hook-up (inlet connection) so the outside fire service can pump into the building network.
- Portable extinguishers of the right class on every floor; fire-fighting tools and PPE (personal protective equipment).
- **Firemen lift** (fire-fighter's lift) on emergency power; trained auxiliary fire fighters; clear chain of command; regular fire drills.

C. Evacuation

- Escape routes: at least two protected staircases with fire-proof self-closing isolation doors, pressurised against smoke, kept clear; exit signs and emergency lighting.
- **Never use the lift** during a fire; crawl low under smoke; feel doors before opening; if clothes catch fire, stop-drop-roll.
- If trapped, go to an area of refuge, seal door gaps with wet towels, signal from the window, do not jump.
- Trained evacuation team (floor wardens), assembly point outside, head count, first aid and ambulance.
- Practise through regular drills under the emergency preparedness plan (EPP).



Slide: A fire prevention, B fire fighting, C evacuation in high-rise buildings



Fire hazard and control tree: plant design, installation, operation, after-fire assessment, fire fighting

Memory trick: High-rise = **"P-F-E"**: **P**revent (design, detect, alarm), **F**ight (water, pumps, sprinklers, firemen lift), **E**vacuate (fire-proof stairs, no lift, assembly point).

A3 Numericals: hazard matrix rating, fractional dead time and hazard rate of a protective system

Good to have · Possible · 33% · Medium

PAST-PAPER QUESTIONS

- **2022 Q1(b)** [15] A safety valve has a failure rate of once every two years (0.5/year), the test interval is 1 week and the system pressure is expected to rise once a year. Calculate the fractional dead time and hazard rate.
- **2021 Q1(c)** [8] How could you assess the impacts of hazard with the help of a hazard matrix?

UNDERSTAND **Read this to understand; you do not need to write it**

A safety valve can break silently (stuck shut). Nobody knows until the next test. On average a hidden fault waits **half the test interval**. So the fraction of time the valve is useless is the **fractional dead time**, FDT (fractional dead time) = f T / 2.

A real accident needs two things at once: a **demand** (pressure rises) **and** a dead valve. Hazard rate H = demand rate × FDT.

Symbols used here

Symbol	Name and meaning	Unit
f	Failure rate of the protective device (fail-danger faults)	per year
T	Test interval (convert to years)	year
FDT	Fractional dead time = fT/2	–
D	Demand rate (how often the device is needed)	per year
H	Hazard rate = D × FDT	per year

CALCULATOR **Fractional dead time and hazard rate**

Change any number; every step is recalculated and shown with the numbers put in.

1. Test interval in years: **7** / 365 = **0.01918** yr
2. Fractional dead time: FDT = fT/2 = **0.5** × **0.01918** / 2 = **0.00479**
3. Hazard rate: H = D × FDT = **1** × **0.004795** = **0.00479** per year
4. Once in: 1 / **0.004795** = **209** years

CALCULATOR **Hazard matrix rating**

Change any number; every step is recalculated and shown with the numbers put in.

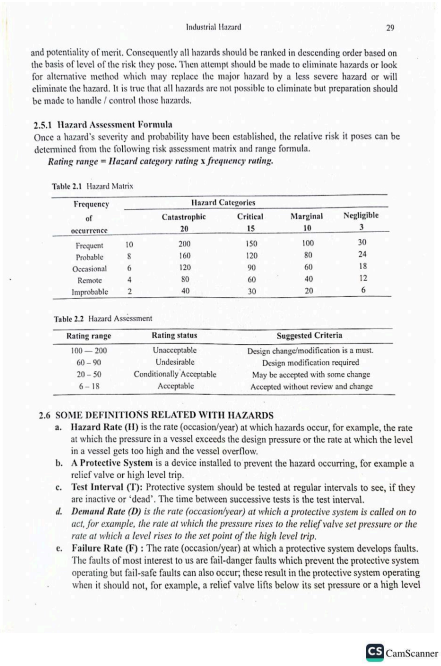
1. Rating: Rating = severity × frequency = **20** × **6** = **120**
2. Status: **Unacceptable: design change a must**

Before you start: two simple numbers

1. Hazard matrix. Give each hazard a score for **how bad** (severity) and a score for **how often** (frequency). Multiply them. A big product means a dangerous hazard that must be fixed first.

2. Hazard rate of a protective device. A safety valve or trip can fail silently (stuck shut) and nobody knows until the next test. On average it stays broken for **half** the test interval. So the fraction of time it is "dead" = failure rate × test interval / 2. A real danger (hazard) happens only if a **demand** (pressure rise) arrives while it is dead: hazard rate = demand rate × dead fraction.

Hazard matrix (2021 1c)



Industrial Hazard 29

and potentiality of merit. Consequently all hazards should be ranked in descending order based on the basis of level of the risk they pose. Then attempt should be made to eliminate hazards or look for alternative method which may replace the major hazard by a less severe hazard or will eliminate the hazard. It is true that all hazards are not possible to eliminate but preparation should be made to handle / control those hazards.

2.5.1 Hazard Assessment Formula

Once a hazard's severity and probability have been established, the relative risk it poses can be determined from the following risk assessment matrix and range formula.

Rating range = Hazard category rating x frequency rating.

Table 2.1 Hazard Matrix

Frequency of occurrence		Hazard Categories: Catastrophic 20	Critical 15	Marginal 10	Negligible 3
Frequent	10	200	150	100	30
Probable	8	160	120	80	24
Occasional	6	120	90	60	18
Remote	4	80	60	40	12
Improbable	2	40	30	20	6

Table 2.2 Hazard Assessment

Rating range	Rating status	Suggested Criteria
100 — 200	Unacceptable	Design change/modification is a must.
60 – 90	Undesirable	Design modification required
20 – 50	Conditionally Acceptable	May be accepted with some change
6 – 18	Acceptable	Accepted without review and change

2.6 SOME DEFINITIONS RELATED WITH HAZARDS

a. **Hazard Rate (H)** is the rate (occasion/year) at which hazards occur, for example, the rate at which the pressure in a vessel exceeds the design pressure or the rate at which the level in a vessel gets too high and the vessel overflow.

b. **A Protective System** is a device installed to prevent the hazard occurring, for example a relief valve or high level trip.

c. **Test Interval (T):** Protective system should be tested at regular intervals to see, if they are inactive or 'dead'. The time between successive tests is the test interval.

*d. **Demand Rate (D)** is the rate (occasion/year) at which a protective system is called on to act, for example, the rate at which the pressure rises to the relief valve set pressure or the rate at which a level rises to the set point of the high level trip.*

e. **Failure Rate (F) :** The rate (occasion/year) at which a protective system develops faults. The faults of most interest to us are fail-danger faults which prevent the protective system operating but fail-safe faults can also occur; these result in the protective system operating when it should not, for example, a relief valve lifts below its set pressure or a high level

CamScanner

Textbook hazard matrix (Table 2.1), hazard assessment (Table 2.2) and definitions of hazard rate terms

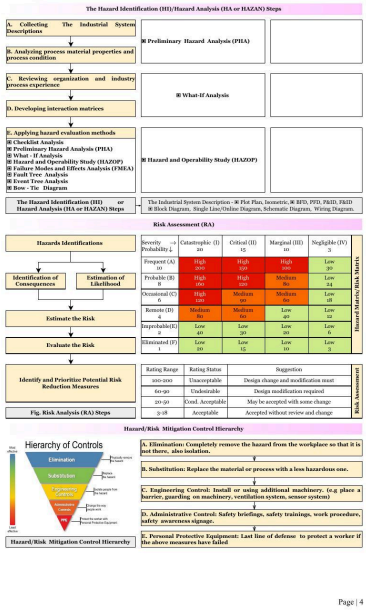
```
Rating = Hazard category (severity) rating × Frequency rating
```

Frequency ↓ / Severity →	Catastrophic 20	Critical 15	Marginal 10	Negligible 3
Frequent 10	200	150	100	30
Probable 8	160	120	80	24
Occasional 6	120	90	60	18
Remote 4	80	60	40	12
Improbable 2	40	30	20	6

Rating range	Status	What to do
100-200	Unacceptable	Design change/modification is a must
60-90	Undesirable	Design modification required
20-50	Conditionally acceptable	May be accepted with some change
6-18	Acceptable	Accepted without review and change

How to assess impacts with it: (1) list all hazards, (2) give each a severity class from its worst credible consequence, (3) give a frequency class from records/experience, (4) multiply to get the rating, (5) read the status: unacceptable ones are redesigned first, then undesirable ones, and so

on. Example: a gas leak that could cause an explosion (catastrophic, 20) expected occasionally (6) → 120 → unacceptable, redesign. A slippery floor (marginal 10, probable 8) → 80 → undesirable, modify.



Teacher's risk assessment steps, hazard/risk matrix and hierarchy of controls

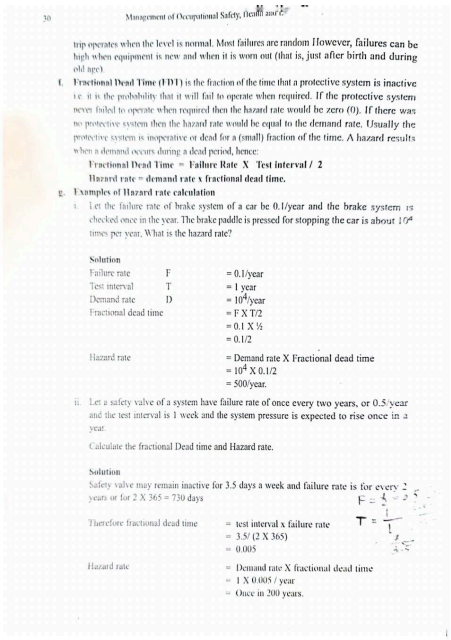
Terms for the numerical

- **Protective system**: device that prevents the hazard (relief valve, high-level trip).
- **Failure rate F** (per year): how often the protective device develops a fail-danger fault.
- **Test interval T**: time between two tests of the device.
- **Demand rate D** (per year): how often the protective system is called on to act (pressure rises to set point).
- **Fractional dead time FDT** = fraction of time the device is inactive = F T / 2.
- **Hazard rate H** (per year) = D × FDT.

Numerical (2022 1b): safety valve

Step (write this)	Why / what it means
`F = 0.5 /year (once in 2 years); T = 1 week = 7 days; D = 1 /year`	Write the data. Keep F and T in the same time unit.
`T in years = 7/365 = 0.01918 yr`	Convert the week into years because F is per year.
`FDT = F T / 2 = 0.5 × 0.01918 / 2 =` **`0.0048 ≈ 0.005`**	A fault appears at a random moment between tests, so on average it stays hidden for half the interval (3.5 days).
`Same thing in days: FDT = 3.5 days / 730 days = 0.0048`	730 days = mean time between failures (2 years); the valve is dead 3.5 days in each failure.
`H = D × FDT = 1 × 0.0048 =` **`0.0048 per year`**	A hazard needs a demand AND a dead valve at the same time.
`1/H = 1/0.005 ≈` **`once in 200 years`**	Easier way to say it in the answer.

Side note: The textbook rounds FDT to 0.005, giving "once in 200 years". With 0.0048 you get once in 208 years: both are accepted. Example in the book: brake failure 0.1/yr tested yearly, pressed 10^4 times a year: FDT = 0.1 × 1/2 = 0.05, H = 10^4 × 0.05 = 500 per year (very unsafe, test more often).



30 Management of Occupational Safety, Health and E

trip operates when the level is normal. Most failures are random However, failures can be high when equipment is new and when it is worn out (that is, just after birth and during old age).

f. **Fractional Dead Time (FDT)** is the fraction of the time that a protective system is inactive i.e. it is the probability that it will fail to operate when required. If the protective system never failed to operate when required then the hazard rate would be zero (0). If there was no protective system then the hazard rate would be equal to the demand rate. Usually the protective system is inoperative or dead for a (small) fraction of the time. A hazard results when a demand occurs during a dead period, hence:

Fractional Dead Time = Failure Rate X Test interval / 2

Hazard rate = demand rate x fractional dead time.

g. **Examples of Hazard rate calculation**

i. Let the failure rate of brake system of a car be 0.1/year and the brake system is checked once in the year. The brake paddle is pressed for stopping the car is about 10^4 times per year. What is the hazard rate?

Solution

Failure rate F = 0.1/year
Test interval T = 1 year
Demand rate D = 10^4/year
Fractional dead time = F X T/2
= 0.1 X ½
= 0.1/2

Hazard rate = Demand rate X Fractional dead time
= 10^4 X 0.1/2
= 500/year.

ii. Let a safety valve of a system have failure rate of once every two years, or 0.5/year and the test interval is 1 week and the system pressure is expected to rise once in a year.

Calculate the fractional Dead time and Hazard rate.

Solution

Safety valve may remain inactive for 3.5 days a week and failure rate is for every 2 years or for 2 X 365 = 730 days

Therefore fractional dead time = test interval x failure rate
= 3.5/ (2 X 365)
= 0.005

Hazard rate = Demand rate X fractional dead time
= 1 X 0.005 / year
= Once in 200 years.

CamScanner

Textbook definitions of FDT and hazard rate with the same safety-valve example

Memory trick: "Fail, Test, Halve; Demand times Dead": FDT = F T / 2, H = D × FDT.

WRITE IN EXAM **Write this in the exam (this much is enough for full marks)**

2022 1(b) Given: f = 0.5 /yr, test interval T = 1 week = 7/365 = 0.01918 yr, demand D = 1 /yr

Fractional dead time: FDT = fT/2

= 0.5 × 0.01918 / 2

= **0.0048 (≈ 0.005)**

Hazard rate: H = D × FDT

= 1 × 0.0048

= **0.0048 per year**

i.e. the pressure vessel may burst about **once in 200 years**.

Section B: Occupational health, chemicals, pollution and environmental management

B8 MSDS (material safety data sheet): purpose, 16 headings, key sections, responsibilities of supplier, employer, worker Must study

Very likely · 100% Easy

PAST-PAPER QUESTIONS

- **2024 Q6(c)** [12] Discuss the purpose of MSDS (material safety data sheet). Also discuss six crucial contents to prepare an MSDS.
- **2023 Q6(c)** [6] What is the primary purpose of a Material Safety Data Sheet (MSDS) and why is it important for work safety?
- **2023 Q6(d)** [14] Discuss these terms to prepare an MSDS: (i) identification of the substance, (ii) hazards identification, (iii) first aid measures, (iv) handling and storage, (v) toxicological information, (vi) regulatory information.
- **2022 Q7(a)** [12] Briefly discuss MSDS. Explain the responsibilities of the (i) supplier, (ii) employer, (iii) worker.
- **2021 Q6(a)** [8] What is meant by hazardous chemicals? What are the responsibilities of a supplier and an employer related to MSDS?
- **2021 Q6(c)** [11] What is meant by MSDS? Discuss: (i) identification of substance and company, (ii) hazards identification, (iii) first aid measures, (iv) handling and storage, (v) regulatory information.
- **2020 Q5(c)** [12] What are the crucial contents of an MSDS?

★ REMEMBER IT

- MSDS has 16 sections. First six to remember: **Identification, Hazards, Composition, First aid, Fire-fighting, Accidental release** ("I Have Cool First-aid For Accidents"). Then handling & storage, exposure controls/PPE, physical properties, stability, toxicology, ecology, disposal, transport, regulatory, other.
- Supplier writes it, employer makes it available and trains, worker reads it and follows it.

Before you start: the chemical's instruction sheet

Every chemical you buy should come with an **MSDS** (material safety data sheet; today called SDS (safety data sheet) under GHS (Globally Harmonised System of classification and labelling of chemicals), the Globally Harmonised System). It tells, in 16 fixed sections, what the chemical is, how it can hurt you, how to store and handle it, what PPE (personal protective equipment) to wear, and what to do if it spills, catches fire or someone is exposed.

MSDS and its purpose (2024 6c, 2023 6c, 2022 7a, 2021 6c)

A **material safety data sheet (MSDS)** is a document prepared by the manufacturer/supplier that gives detailed information on the potential hazards (health, fire, reactivity and environmental) of a chemical product and how to work with it safely. It is the starting point of a chemical health and safety programme.

- Tells what the hazards of the product are.
- Tells how to use, handle and store it safely.
- Tells what to expect if the recommendations are not followed.
- Tells how to recognise symptoms of overexposure and what to do.
- Tells what to do in an accident: first aid, fire fighting, spill clean-up.
- Helps select engineering controls and PPE, train workers and meet legal requirements (hazard communication).

Why important for work safety (2023 6c): workers and supervisors can only protect themselves from hazards they know; the MSDS gives emergency responders and doctors the information they need quickly; it prevents mixing of incompatible chemicals; it is legally required. It must be reviewed at least every 3 years.

The 16 headings (2020 5c)

#	Section	#	Section
1	Identification of the substance and the company	9	Physical and chemical properties
2	Hazards identification	10	Stability and reactivity
3	Composition / information on ingredients	11	Toxicological information
4	First-aid measures	12	Ecological information
5	Fire-fighting measures	13	Disposal considerations
6	Accidental release measures	14	Transport information
7	Handling and storage	15	Regulatory information
8	Exposure controls / personal protection	16	Other information (date prepared/revised)

Crucial contents explained (2024 6c, 2023 6d, 2021 6c)

Section	What it contains
1. Identification of substance and company	Product identifier (name, code), synonyms; supplier's name, address, phone; **emergency phone number**; recommended use and restrictions
2. Hazards identification	GHS classification (health, physical, environmental hazards); signal word (Danger/Warning); hazard statements; pictograms; precautionary statements; other hazards; % of ingredients of unknown toxicity
3. Composition	Chemical names, CAS (Chemical Abstracts Service (registry number of a chemical)) numbers, concentrations of hazardous ingredients
4. First-aid measures	First aid for each route (inhalation, skin, eye, ingestion); most important symptoms (acute and delayed); need for immediate medical care and special treatment
5. Fire-fighting measures	Suitable and unsuitable extinguishing media; hazardous combustion products; PPE for fire fighters
6. Accidental release	Personal precautions, evacuation, containment and clean-up methods
7. Handling and storage	Safe handling precautions; incompatible materials; hygiene (no eating, drinking, smoking); storage conditions (temperature, ventilation, containers)
8. Exposure controls / PPE	Exposure limits (PEL, TLV (threshold limit value), OEL (occupational exposure limit)); engineering controls (local exhaust ventilation); PPE type, glove material and breakthrough time
11. Toxicological information	Likely routes of exposure; symptoms; delayed, immediate and chronic effects; numerical toxicity (LD_{50}, LC_{50}); carcinogenicity
15. Regulatory information	Safety, health and environmental regulations specific to the product (OSHA, EPA (Environmental Protection Agency (USA)), DOT (Department of Transportation (USA)), CPSC (Consumer Product Safety Commission (USA)) in the USA; national laws, e.g. Bangladesh Labour Act, ECR (Environment Conservation Rules (Bangladesh, 1997; new rules 2023)))

Responsibilities (2022 7a, 2021 6a)

Supplier	Employer	Worker
Develop or obtain an MSDS for every controlled product it sells	Obtain an up-to-date MSDS the first time a product is received	After training, follow the safe work measures in the MSDS
Make sure the purchaser has a current MSDS at or before receipt of the product	Check it is dated within the last 3 years; get updates	Know where the MSDSs are kept
Keep the MSDS current (revise when new information appears)	Keep copies easily accessible to workers at the workplace	Know how to find safe-use, PPE and first-aid information in it
Disclose confidential (trade-secret) information to a doctor or nurse when needed for diagnosis or treatment	Train workers on the MSDS contents; may prepare its own sheet with no less information; give confidential information to medical staff in an emergency	Report hazards, missing or outdated MSDSs

Memory trick: Key sections: **"1 Who, 2 Danger, 4 Aid, 5 Fire, 7 Store, 8 PPE, 11 Toxic, 15 Law"**.

B11 Greenhouse effect and global warming, acid rain, smog: causes, effects, control

Must study · Very likely · 100% · Easy

PAST-PAPER QUESTIONS

- **2024 Q7(a)** [8] What are the major causes and environmental effects of (i) acid rain, (ii) SMOG?
- **2023 Q7(d)** [10] What are the major greenhouse gases responsible for global warming? Explain the mechanism of the greenhouse effect.
- **2023 Q8(a)** [6] Short notes on (i) acid rain, (ii) SMOG.
- **2022 Q7(d)** [10] How is global warming caused by greenhouse gases? What are the measures to prevent (i) acid rain, (ii) SMOG?
- **2021 Q7(c)** [10] What is the greenhouse effect? How does CO_2 in air increase global warming?
- **2021 Q7(d)** [10] Write the causes and effects of (i) acid rain, (ii) SMOG.
- **2020 Q6(c)** [12] Explain the greenhouse effect, smog and acid rain.

LEARN IT FROM ZERO

Greenhouse effect, acid rain and smog, from zero

Sunlight (short waves) passes through the air and warms the ground. The warm ground sends heat back up as **infrared** (long waves). CO_2, CH_4, N_2O, water vapour and CFCs **absorb that infrared and send part of it back down**, like the glass of a greenhouse. Without them the earth would be −18 °C; with more of them (from burning fossil fuel) it warms further: global warming.

Acid rain: SO_2 and NO_x from power plants and vehicles react with water in clouds to make sulphuric and nitric acid (pH below 5.6). It kills fish and forests and eats buildings. **Smog** = smoke + fog: classic (London) smog is soot + SO_2 in cold damp air; photochemical (Los Angeles) smog is NO_x + hydrocarbons cooked by sunlight into ozone.

★ REMEMBER IT

- Greenhouse gases: **"Cows Make Nasty Water Clouds"** = CO_2, Methane, N_2O, Water vapour, CFCs.

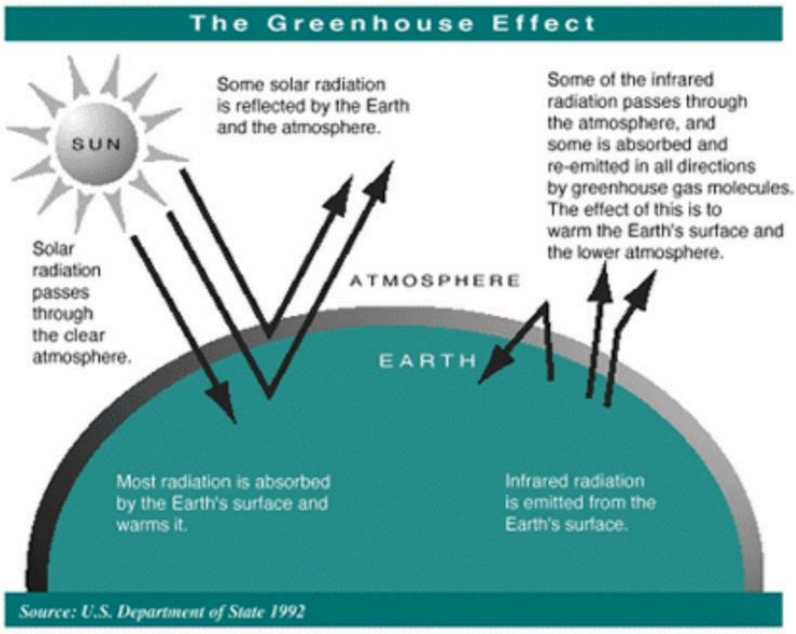
Before you start: three famous air-pollution effects

Greenhouse effect: like the glass of a greenhouse, certain gases let sunlight in but trap the heat the Earth gives back, so the Earth gets warmer. Natural greenhouse effect keeps Earth at about +15 °C (otherwise −18 °C); extra gases from human activity make it too warm (global warming).

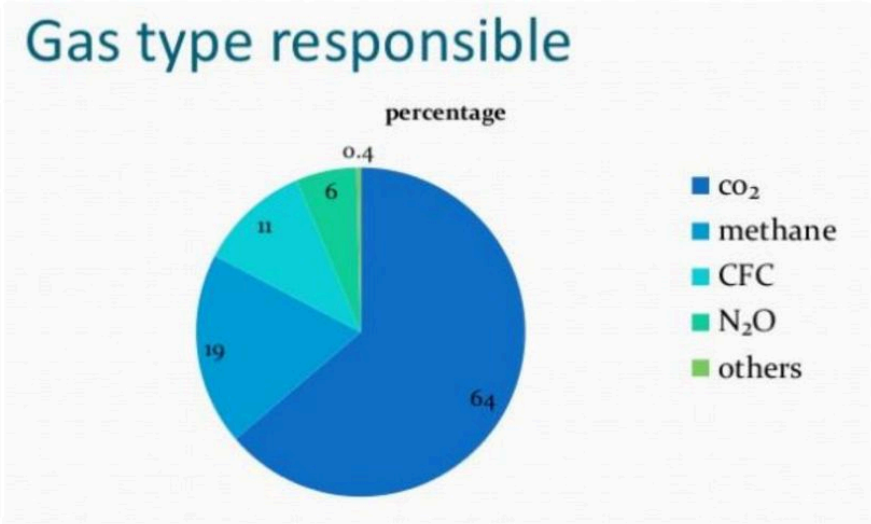
Acid rain: SO_2 and NO_x dissolve in rain to make sulphuric and nitric acid. pH is the acidity scale: 7 neutral, lower = more acidic; normal rain is about 5.6.

Smog = **sm**oke + f**og**: a dirty haze over cities.

Greenhouse effect: mechanism and gases (2023 7d, 2022 7d, 2021 7c, 2020 6c)



Greenhouse effect: incoming solar radiation, infrared radiation from the Earth trapped by greenhouse gases



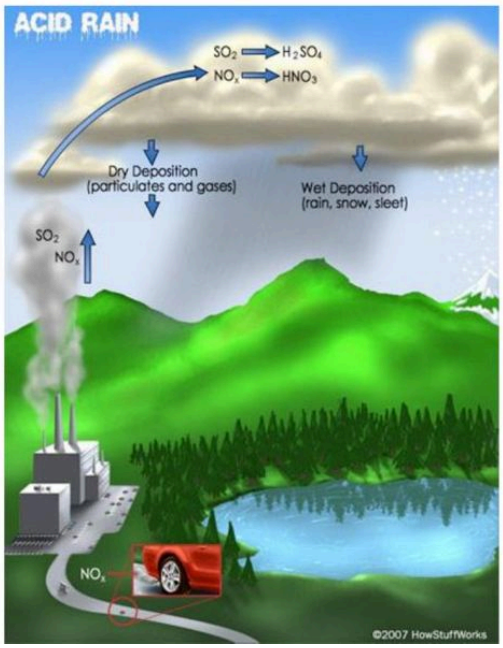
Share of gases responsible for the greenhouse effect: CO_2 largest, then methane, CFCs, N_2O

1. The sun emits short-wave radiation (visible light, UV); about 30 % is reflected by clouds and the surface, the rest passes through the atmosphere.
2. The Earth's surface absorbs it and warms up.
3. The warm Earth emits long-wave **infrared (heat) radiation** back towards space.
4. **Greenhouse gases** in the atmosphere absorb much of this infrared radiation and re-emit it in all directions, including back to the surface.
5. So heat is trapped in the lower atmosphere and the Earth's average temperature rises. More greenhouse gases → more heat trapped → **global warming**.

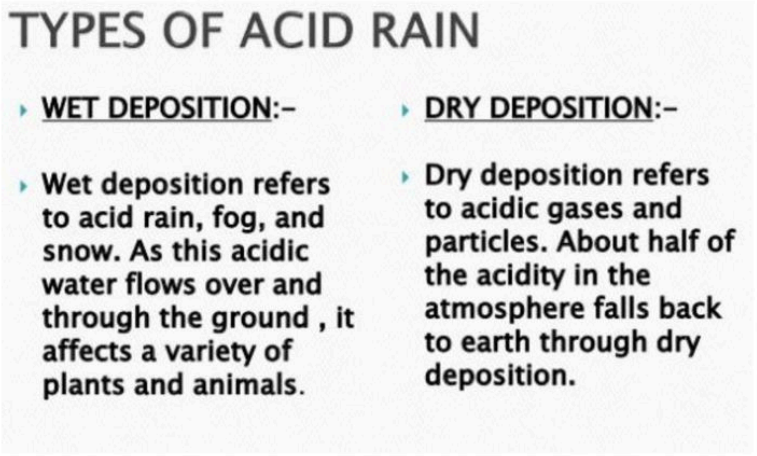
Greenhouse gas	Main sources
Water vapour (most abundant)	Evaporation; increases as the air warms (feedback)
Carbon dioxide CO_2 (largest human contribution)	Burning coal, oil, gas; deforestation; cement
Methane CH_4	Rice paddies, livestock, landfills, gas leaks, coal mines
Nitrous oxide N_2O	Fertilisers, combustion
CFCs (chlorofluorocarbons) and HFCs	Synthetic: refrigerants, aerosol propellants, foam (also destroy ozone)
Ozone (lower atmosphere)	Photochemical smog

How CO_2 increases global warming (2021 7c): CO_2 molecules absorb infrared radiation strongly (at about 15 µm, a window where water vapour does not absorb). Burning fossil fuels and cutting forests have raised CO_2 from about 280 ppm (pre-industrial) to over 420 ppm; CO_2 stays for centuries, so it keeps trapping more heat, raising temperature, which also increases water vapour (amplifying feedback) → melting ice, sea-level rise (a big threat for Bangladesh), extreme weather. Prevention: energy efficiency, renewables, fuel switching, afforestation, carbon capture.

Acid rain: causes, effects, prevention (2024 7a, 2023 8a, 2022 7d, 2021 7d)



Formation of acid rain from SO_2 and NO_x

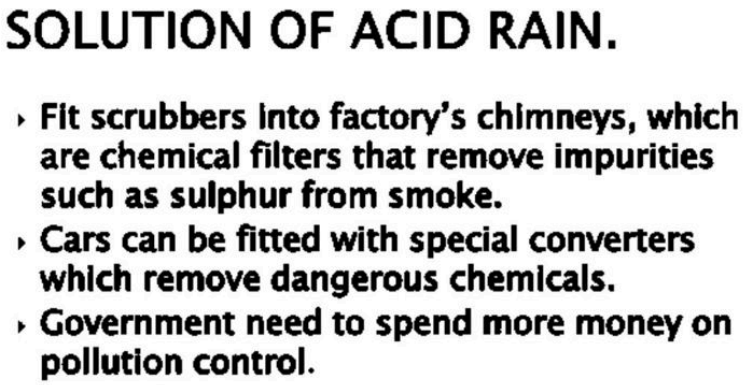


Types of acid rain: wet and dry deposition

Acid rain is rain (or snow, fog, dry particles) with pH below 5.6 (normal rain, from dissolved CO_2), as low as 2.4. **Causes:** SO_2 (coal power plants, smelters, refineries) and NO_x (vehicles, power plants), plus HCl, react with water and oxygen in the air:

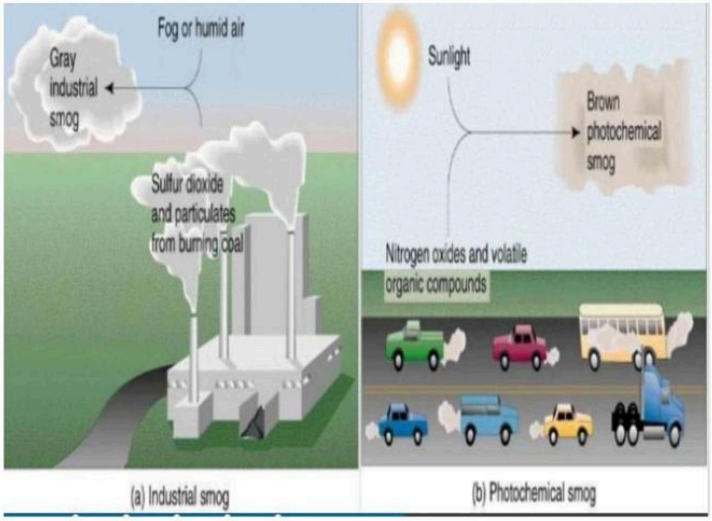
$$SO_2 + ½O_2 + H_2O \rightarrow H_2SO_4; \quad 4NO_2 + O_2 + 2H_2O \rightarrow 4HNO_3$$

They fall as **wet deposition** (rain, snow, fog) or **dry deposition** (acidic gases and particles). **Effects:** acidifies lakes and rivers, fish die (low pH + released aluminium); kills soil micro-organisms and leaches nutrients; damages forests and crops; corrodes metals and dissolves limestone and marble buildings and monuments (Taj Mahal); health effects (respiratory problems). **Prevention:** (see the slide solution below) low-sulphur fuels, coal washing, flue-gas desulphurisation (scrubbers), low-NO_x burners, catalytic converters in vehicles, renewable energy, energy saving; liming of acidified lakes.



Solution of acid rain

Smog: types, causes, effects, prevention (2024 7a, 2023 8a, 2022 7d, 2021 7d)



(a) Industrial (London) smog and (b) photochemical smog

	London (sulphurous, classical) smog	**Los Angeles (photochemical) smog**
Cause	Smoke and SO_2 from burning coal + fog, in cold humid weather with temperature inversion	NO_x + hydrocarbons (VOCs) from vehicle exhaust + strong sunlight → ozone, PAN (peroxyacetyl nitrate), aldehydes
Nature	Reducing, grey-yellow, contains sulphuric acid droplets and soot	Oxidising, brown haze, warm sunny days
Example	London December 1952: about 4000 deaths	Los Angeles, Delhi, Dhaka in winter

Effects: breathing difficulty, asthma, bronchitis, eye and throat irritation, premature deaths; reduced visibility (traffic accidents); damage to crops, rubber and materials; ozone harms plants.
Prevention: reduce vehicle emissions (catalytic converters, CNG, public transport), control industrial emissions, avoid burning coal and garbage, plant trees, use cleaner fuels and renewable energy, monitor air quality and restrict traffic on bad days.

> **Memory trick:** Acid rain = **"SO-NO + water = acid"**, pH < 5.6. Smog: **London = coal + fog (SO_2)**; **LA = cars + sun (ozone)**.

B6 Hazardous chemicals: categories and health effects; hazardous vs toxic; routes of entry; local/systemic and acute/chronic effects Must study

Very likely · 100% Easy

PAST-PAPER QUESTIONS

- **2024 Q6(a)** [8] Distinguish with examples: (i) systemic and local effects, (ii) acute and chronic effects.
- **2024 Q6(b)** [9] What are the differences between hazardous chemicals and toxic chemicals? Write the effects of common categories of hazardous chemicals on human health.
- **2023 Q5(c)** [6] Distinguish between inhalation, ingestion and injection.
- **2022 Q6(c)** [9] Write the common categories of hazardous chemicals and explain their effects on health. How do toxic substances affect the surroundings?
- **2022 Q6(d)** [12] Define and explain with examples: (i) inhalation, (ii) ingestion, (iii) systemic effects, (iv) chronic effects.
- **2021 Q6(d)** [8] Write the occupational effects of: (i) toxic, (ii) corrosive, (iii) reactive, (iv) flammable chemicals.
- **2020 Q5(b)** [8] Explain local, systemic, acute and chronic effects.
- **2020 Q5(d)** [4] What are the types of hazardous chemical exposure process?

★ REMEMBER IT

- Routes of entry, **"I I I + skin"**: Inhalation (breathing, most common), Ingestion (swallowing), Injection (needle/cut), plus skin absorption. **Local** = harm where it touches (acid burn); **systemic** = travels in blood and harms other organs (lead → brain). **Acute** = big dose, quick effect; **chronic** = small doses, years later.

Before you start: chemicals and the body

A chemical harms you only after it **gets in**. There are four doors: breathing it in (inhalation), swallowing it (ingestion), through a cut or needle (injection), or through skin and eyes (absorption).

Once in, it may hurt the **spot it touched** (local effect, e.g. acid burn) or travel in the blood and hurt **another organ** (systemic effect, e.g. CO poisoning the blood). It may act **fast** after one big dose (acute) or **slowly** after years of small doses (chronic).

Hazardous vs toxic chemicals (2024 6b)

Hazardous chemical	Toxic chemical
Any chemical that can cause **health** deterioration **or physical** catastrophe (fire, explosion, corrosion, reactivity)	A chemical that has the inherent ability to **poison** or harm living organisms (health effect)
Wide group: flammable, explosive, corrosive, oxidising, reactive, radioactive and toxic	A **subset** of hazardous chemicals
Example: petrol (flammable, low toxicity), concentrated H_2SO_4 (corrosive), LPG (explosive)	Example: H_2S, CO, cyanide, lead, mercury, benzene, pesticides
All toxic chemicals are hazardous	Not all hazardous chemicals are toxic

Common categories of hazardous chemicals and their health effects (2024 6b, 2022 6c, 2021 6d)

Category	Examples	Effect on health
Toxic	Metals Hg, Pb, Cd; asbestos, pesticides, PCBs, phenols; gases Cl_2, NH_3, H_2S, SO_2; cyanide; benzene, toluene, xylene	Poisoning of organs: brain/nerves, liver, kidney, blood; cancer; death at high dose
Corrosive	Strong acids (H_2SO_4, HCl) and bases (NaOH)	Destroy skin, eyes and respiratory lining on contact; severe burns, blindness (wash 15 min)
Oxidising	Perchlorates, permanganates, peroxides	Burns; intensify fires
Dehydrating	Conc. H_2SO_4, NaOH, lime, P_2O_5	Remove water from tissue, release heat → thermal burns
Reactive (sensitisers)	Mercury salts, formaldehyde, organometallics, isocyanates	Allergies, asthma, dermatitis; violent reactions
Radioactive	Isotopes of Co, Ir, U (radiography, NDT (non-destructive testing))	Irreversible cell damage, cancer, genetic effects
Flammable	Hydrocarbons, solvents, Na, Mg	Burns from fire/explosion; vapour narcosis
Asphyxiant / irritant / anaesthetic gases	H_2S, CO, HCN / NH_3, Cl_2, SO_2, O_3, NO_2 / N_2O, halothane	Stop oxygen use / irritate eyes, nose, lungs / depress nervous system

By target organ: **hepatotoxins** liver (CCl_4, As, Cd, Pb, Hg); **nephrotoxins** kidney (CCl_4, heavy metals); **neurotoxins** nerves (Hg, Pb, pesticides); **haematotoxins** blood (CO, aniline, benzene); **respiratory toxins** (NH_3, Cl_2, asbestos, silica); **cardiotoxins** (CO); **skin toxins** (nickel, latex).

How toxic substances affect the surroundings (2022 6c)

A toxic cloud released from a plant affects the site and the surrounding community: people downwind must be evacuated; it can cause a **domino effect** on nearby plants (loss of operators, fires); fires deplete oxygen; it pollutes air, water and soil, kills animals and crops and enters the food chain. Chlorine and ammonia are the most common major toxic hazards (chlorine: 1 volume of liquid gives 457 volumes of gas, heavier than air; 1000 ppm is fatal). H_2S above 1000 ppm kills instantly.

Routes of entry (exposure process) (2023 5c, 2022 6d, 2020 5d)

Route	Meaning	Example / prevention
Inhalation	Breathing in gases, vapours, mists, dust, fumes - the most common route	Petroleum vapour narcosis, CO into the blood, ozone and ammonia irritate lungs; prevent with ventilation and respirators
Ingestion	Entering through the mouth and swallowing (and aspiration into the lungs)	Eating/smoking with contaminated hands, siphoning fuel by mouth, children; prevent by hygiene, no food in work areas
Injection	Entering directly into the bloodstream through a puncture	Needle-stick, cuts by contaminated glass/metal, high-pressure grease guns, explosions; prevent with care with sharps, guards
Skin / eye contact (absorption)	Passing through skin or eyes	Phenol, solvents, pesticides, ammonia; local or systemic effects; wash eyes 10-15 min; gloves, goggles

Local vs systemic and acute vs chronic effects (2024 6a, 2022 6d, 2020 5b)

	Meaning	Examples
Local effect	Harm at the point of contact	H_2SO_4 burns skin; ammonia, ozone, phosgene irritate nose, throat, lungs
Systemic effect	Chemical absorbed into blood and harms organs away from the entry point	Inhaled CO binds haemoglobin in the blood; lead damages nerves and blood; cadmium fume: local lung damage + systemic kidney damage
Acute effect	From a single, short (minutes to days), usually high exposure; effect appears quickly (within 1-3 days)	Chlorine gas causing choking; benzene at high dose causing narcosis (drowsiness, unconsciousness); H_2S causing collapse
Chronic effect	From repeated low exposures over months to years; effect develops slowly and is often permanent	Benzene → leukaemia; silica → silicosis; lead → anaemia and nerve damage; asbestos → cancer

Memory trick: Routes = **"3 I + S"**: Inhale, Ingest, Inject, Skin. **Local = Location of contact; Systemic = System-wide. Acute = Abrupt; Chronic = Continuous**.

B1 Occupational health, OSH Act and OSHA, occupational health service (functions, how to provide) Must study Very likely · 100% Easy

PAST-PAPER QUESTIONS

- **2024 Q5(a)** [12] Short notes: (i) occupational health, (ii) occupational safety act, (iii) occupational diseases, (iv) occupational exposure.
- **2023 Q5(a)** [12] Define occupational health and occupational diseases. What are the functions of an occupational health service?
- **2022 Q5(a)** [6] Short notes: (i) occupational health, (ii) Occupational Safety and Health Act.
- **2021 Q5(a)** [9] Short notes: (i) occupational health, (ii) Occupational Safety and Health Act, (iii) Occupational Safety and Health Administration.
- **2020 Q4(a)** [10] Define occupational health, occupational diseases, occupational exposure and TLV (threshold limit value).
- **2020 Q4(b)** [10] How can proper occupational health be provided?

★ REMEMBER IT

- Occupational **health** = keeping workers physically, mentally and socially well (WHO/ILO). Occupational **disease** = caused mainly by exposure at work (silicosis, asbestosis, noise deafness). OSHA (Occupational Safety and Health Administration (US government agency)) = the US agency that enforces the OSH (occupational safety and health) Act of 1970.

Before you start: health at work

Safety is about sudden injuries; **occupational health** is about the slow harm that work does to the body and mind (dust in the lungs, noise in the ears, stress). An **occupational disease** is an illness that comes from the job. **Exposure** means how much of a hazard reaches the worker and for how long.

In the USA the **OSH Act** (a law of 1970) created **OSHA** (the government agency that writes and enforces workplace rules) and NIOSH (the research institute). In Bangladesh the equivalent law is the Bangladesh Labour Act 2006, enforced by DIFE (Department of Inspection for Factories and Establishments).

Short notes / definitions (2024 5a, 2023 5a, 2022 5a, 2021 5a, 2020 4a)

- **Occupational health** (ILO/WHO joint committee 1950, revised 1995): the promotion and maintenance of the highest degree of physical, mental and social well-being of workers in all occupations; prevention of departures from health caused by working conditions; protection from risks of agents adverse to health; placing and keeping the worker in an environment adapted to his physiological and psychological capabilities - **adapting work to man and each man to his job**.
- **Occupational disease**: a disease resulting from exposure, in the course of employment, to a hazardous material, physical agent, biological organism or ergonomic stress (in law: a disease resulting from exposure during employment to conditions or substances unfavourable to health). Examples: silicosis, asbestosis, noise-induced hearing loss, lead poisoning, byssinosis.
- **Occupational exposure**: the contact of a worker with a health hazard (chemical, physical, biological agent or ergonomic factor) while carrying out work; measured by its **intensity (concentration)** and its **duration (time)**.

- **TLV (threshold limit value)**: the airborne concentration of a substance to which nearly all workers may be repeatedly exposed, day after day, over a working lifetime, without adverse health effects (ACGIH). See B4.

Occupational Safety and Health Act and Administration (2024 5a, 2022 5a, 2021 5a)

	OSH Act (Occupational Safety and Health Act)	OSHA (Occupational Safety and Health Administration)
What	US federal **law** passed in 1970 (signed by President Nixon, effective 1971)	The federal **agency** under the US Department of Labor created by that Act
Purpose	"To assure so far as possible every working man and woman in the nation safe and healthful working conditions"	To set and enforce the standards required by the Act
Key points	**General duty clause**: every employer must keep the workplace free from recognised hazards likely to cause death or serious harm and comply with OSHA standards; employees must follow rules; also created NIOSH (research) and OSHRC (review commission)	Writes standards (PELs, PPE (personal protective equipment), lockout, hazard communication), inspects workplaces, issues citations and fines, requires injury/illness records (OSHA 300 log), provides training and consultation, protects whistle-blowers

Functions of an occupational health service (2023 5a)

ILO (International Labour Organization) Convention 161 (1985): **occupational health services** are services entrusted with essentially **preventive** functions, responsible for advising the employer, workers and their representatives on (a) the requirements for a safe and healthy working environment for optimal physical and mental health, and (b) adapting work to the capabilities of workers.

1. Identification and assessment of health risks at the workplace.
2. Surveillance (monitoring) of the work environment: dust, noise, vibration, heat, chemicals.
3. Advice on planning and organisation of work, workplace design, equipment and substances (ergonomics).
4. Advice and training on occupational health, safety, hygiene, ergonomics and PPE.
5. Surveillance of workers' health: **pre-employment**, **periodic** and special medical examinations.
6. First aid and emergency treatment.
7. Supervision of dangerous substances and processes.
8. Rehabilitation and return to work of injured/sick workers.
9. Sanitation, hygiene, canteen and nutrition supervision.
10. Medical and health records.
11. Health education, counselling, family welfare and social services.

How proper occupational health can be provided (2020 4b)

- **Medical services**: pre-employment, periodic and special examinations; treatment and first aid; health records.
- **Environmental control**: hazard identification and monitoring (dust, noise, heat, chemicals); engineering control, substitution, ventilation; PPE.
- **Health and first-aid education**; counselling and stress management; mental and physical fitness programmes.
- **Rehabilitation**: medical, social, educational and vocational.
- **Family welfare**, nutrition, sanitation, disaster management.

- **Staff**: occupational health nurse and doctor (part-time or full-time), specialists (hygienists, ergonomists, physiotherapists, psychologists).
- **Effectiveness factors**: nature, size and location of industry; infrastructure; trained manpower; worker involvement and employer commitment; epidemiological investigation.

> **Memory trick: Act = law, Administration = agency**. Health service = **"Assess, Monitor, Examine, Treat, Train, Record, Rehabilitate"**.

B2 Occupational diseases: definitions, vs normal disease; lung disease, blood disorders, urticaria, pigmentary changes, sore eye, stress Must study

Very likely · 100% Easy

PAST-PAPER QUESTIONS

- **2024 Q5(b)** [9] What are the major occupational causes of: (i) lung disease, (ii) blood disorders, (iii) pigmentary changes?
- **2023 Q5(b)** [8] Explain: (i) urticaria, (ii) pigmentary changes, (iii) blood disorders, (iv) sore eye.
- **2022 Q5(b)** [15] Discuss the differences between occupational diseases and normal diseases. Write the major causes of: (i) lung disease, (ii) sore eye, (iii) blood disorders, (iv) urticaria.
- **2021 Q5(b)** [11] Provide two definitions of occupational disease. Write the occupational causes of: (i) stress reactions, (ii) lung diseases, (iii) blood disorders.
- **2020 Q4(c)** [10] Explain urticaria and pigmentary changes.

★ REMEMBER IT

- Lung: dust (silica, coal, asbestos, cotton = byssinosis). Blood: benzene, lead, radiation. Skin: **urticaria** = itchy raised rash (hives) from allergens; **pigmentary changes** = skin darkens or loses colour (tar, hydroquinone, burns). Sore eye: dust, welding arc, chemical splash.

Before you start: work-caused disease

An ordinary disease (flu, diabetes) can come from anywhere. An **occupational disease** has a clear cause **in the job**: dust, chemicals, noise, germs, posture or stress. That is why it can be prevented by controlling the workplace, and why the worker is legally entitled to compensation.

The lungs are the main door (we breathe 10 m^3 of air in a shift), the skin is the largest organ (1.5-2 m^2), and the blood carries poisons everywhere.

Definitions (2021 5b)

1. **General definition**: a disease that is the result of exposure to a hazardous material, physical agent, biological organism or ergonomic stress in the course of work.
2. **Legal definition**: a disease resulting from exposure during employment to conditions or substances that are unfavourable to health, and which is listed in the law as compensable.

Occupational vs normal (general) diseases (2022 5b)

Occupational disease	Normal disease
Caused by exposure at the workplace to a specific agent	Caused by general factors: germs in the community, genes, lifestyle, age
Clear link to a job; affects a group of workers doing the same work	No link to occupation; affects anyone
Preventable by controlling workplace exposure, PPE (personal protective equipment), medical surveillance	Prevented by public health measures, vaccines, lifestyle
Often slow, cumulative, irreversible (silicosis, asbestosis, hearing loss)	Many are acute and curable
Legally notifiable and compensable by the employer	Not compensable by the employer

Causes of Disease:

Hazard	Sources
Psychological	Other people, patterns of work, domestic problems
Physical	Materials, other people, noise, heat, radiation, vibration
Chemical	Fumes, gases, minerals, drugs, materials
Biological	Bacteria, viruses, allergens, other animals and plants

The diseases have been grouped generally according to organ systems, and reflect those systems that are most usually the target of occupational disease.

- Thus, the skin and the lung, both of which are in direct contact with the outside environment.
- The ear and eye may also quite commonly be affected at work by physical hazards.
- Absorbed chemical substances may injure the brain and nervous system or the liver and urinary tract in the process of biotransformation *(chemical alteration of chemicals such as nutrients, amino acids, toxins)* and excretion *(the process of eliminating or expelling waste matter)*.

Causes of occupational ill health: psychological, physical, chemical and biological

Lung diseases (2024 5b, 2022 5b, 2021 5b)

Some occupational causes of lung disease

Syndrome	Main causes
Allergic asthma	Animals, including crustaceans, rats and mice; vegetables such as grains, flour, hardwoods; chemicals such as isocyanates, colophony, latex, acid anhydrides, vanadium and cobalt
Irritant asthma (RADS)	Any irritant gas or fume
Allergic alveolitis	Mouldy hay
Chronic airflow obstruction	Coal dust, cigarette smoke, cadmium
Bronchial carcinoma	Cigarette smoke, radiation, asbestos, *bis*-chloromethyl ether
Nodular fibrosis	Coal, quartz
Diffuse fibrosis	Asbestos
Sarcoidosis	Beryllium
Hilar lymph node enlargement	Quartz
Tuberculosis	*Mycobacterium tuberculosis* in healthcare workers
Pneumonitis, bronchiolitis	Irritant fumes and gases
Pneumonia	*Legionella* spp.
Pleural fibrosis	Asbestos, quartz
Mesothelioma	Asbestos

Occupational lung diseases and their causes

Disease	Occupational cause
Allergic (occupational) asthma	Animals, flour, grain, isocyanates (TDI), colophony (solder flux), latex, cotton dust
Irritant asthma	Irritant gases and fumes (chlorine, SO_2, ammonia)
Allergic alveolitis (farmer's lung)	Mouldy hay, fungal spores
Chronic airflow obstruction (bronchitis)	Coal dust, cadmium fume
Nodular fibrosis: **silicosis**, coal workers' pneumoconiosis	Quartz/silica (sand blasting, stone crushing, glass), coal dust
Diffuse fibrosis: **asbestosis**; berylliosis	Asbestos; beryllium
Bronchial carcinoma, mesothelioma, pleural fibrosis	Asbestos, ionising radiation
Byssinosis	Cotton, flax dust (textile mills)
Tuberculosis	Health-care workers

Symptoms: breathlessness, cough, chest pain, abnormal chest X-ray. MIC (methyl isocyanate) at Bhopal 1984 killed more than 2000 people.

Blood disorders (2024 5b, 2023 5b, 2022 5b, 2021 5b)

Blood disorders:

Toxic effects of workplace substances causing a wide range of haematological disorders.

Table 2.10 *Occupational blood disorders*

Disorder	Features	Causes
Marrow aplasia	Normocytic anaemia, neutropenia	Benzene, gamma radiation
Anaemia	Impaired haem synthesis, stippled red cells	Lead
Methaemoglobinaemia	Cyanosis reversed by methylene blue	Aniline and analogues, nitrites
Haemolysis	Intravascular haemolysis, haemoglobinuria	Arsine, trimellitic anhydride
Leukaemia	Usually chronic myeloid, occasionally acute	Benzene

Blood disorders with features and occupational causes

Disorder	Feature	Cause
Marrow aplasia (bone marrow stops making cells)	Normocytic anaemia, neutropenia	Benzene, gamma radiation
Anaemia	Impaired haem synthesis, stippled red cells	Lead
Methaemoglobinaemia (blood cannot carry oxygen)	Cyanosis (blue skin), reversed by methylene blue	Aniline and analogues, nitrites
Haemolysis (red cells burst)	Intravascular haemolysis	Arsine, trimellitic anhydride
Leukaemia (blood cancer)	Chronic myeloid leukaemia	Benzene, ionising radiation

Skin: urticaria and pigmentary changes (2024 5b, 2023 5b, 2022 5b, 2020 4c)

- **Urticaria** (hives): a rash of round, raised red welts on the skin that itch intensely, sometimes with dangerous swelling (angio-oedema); an allergic reaction to substances touched or inhaled. Occupational causes: natural rubber latex gloves in hospitals (also gives dermatitis and asthma), foods, plants, animals, some chemicals and drugs. Prevent: substitute (non-latex gloves), avoid contact, antihistamines.
- **Pigmentary changes**: changes in skin colour. *Hyperpigmentation* (hypermelanosis) = darker patches; *hypopigmentation* (hypomelanosis, leucoderma) = lighter or white patches. Causes: **tar and pitch** (road work, roofing), mineral oils, plants such as fennel and dill causing photosensitisation (skin reacts to sunlight), UV radiation, some chemicals (phenols, hydroquinone cause white patches), metals (arsenic, silver).
- **Contact dermatitis** (about 80 % of reported occupational disease cases): irritant (acids, alkalis, solvents) or allergic (nickel, chromium, cobalt, dyes, rubber additives, pesticides).



Occupational skin disease: contact dermatitis and sensitisers

Sore eye (2023 5b, 2022 5b)

- Long hours at **VDU** (visual display unit (computer screen)) (visual display unit, computer screens): poor lighting, screen reflection/glare, wrong posture and distance.
- Arc welding without eye protection (UV → "arc eye", conjunctivitis, keratitis).

- Foreign bodies (grinding, chipping) and chemical burns.
- Itchy eyes from flour and grain dust; irritation by chlorine, formaldehyde, ammonia, smoke.

Stress reactions (2021 5b)

Stress reactions: The majority of patients seen in occupational medical practice with neuropsychiatric symptoms are suffering from psychological rather than physical or chemical stressors. Such patients commonly present with a picture of anxiety and depression, and an array of minor physical symptoms. The causes are usually multiple, involving home and family as well as work. Stress-related some factors are in the workplace are shown in Table.

Table 2.9 *Common adverse factors in the workplace leading to psychological breakdown*

Recent promotion beyond capabilities
Conflicts resulting from multiple responsibilities
Too many demands on time
Tiring shift pattern, excess overtime
Too little or boring work
New technology, inadequate training for changed role
A new or unreasonable boss, macho management
Increased, unrealistic productivity targets
Threat of redundancy
Sexual harassment or bullying
High sickness absence in colleagues
Lazy colleagues

Occupational causes of stress

Psychological stressors at work (often combined with home problems) cause anxiety, depression and minor physical symptoms (headache, sleeplessness, high blood pressure). Causes: promotion beyond capability; conflicting responsibilities; too many demands on time; shift pattern and excessive overtime; too little or boring work; new technology with inadequate training; a new or unreasonable boss, "macho" management; unrealistic productivity targets; threat of redundancy (job loss); sexual harassment or bullying; high sickness absence; lazy colleagues.

> **Memory trick:** Blood: **"Benzene Breaks marrow, Lead Lowers haem, Aniline turns you blue, Arsine bursts cells"**. Lungs: **silica → silicosis, asbestos → asbestosis, cotton → byssinosis**.

B3 Occupational ergonomics: definition, importance, five examples of good ergonomics

Must study | Very likely · 100% | Easy

PAST-PAPER QUESTIONS

- **2024 Q5(c)** [8] Define occupational ergonomics. Mention five examples of good occupational ergonomics and explain them.
- **2023 Q5(d)** [9] Define occupational ergonomics. Mention five examples of good occupational ergonomics and explain them.
- **2022 Q5(c)** [7] Define occupational ergonomics. Explain its importance in the workplace.
- **2021 Q5(d)** [5] Briefly discuss occupational ergonomics.
- **2020 Q6(a)** [5] Why is ergonomics important?

★ REMEMBER IT

- Ergonomics = fit the job to the worker. Five examples: adjustable chair with lumbar support, monitor at eye level, keyboard at elbow height, lift with knees (and mechanical aids), job rotation and breaks, good lighting.

Before you start: fit the job to the person

Ergonomics comes from Greek *ergon* (work) + *nomos* (law): the laws of work. Instead of forcing the worker to twist, stretch and strain, we design the chair, desk, tool and task to fit the human body. Bad ergonomics gives back pain, neck and wrist pain (musculoskeletal disorders) and more mistakes.

Definition (2024 5c, 2023 5d, 2022 5c, 2021 5d)

Occupational ergonomics is the study and practice of human factors at the workplace - **"the science of making the job fit the worker"**. ILO (International Labour Organization): the application of human biological sciences together with engineering sciences to the worker and his working environment so as to obtain maximum satisfaction for the worker, which at the same time enhances productivity. Its objective is the best mutual adjustment of man and his work (it grew from British studies in the Second World War).

Importance in the workplace (2022 5c, 2020 6a)

1. **Prevents musculoskeletal disorders** (back pain, neck, shoulder, wrist injuries, carpal tunnel) caused by poor posture, forceful movement, repetition and lifting.
2. **Reduces fatigue and discomfort**, so fewer errors and accidents.
3. **Increases productivity and quality**: comfortable workers work faster and better.
4. **Reduces absenteeism, compensation and medical costs.**
5. **Improves morale and job satisfaction**; accommodates older, disabled and differently sized workers.
6. Helps comply with health and safety law.

Five examples of good occupational ergonomics (2024 5c, 2023 5d)

Example	Explanation
1. **Adjustable chair with lumbar support**	Seat height adjustable so feet rest flat and knees are at 90°; back support keeps the natural curve of the spine, preventing back pain
2. **Monitor at eye level and arm's length**	Top of the screen at or slightly below eye level, 50-70 cm away; prevents neck strain and sore eyes; reduce glare
3. **Keyboard and mouse at elbow height, wrists straight**	Prevents wrist and forearm strain (repetitive strain injury, carpal tunnel syndrome)
4. **Safe lifting and lifting aids**	Lift with legs, not back; keep the load close; use trolleys, hoists, conveyors for heavy loads; prevents back injury
5. **Tools and materials within easy reach; work surface at correct height**	Avoids over-reaching and twisting; standing work at elbow height; anti-fatigue mats for standing workers
6. **Job rotation, micro-breaks and good lighting**	Breaks the repetition, reduces fatigue and eye strain; lighting suited to the task without glare
7. **Ergonomic hand tools**	Padded, bent handles keep the wrist straight; anti-vibration tools prevent white finger

Memory trick: Ergonomics = **"Fit the job to the worker, not the worker to the job"**. Five examples = **Chair, Screen, Keyboard, Lift, Reach**.

B12 Sustainable development: definition, environmental/economic/socio-political sustainability, pollution prevention and clean technology, challenges

Must study · Very likely · 93% · Easy

PAST-PAPER QUESTIONS

- **2024 Q7(b)** [12] Define sustainable development. Discuss: (i) environmental sustainability, (ii) economic sustainability, (iii) socio-political sustainability.
- **2023 Q7(c)** [8] What is meant by sustainable development? Briefly discuss the challenges of sustainable development.
- **2022 Q8(a)** [10] "Sustainable development calls for pollution prevention and clean technology" - explain. How to overcome the challenges of sustainable development?
- **2021 Q8(a)** [8] Why does sustainable development call for pollution prevention and clean technology? What are the environmental challenges of sustainable development?

★ REMEMBER IT

- Brundtland (1987): development that meets the needs of the present without compromising future generations. Three pillars: **environmental, economic, socio-political** (planet, profit, people).

Before you start: using resources without stealing from the future

If we cut forests faster than they grow, pump groundwater faster than rain refills it, or pollute rivers faster than they clean themselves, our children inherit less. **Sustainable development** is growth that keeps the environment, the economy and society healthy at the same time, so that it can continue for ever.

Definition (2024 7b, 2023 7c)

Sustainable development is "development that meets the needs of the present without compromising the ability of future generations to meet their own needs" (Brundtland Commission report "Our Common Future", 1987). Development must be sustained environmentally, economically, socially and politically.

Three dimensions (2024 7b)

Sustainability	Meaning	Examples
Environmental	Use renewable resources no faster than they regenerate, non-renewables no faster than substitutes are found, and emit wastes no faster than nature can absorb them; protect biodiversity and ecosystems (sustainable yield)	Renewable energy, 3R (reduce, reuse, recycle), ETPs, afforestation, ISO (International Organization for Standardization) 14001
Economic	An economy that produces goods and services continuously and generates surplus without running down natural, human or financial capital; viable and profitable over the long term	Efficient use of resources, clean technology that cuts cost, fair trade, long-term investment
Socio-political	Fair distribution of benefits, meeting basic needs, equity between rich and poor and between generations, participation, good governance, human rights, peace and stability	Safe jobs and fair wages, education and health, democratic decision-making, CSR (corporate social responsibility)

Requirements (Brundtland): a political system with effective citizen participation; an economic system generating surplus and technical knowledge self-reliantly; a social system resolving tensions; a production system respecting the ecological base; a technological system searching for new solutions; an international system fostering sustainable trade and finance; a flexible administrative system that corrects itself.

"Sustainable development calls for pollution prevention and clean technology" (2022 8a, 2021 8a)

End-of-pipe treatment (filters, ETPs) only moves pollution from one medium to another, costs money and still consumes resources. Sustainable development requires that development does not degrade the environment for future generations, so pollution must be **prevented at the source** and processes must use **clean technology**:

- **Pollution prevention**: reduce or eliminate waste at the source by changing raw materials, processes and products (substitution of toxic inputs, good housekeeping, leak prevention, recycling within the process).
- **Clean technology**: processes that use less energy, water and raw material and produce less waste and emissions (high-efficiency combined cycle, renewables, low-NO_x burners, waste-heat recovery, closed-loop water).
- Benefits: conserves resources for the future; lowers production and treatment costs (economic sustainability); protects health of workers and communities (social sustainability); reduces liability and meets laws; improves competitiveness and public image.
- So both are the practical tools by which the environmental, economic and social goals are achieved together (also ICC (International Chamber of Commerce)'s 16 Business Charter principles, 1991).

Challenges of sustainable development (2023 7c, 2021 8a)



Global challenges of sustainable development

1. Rapid population growth increasing demand for food, water and energy.
2. Urbanisation and industrialisation: pollution, slums, waste, congestion.
3. Intensive agriculture: fertiliser and pesticide pollution, soil degradation, groundwater depletion.
4. Declining natural resources: forests, fish, fossil fuels, freshwater.
5. Altered biogeochemical cycles: climate change from CO_2, nitrogen and phosphorus overload.
6. Loss of biodiversity and ecosystems.
7. Poverty and inequality; social disintegration; weak governance and corruption.
8. Lack of finance, technology and awareness in developing countries.

How to overcome the challenges (2022 8a)

- Population control, education and health care.
- Planned urbanisation, public transport, waste management and 3R.
- Sustainable agriculture: organic and bio-fertilisers, integrated pest management, efficient irrigation.
- Renewable energy, energy efficiency and clean technology.
- Strong environmental laws and enforcement (EIA, ECA (Environment Conservation Act (Bangladesh, 1995))/ECR), economic instruments (pollution charges).
- Afforestation and conservation of biodiversity.
- Good governance, public participation, international cooperation and technology transfer.

> **Memory trick:** Definition key words: **"present needs, future generations"** (Brundtland 1987). Three pillars: **Planet, Profit, People** = environmental, economic, socio-political.

B4 Occupational exposure, TLV and PEL, TLV-TWA, TLV-STEL, TLV-C

Must study | Very likely · 93% | Easy

PAST-PAPER QUESTIONS

- **2024 Q5(d)** [6] Write down the information that can be obtained from: (i) TLV-TWA, (ii) TLV-STEL, (iii) TLV-C.
- **2023 Q6(b)** [7] Why is it necessary to study PEL (permissible exposure limit) and TLV (threshold limit value)? Explain the relation between them.
- **2022 Q6(a)** [8] What is meant by occupational exposure? What are the two terms that reflect the level of exposure? Differentiate between them.
- **2022 Q6(b)** [6] Write down the information that can be obtained from: (i) TLV-TWA, (ii) TLV-STEL, (iii) TLV-C.
- **2021 Q5(c)** [10] Write the differences between TLVs and PELs. Explain the different ways of calculating the TLVs.

LEARN IT FROM ZERO

Exposure limits, from zero

A worker breathes some chemical vapour every day. How much is safe? Experts set a **threshold limit value** (TLV, by ACGIH (American Conference of Governmental Industrial Hygienists), a guideline) and the law sets a **permissible exposure limit** (PEL, by OSHA (Occupational Safety and Health Administration (US government agency)), enforceable). Both are concentrations (ppm or mg/m³) that nearly all workers can face day after day without harm.

- **TLV-TWA** (time-weighted average): average over an 8-hour day, 40-hour week. "How much on average."
- **TLV-STEL** (short-term exposure limit): average over any 15 minutes, max 4 times a day, 60 min apart. "How much in a short burst."
- **TLV-C** (ceiling): must **never** be exceeded, even for an instant.

★ REMEMBER IT

- TWA (time-weighted average) = the day, STEL (short-term exposure limit) = 15 minutes, C = never cross the line.

Before you start: how much is too much

Almost any substance is harmful if there is enough of it. So for each chemical, expert bodies publish a **limit** on the concentration in the air a worker breathes. ppm = parts per million (1 ppm = 1 litre of gas in 1,000,000 litres of air); mg/m^3 = milligrams per cubic metre.

TLV (threshold limit value) is published by ACGIH (American Conference of Governmental Industrial Hygienists, a professional body) as advice. **PEL** (permissible exposure limit) is set by OSHA as law. Three kinds of limit cover three time spans: a full day (TWA), 15 minutes (STEL), and any instant (Ceiling).

Occupational exposure and the two terms that reflect it (2022 6a)

Occupational exposure is the contact of a worker with a chemical, physical or biological agent or ergonomic factor while carrying out work. Its level depends on two terms: **concentration (intensity)** of the agent and **duration (time)** of exposure. Dose ≈ concentration × time.

Concentration (intensity)	Duration (time)
How much of the agent is present in the breathing zone (ppm, mg/m^3, dB, rem)	How long the worker is exposed (minutes, hours per day, years)
High concentration for a short time → acute effects (irritation, narcosis, death)	Low concentration for a long time → chronic effects (cancer, silicosis)
Controlled by ventilation, enclosure, substitution	Controlled by job rotation, shorter shifts, breaks
Limits: TLV-C, STEL	Limits: TLV-TWA (8 h/day, 40 h/week)

TLV vs PEL (2023 6b, 2021 5c)

TLV (threshold limit value)	PEL (permissible exposure limit)
Set by **ACGIH** (professional, non-government)	Set by **OSHA** (US government)
Guidelines / recommendations, not legally binding	**Legal limits**, enforceable with fines
Updated every year with the latest science; often stricter	Updated rarely (many date from 1971), so often less strict
Health-based only	Considers health and also technical/economic feasibility
Types: TLV-TWA, TLV-STEL, TLV-C	Mostly 8-hour TWA, also ceiling and STEL values

Relation: both state the airborne concentration a worker can be exposed to without harm, expressed in the same way (8-h TWA, STEL, ceiling). Many PELs were originally adopted directly from the 1968 ACGIH TLVs, so the TLV is the scientific basis and the PEL is the legal version. (NIOSH also gives RELs, recommended exposure limits; all are kinds of OEL (occupational exposure limit), occupational exposure limit.)

Why study them: to know safe exposure levels when designing ventilation and processes; to monitor workers and judge if exposure is acceptable; to select PPE (personal protective equipment); to comply with the law and avoid penalties; to prevent occupational disease; to compare with measured TWA (B5) and to write MSDS (material safety data sheet) section 8.

Information from TLV-TWA, TLV-STEL, TLV-C; ways of expressing TLV (2024 5d, 2022 6b, 2021 5c)

Threshold Limit Values (TLVs) can be measured in one of the three ways: TLV-TWA, TLV-STEL, and TLV-C.

Threshold limit value – time-weighted average (TLV-TWA)

TLV-TWA (time-weighted average) represents the concentration to which nearly all workers worker can be exposed day after day for 8 hours per day, 40 hours per week without suffering adverse health effects.

Most TLVs are expressed as 8 hours' time weighted average (TWA). It is determined by sampling the breathing zone of the worker for 8 hours. It is expressed as follows when a series of successive samples have been taken:

$$TWA = \sum_i (C_i t_i)/(\sum t_i)$$

where:

t_i the period of time during which one sample is taken
C_i the average concentration over time period t_i.

TLV-TWA formula and the three types of TLV

Type	Information it gives
TLV-TWA (time-weighted average)	The average concentration over a normal **8-hour day and 40-hour week** to which nearly all workers may be exposed day after day for a working lifetime without adverse effects. TWA = $\Sigma C_i t_i$ / 8 h. Short peaks above it are allowed if the average stays below.
TLV-STEL (short-term exposure limit)	The maximum concentration to which workers can be exposed **continuously for up to 15 minutes** without irritation, chronic or irreversible tissue damage, or narcosis that increases accident risk. Not more than **4 times a day**, at least **60 minutes apart**, and the daily TWA must not be exceeded (STEL ≤ about 3 × TWA).
TLV-C (ceiling)	The concentration that must **never be exceeded, even for an instant**; used for fast-acting irritants and poisons.

$$TWA = (C_1t_1 + C_2t_2 + \ldots + C_nt_n) / 8\ h$$

Ways of calculating TLVs: from industrial experience, human volunteer studies, animal experiments and epidemiology; expressed as TWA (full-shift sampling or several samples averaged), as 15-min STEL samples, as instantaneous ceiling readings; for mixtures the additive formula $C_1/T_1 + C_2/T_2 + \ldots$ must not exceed 1.

> **Memory trick: TWA = Whole day, STEL = Short 15 min, C = Ceiling never touch. TLV = Tells (advice), PEL = Police (law).**

B15 ISO 14001: purpose and major components of an environmental management system

Must study · Very likely · 93% · Easy

PAST-PAPER QUESTIONS

- **2024 Q8(b)** [12] Mention the purpose of ISO (International Organization for Standardization) 14001. What are the major components of ISO 14001?
- **2023 Q8(e)** [6] Briefly discuss ISO 14001.
- **2022 Q8(d)** [7] Mention the purposes of ISO 14001. What are the major components of ISO 14001?
- **2021 Q8(d)** [5] Briefly discuss ISO 14001.

★ REMEMBER IT

- ISO 14001 = Plan-Do-Check-Act for the environment: **policy → planning (aspects, legal, objectives) → implementation (resources, training, documents, emergency) → checking (monitoring, audit) → management review**.

Before you start: a management system for the environment

ISO (International Organization for Standardization, Geneva, 1947) publishes standards. ISO 9001 is for quality, **ISO 14001 for environmental management**, ISO 45001 for occupational health and safety. ISO 14001 does not set pollution limits; it sets up a **system** (an EMS (environmental management system), environmental management system) through which a company keeps finding and reducing its environmental impacts, following the cycle **Plan-Do-Check-Act** (PDCA).

What ISO 14001 is (2023 8e, 2021 8d)

ISO 14001 (latest version 2015) is the international standard that specifies requirements for an **environmental management system (EMS)**. It is part of the ISO 14000 family (14004 guidance, 14031 performance evaluation, 14040 life-cycle assessment, ...), but it is the **only one an organisation can be certified to**. It gives a framework for the systematic identification, control and continual improvement of the environmental **aspects** (activities that interact with the environment, e.g. emissions, effluent, waste) and **impacts** of an organisation. Any organisation of any size or sector can apply it, and certification is by an independent third-party body.

Purposes (2024 8b, 2022 8d)

- Identify and control the environmental aspects and impacts of activities, products and services.
- Ensure compliance with environmental laws and regulations (e.g. ECA (Environment Conservation Act (Bangladesh, 1995))/ECR).
- Prevent pollution and reduce waste, emissions and resource use (efficient resources, lower cost).
- Continually improve environmental performance.
- Provide competitive advantage, access to export markets, and the trust of customers, investors, regulators and the community.
- Reduce environmental incidents, liability and insurance costs; integrate with ISO 9001 and ISO 45001.

Major components of ISO 14001 (2024 8b, 2022 8d)

Major components of a strong environmental management system that are reflected in the ISO 14001

(1) top management leadership, policy setting, and review;

(2) identifying environmental issues/risks (or "aspects" in the vocabulary of ISO 14001) and legal and other requirements;

(3) establishing objectives to successfully manage these risks and requirements in line with policy;

(4) implementing programs and procedures (including for emergencies) that establish how, by whom, and when those objectives will be met;

(5) training people so that they know their risks and what to do to best manage them;

(6) monitoring, measuring, and auditing to track performance and verify implementation;

(7) taking preventative and corrective action;

(8) continual improvement; and

(9) documentation and record keeping.

ISO 14001 environmental management system components

PDCA (Plan-Do-Check-Act) **stage**	**Component**	**What it means**
Plan	1. **Top management leadership and environmental policy**	Policy appropriate to the nature and scale of impacts, committing to compliance, prevention of pollution and continual improvement; documented and communicated
Plan	2. **Environmental aspects and legal requirements**	Identify aspects with significant impacts and all legal and other obligations
Plan	3. **Objectives, targets and programmes**	Measurable environmental objectives and plans to achieve them
Do	4. **Implementation and operation**	Resources, roles and responsibilities; operational control procedures
Do	5. **Competence, training and awareness; communication**	Train staff; internal and external communication
Do	6. **Documentation and control of documents/records**	Manual, procedures, records kept up to date
Do	7. **Emergency preparedness and response**	Plans for spills, fires and releases
Check	8. **Monitoring, measurement, evaluation of compliance and internal audit**	Measure key characteristics (emissions, effluent, waste), audit the EMS
Check/Act	9. **Nonconformity, corrective and preventive action**	Fix problems and their causes
Act	10. **Management review and continual improvement**	Top management reviews performance and sets new targets

Memory trick: ISO numbers: **9001 Quality, 14001 Environment, 45001 Safety**. Components follow **Plan-Do-Check-Act**: Policy → Aspects → Objectives → Operate → Train → Document → Emergency → Monitor → Correct → Review.

B10 Local, regional and global environmental impacts and their causes (thermal power plants, petroleum industry)

Must study · Very likely · 93% · Easy

PAST-PAPER QUESTIONS

- **2024 Q6(d)** [6] Explain the following terms of environmental impact and their main causes: (i) local, (ii) regional, (iii) global.
- **2023 Q7(b)** [9] Explain local, regional and global environmental impacts and mention their main causes.
- **2022 Q7(c)** [6] Explain the following terms of environmental impact: (i) local, (ii) regional, (iii) global.
- **2021 Q7(b)** [5] What are the local and regional environmental impacts posed by thermal power plants and petroleum industries?

★ REMEMBER IT

- Scale ladder: **local** (around the plant: noise, dust, thermal water, smell) → **regional** (hundreds of km: acid rain, smog, river pollution) → **global** (whole planet: greenhouse warming, ozone depletion).

Before you start: how far pollution travels

Some pollution stays near its source (smoke over a city), some is carried hundreds of km across borders (acid rain), and some changes the whole planet (greenhouse gases warming the Earth). So impacts are grouped as **local**, **regional** and **global**.

Local, regional and global impacts and their causes (2024 6d, 2023 7b, 2022 7c)

Scale	Impacts	Main causes
Local (city, plant surroundings, a river)	Urban air pollution (SO_2, CO, NO_x, particulates, ground-level ozone, lead); indoor air pollution; polluted rivers, groundwater and soil; noise; solid waste; land degradation	Vehicles, power plants, industries, brick kilns, burning of biomass/fuelwood for cooking (indoor), untreated effluent and sewage, poor waste disposal
Regional (several countries/a sea)	Acid rain (forests, lakes, buildings); pollution of seas and cross-border rivers; regional haze/smog; eutrophication	Long-range transport of SO_2 and NO_x from tall stacks of coal power plants and smelters; oil spills; agricultural runoff into rivers and seas
Global (the whole Earth)	Greenhouse effect and climate change (sea-level rise, extreme weather); ozone-layer depletion; persistent organic pollutants and radioactive waste; loss of biodiversity; ocean acidification; desertification	Burning of fossil fuels (CO_2), deforestation, methane from agriculture and gas leaks, CFCs, nuclear accidents

Local and regional impacts of thermal power plants and petroleum industries (2021 7b)

Thermal power plants	Petroleum industries (refineries, oil and gas fields)
Air: SO_2, NO_x, CO, particulates and fly ash from stacks → respiratory disease, smog, acid rain (regional)	Air: VOCs, H_2S, SO_2, flaring smoke, hydrocarbons → smog and odour
Thermal pollution: hot cooling water discharged to rivers kills fish, lowers dissolved oxygen	Water: oily effluent, produced water, spills and leaks contaminate rivers, sea and groundwater
Ash disposal: fly ash and bottom ash ponds pollute land and groundwater (heavy metals)	Soil contamination from leaks of tanks and pipelines; sludge and spent catalyst waste
Noise, coal dust from handling; land use; water withdrawal	Fire, explosion and toxic release risk to communities; noise; land use
Regional: acid rain on forests and lakes, regional haze	Regional: marine oil spills, coastal pollution, acid rain

Table Environmental impacts, dimensions and causes

Impact and magnitude	*Main cause*
LOCAL	
Urban air pollution	Emissions of sulphur dioxide (SO_2), carbon monoxide (CO), nitrogen oxides (NO_x) and particulate matter (PM) in fossil fuels burning, especially oil and coal. Evaporative emissions of hydrocarbons (HCs) and other volatile organic compounds (VOCs) by solvents and fuel transfer operations. Formation of low-altitude ozone (O_3) by the solar light action on NOx and HCs.
	Emissions of heavy metals such as lead (Pb), cadmium (Cd), mercury (Hg) and other toxic and carcinogenic substances (such as dioxins and furans) when burning coal, oils and solid wastes.

Environmental impacts: local (textbook table)

REGIONAL	
Acid rain	Deposition of sulphuric (H_2SO_4) and nitric (HNO_3) acids, formed by the reaction of water (rain, snow, etc.) with SO_2 and NO_2 generated by fossil fuel burning.
Pollution of seas and transboundary waterbodies	Oil leakages and other leakages in interstate or international waters. Contamination of underground aquifers by the percolation of toxic substances.

Regional impacts

GLOBAL	
Greenhouse effect	Emissions of carbon dioxide (CO_2) by fossil fuels burning and by deforestation of native forests. Methane emissions (CH_4) by incomplete fuel burning and by (anaerobic) digestion of waste.
Persistent organic pollutants (POPs) and radioactive waste	Accumulation of heavy metals (such as the mercury emitted by coal thermopower plants, entering in the food chain) in living organisms, of man-made toxic compounds (such as PCBs – polychlorinated biphenyls found in fluids utilized in electric equipment) and of radioactive substances (deriving from nuclear accidents, tests and leakages).
Loss of biodiversity, changes in the oceans and desertification	Deforestation for producing fuelwood and charcoal, as well as land cleaning for agriculture, coastal and marine degradation due to oil leakages into the sea, carbon dioxide (CO_2) emissions causing acidification of the oceans. Ecosystems flooded by hydropower reservoirs. Use of agrochemicals by energy or food monoculture.

Global impacts

Memory trick: Local = Lungs of the city, Regional = Rain (acid) across borders, Global = Greenhouse.

B13 IEE and EIA: why needed, differences, EIA process (flow chart), red category, EMP

Must study · Very likely · 100% · Medium

PAST-PAPER QUESTIONS

- **2024 Q7(c)** [10] Why are IEE (initial environmental examination) and EIA (environmental impact assessment) studies required for a project? Explain the EIA process.
- **2023 Q8(b)** [10] When are IEE and EIA studies required for a project? Explain the EIA process in detail.
- **2023 Q8(c)** [6] What is an Environmental Management Plan (EMP)? At which stage of a new project is EMP (environmental management plan) necessary and why?
- **2022 Q8(b)** [12] Why are IEE and EIA studies required for a project? What type of assessment is necessary for a red-category project? Explain with a flow chart.
- **2021 Q8(b)** [12] What are the differences between EIA and IEE? With a flow chart, explain the EIA process.
- **2021 Q8(c)** [10] Why is an Environmental Management Plan essential during project implementation? (clearance part in B14)
- **2020 Q6(d)** [8] What are the purposes of IEE and EIA?

LEARN IT FROM ZERO

IEE and EIA, from zero

Before a factory or power plant is built, the government asks: "what will this do to the air, water, land and people around it, and how will you limit it?". A small, low-risk project needs only a short first look, the **Initial Environmental Examination (IEE)**. A big or risky project (Red category: power plants, refineries, cement) needs a full **Environmental Impact Assessment (EIA)**.

EIA flow: **screening** (does it need EIA?) → **scoping** (which impacts matter) → **baseline study** (environment today) → **impact prediction and evaluation** → **alternatives and mitigation** → **EMP** (environmental management plan: who does what to limit and monitor impacts) → **report and public consultation** → **review and decision (ECC)** → **monitoring**.

★ REMEMBER IT

- EIA steps: **"Some Scientists Buy Pretty Maps, Then Review Monitors"** = Screening, Scoping, Baseline, Prediction, Mitigation (EMP), Report/consultation, Review, Monitoring.

Before you start: checking a project's effect on nature before building it

Before a big project (a power plant, factory, road) is built, the government wants to know what it will do to air, water, land, people and wildlife, and how the bad effects will be reduced. First comes a short quick check (**IEE**, initial environmental examination). If serious impacts are possible, a full detailed study follows (**EIA**, environmental impact assessment). The **EMP** (environmental management plan) is the action plan to actually do the mitigation and monitoring.

In Bangladesh: ECA (Environment Conservation Act (Bangladesh, 1995)) = Environment Conservation Act 1995; ECR (Environment Conservation Rules (Bangladesh, 1997; new rules 2023)) = Environment Conservation Rules 1997 (revised 2023); DoE (Department of Environment (Bangladesh)) = Department of Environment.

Why/when IEE and EIA are required; purposes (2024 7c, 2023 8b, 2022 8b, 2020 6d)

IEE is the first stage of environmental assessment, done at the pre-feasibility stage, to identify possible impacts and decide whether a full EIA is needed. **EIA** is a systematic, documented and objective evaluation and reporting of the social, economic and environmental impacts of a proposed project, done at the feasibility stage, with quantified impacts and preventive/remedial measures and alternatives.

Required: by law in Bangladesh (ECA 1995, ECR 1997) before a project can get site clearance and environmental clearance from the DoE: Orange-B projects need IEE; **Red** category projects (power plants, refineries, fertiliser, cement, large industries) need IEE for **site clearance** and full EIA for **environmental clearance**. Also required by lenders (World Bank, ADB).

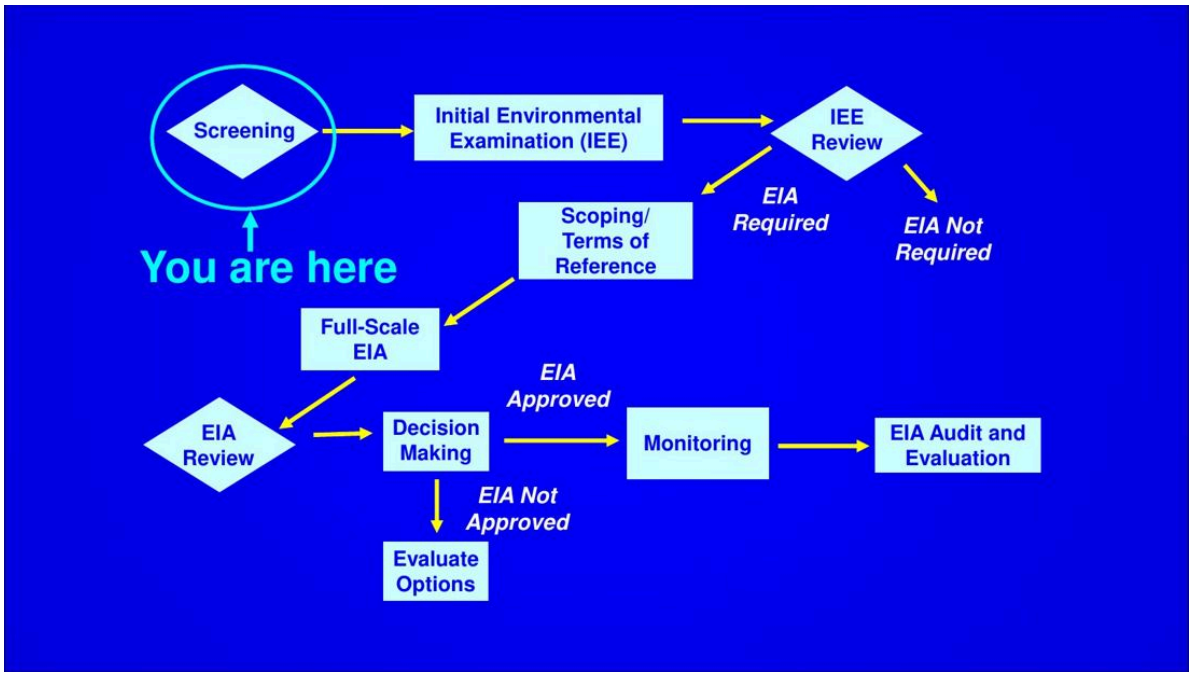
Purposes / objectives:

- Predict environmental, social and economic impacts at an early stage.
- Find ways to reduce adverse impacts and enhance benefits; shape the project to suit the local environment.
- Present alternatives (including the "no project" option) to decision-makers.
- Reduce environmental damage and lower long-term costs (cheaper to design out than to repair).
- Ensure conformity with national standards, policies and international obligations; obtain approval.
- Inform and involve the public.

IEE vs EIA (2021 8b)

IEE (initial environmental examination)	EIA (environmental impact assessment)
Preliminary, screening-level study	Full, detailed study
At pre-feasibility stage	At feasibility stage
Mostly from existing data, site visits, expert judgement; qualitative	Field surveys, baseline monitoring, modelling; quantitative impacts and valuation
Short time and low cost	Longer (months) and costly
Output: brief description of predicted changes, mitigation measures, alternatives, need and ToR (terms of reference) for detailed EIA	Output: EIA report with baseline, impacts, mitigation, alternatives, EMP, monitoring plan, public consultation
Enough for Orange-B; for Red gives site clearance	Required for Red category to get environmental clearance

EIA process with flow chart (2024 7c, 2023 8b, 2022 8b, 2021 8b)



EIA process flow chart: screening, IEE, scoping/ToR, full EIA, review, decision, monitoring

1. **Project proposal** (and request for proposal to consultants).
2. **Screening**: decide from the category list whether the project needs IEE/EIA.
3. **IEE**: identify likely impacts; reviewed by DoE.
4. **Is EIA required?** If no: decision; if yes, go on.
5. **Scoping and ToR** (terms of reference): decide which issues, area and alternatives the EIA must cover; approved by DoE.
6. **Full-scale EIA**: baseline data, impact prediction and evaluation, mitigation measures, alternatives, EMP, public consultation.
7. **Review** of the EIA report by DoE/experts.
8. **Decision making**: approved → environmental clearance with conditions; not approved → evaluate options / redesign and resubmit.
9. **Implementation with monitoring, audit and evaluation** according to the EMP.

Red-category project (2022 8b): needs both **IEE** (with ToR for EIA) to get site clearance and a full **EIA** with EMP and ETP (effluent treatment plant) design to get environmental clearance (detailed steps in B14).

Environmental Management Plan (EMP) (2023 8c, 2021 8c)

An **EMP** is a site-specific plan that sets out the environmental activities to be carried out during project implementation to mitigate adverse impacts and maximise benefits: the mitigation measures, monitoring programme (what, where, how often), responsibilities, schedule, budget, training and reporting. It covers **construction, operation and decommissioning**.

Stage: it is prepared as part of the EIA (feasibility/design stage, before approval) and applied from the **construction stage** onwards through operation. **Why essential:** it turns the EIA's recommendations into actions; ensures contractors and operators understand the environmental risks and their duties; it is a condition of the ECC (DoE checks it on renewal); it allows early detection of problems through monitoring; it demonstrates compliance to regulators, lenders and the public.

> **Memory trick: IEE = Initial (quick), EIA = Entire (detailed), EMP = Execute (the plan)**. EIA chain: **"Screen, IEE, Scope, Study, Review, Decide, Monitor"**.

B14 Environmental clearance in Bangladesh (Green, Orange-A, Orange-B, Red), Department of Environment (DoE) and its history Must study

Very likely · 93% Medium

PAST-PAPER QUESTIONS

- **2024 Q7(d)** [5] Write down the brief history of the Department of Environment (DoE) of Bangladesh.
- **2024 Q8(a)** [10] Explain the general procedure to get environmental clearance for a red category new project according to the regulation Act in Bangladesh.
- **2023 Q8(d)** [7] Discuss the general process of gaining environmental clearance of a new project.
- **2022 Q8(c)** [6] Write the procedure to get environmental clearance for a new project according to the regulation Act in Bangladesh.
- **2021 Q8(c)** [10] Discuss the general process of gaining environmental clearance of a new project.

★ REMEMBER IT

- Bangladesh categories (ECR 1997, now 2023 rules): **Green** (no clearance study) → **Orange-A** → **Orange-B** (IEE) → **Red** (IEE + EIA (environmental impact assessment)). Red: site clearance first, then EIA approval, then ECC (environmental clearance certificate) before operation, renewed yearly.

Before you start: permission from the Department of Environment

In Bangladesh no factory or project may be set up or run without an **Environmental Clearance Certificate (ECC)** from the **DoE** (Department of Environment (Bangladesh)). How much paperwork you need depends on the project's **category**, based on how polluting it is: Green (least), Orange-A, Orange-B, Red (most; power plants, refineries, chemicals).

ECA (Environment Conservation Act (Bangladesh, 1995)) 1995 section 12 requires the ECC (Environmental Clearance Certificate); ECR (Environment Conservation Rules (Bangladesh, 1997; new rules 2023)) 1997 (rule 7) gives the procedure and the category lists. NOC (no objection certificate) = no objection certificate from the local authority.

Brief history of the Department of Environment (DoE) (2024 7d)

Side note: The class materials do not cover DoE history; the points below are from standard sources.

- **1973**: Water Pollution Control Project started under the Water Pollution Control Ordinance (1973).
- **1977**: Environment Pollution Control Ordinance; Environment Pollution Control Board formed.
- **1982**: renamed Department of Environmental Pollution Control (DEPC).
- **1989**: renamed **Department of Environment (DoE)** and placed under the newly created Ministry of Environment and Forest (MoEF).
- **1992**: National Environment Policy. **1995**: Environment Conservation Act (ECA) gave DoE its legal powers. **1997**: Environment Conservation Rules (ECR) with categories and clearance procedure. **2000**: Environment Court Act. **2010**: ECA amended. Ministry renamed MoEFCC (Ministry of Environment, Forest and Climate Change) in 2018. **2023**: new Environment Conservation Rules.
- Functions: environmental clearance, pollution monitoring and enforcement, setting standards, EIA review, awareness, implementing international conventions, laboratory testing.

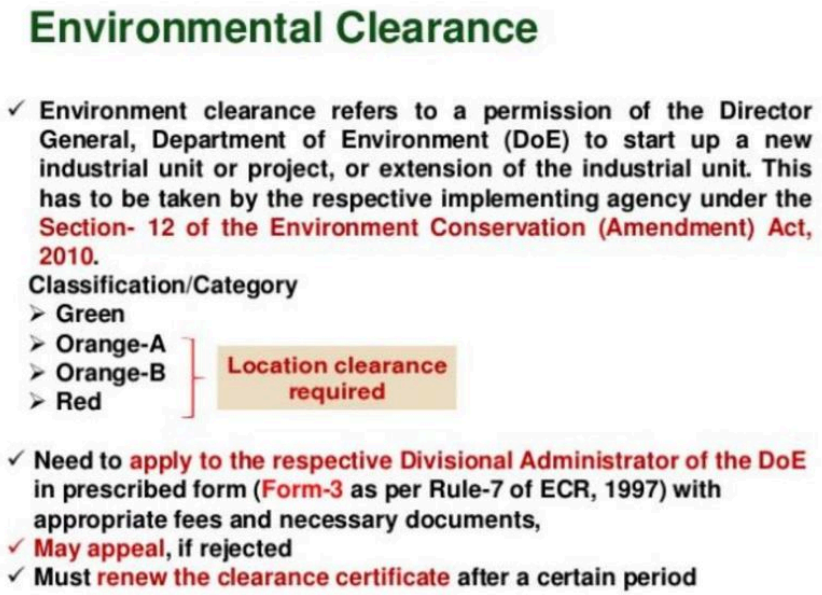
Department of Environment (DoE)	The functions include liasing with related environmental authorities or associations; giving 'environmental clearance' to the newly established industries; conducting enquiry into environmental problems or allegations of industrial pollution; advising and, where necessary, directing any person engaged in the import/export of hazardous materials on "environment-friendly use" of such materials; undertaking any other measures necessary for conservation and improvement of the environment and for the prevention of pollution.

Functions of the Department of Environment

Categories of Industrial Units:

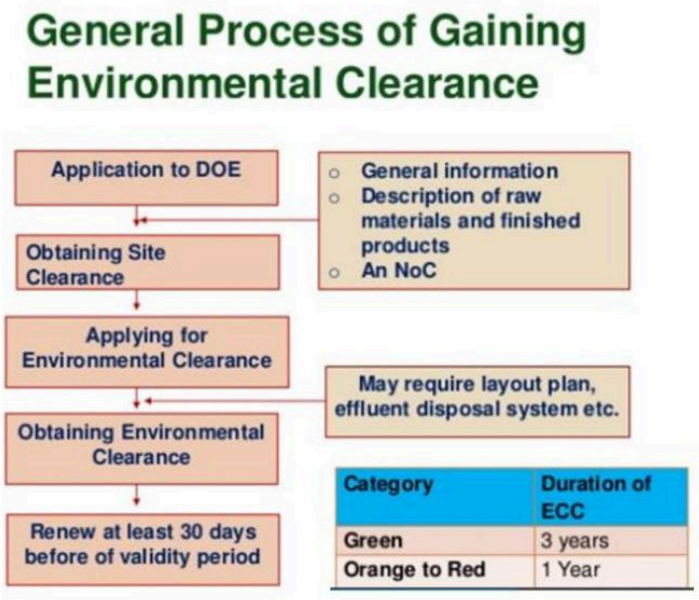
- **Green Category**:
  Examples:
  Television/radio assembling, book-binding, tea packing, sports goods, laundry without washing
- **Orange-A Category**:
  Examples:
  dairy, poultry, saw mill, printing press, restaurant, cinema hall, dry cleaning
- **Orange- B Category:**
  Examples:
  glass factory, edible oil, jute mill, coal tar, hotel, multi-storied apartment, brick field, Ice cream factory
- **Red Category:**
  Examples:
  Textile, tannery, fertilizer factory, power plants, chemical dyes, cement, refinery, paper & pulp, sugar etc.

Categories of industrial units: Green, Orange-A, Orange-B, Red with examples



Environmental clearance: classification and where to apply

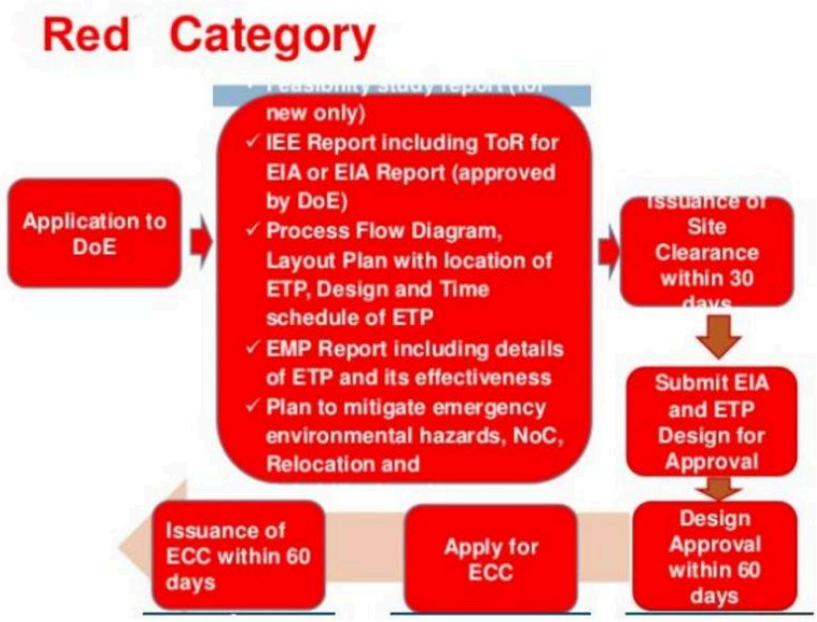
General process of gaining environmental clearance (2023 8d, 2022 8c, 2021 8c)



General environmental clearance procedure

1. **Application** to the DoE divisional/district office in the prescribed form (Form-3) with fee, giving general information on the project, raw materials and products, and the NOC from the local authority.
2. **Site clearance**: for Orange-A, Orange-B and Red projects, DoE inspects and issues location (site) clearance (Green projects get ECC directly).
3. **Apply for environmental clearance**: submit the layout plan, effluent/emission disposal arrangements (ETP), IEE (initial environmental examination)/EIA as required by category.
4. **Obtain the ECC** after DoE review and inspection.
5. **Renew** the ECC at least 30 days before expiry. Validity: **Green 3 years**; Orange-A, Orange-B and Red **1 year**.

Clearance for a Red-category new project (2024 8a)



Red category: steps to obtain site clearance and environmental clearance

Step (write this)	Why / what it means
Apply to DoE with: feasibility report; **IEE report including ToR** (terms of reference) **for EIA** (or EIA if already done); process flow diagram; layout plan with ETP (effluent treatment plant) position; ETP design and time schedule; EMP (environmental management plan) outline; emergency plan; NOC from local authority; relocation/rehabilitation plan if needed; fee	Red projects can cause severe pollution, so DoE needs full project information first.
DoE reviews, inspects the site and approves the ToR → **site (location) clearance** within **30 days**	Permission to start land development and construction, not operation.
Carry out and submit the **full EIA report** (with EMP) and the **ETP design** for approval	Detailed study and treatment design as per the approved ToR.
DoE approves EIA and ETP within **60 days**	Review by experts; conditions may be attached.
Construct the plant and ETP; then apply for the **Environmental Clearance Certificate (ECC)**	Clearance is given only after the ETP is built and works.
DoE inspects and issues the ECC within **60 days**; renew every **1 year**, 30 days before expiry	Annual renewal lets DoE check EMP compliance.

Memory trick: Red route = **"IEE → Site (30 d) → EIA + ETP (60 d) → ECC (60 d) → renew yearly"**. Validity: **Green 3 years, others 1 year**.

B9 Pollution and pollutants: classification, sources of air and water pollution and their control

Must study | Likely · 67% | Easy

PAST-PAPER QUESTIONS

- **2023 Q7(a)** [8] What are the major sources of water pollution? How to control them?
- **2022 Q7(b)** [7] Classify the materials that create pollution, with definition and examples.
- **2021 Q7(a)** [10] Classify and explain the pollutants. Discuss the major sources of air pollution.
- **2020 Q6(b)** [5] Classify pollutants by the nature of disposal.

★ REMEMBER IT

- Pollutants: by form (gas, liquid, solid), by origin (primary straight from source: SO_2, CO; secondary formed in air: O_3, smog), by nature of disposal (biodegradable vs non-biodegradable).

Before you start: pollution in one sentence

Pollution = putting harmful substances (or energy like noise, heat, light) into air, water or soil, causing an undesirable change in their physical, chemical or biological character. The harmful substance is the **pollutant**.

Pollutants are sorted by how nature handles them: some break down quickly (biodegradable), some very slowly (persistent), some never (non-biodegradable).

Classification of pollutants (2022 7b, 2021 7a, 2020 6b)

A. By nature of disposal

Type	Definition	Examples
Biodegradable (non-persistent)	Broken down rapidly by natural processes (microbes)	Domestic waste, garbage, sewage, paper, food waste
Slowly degradable (persistent)	Remain in the environment unchanged for decades	DDT (dichloro-diphenyl-trichloroethane (an insecticide)), pesticides, aerosols, some plastics
Non-biodegradable	Never degraded by natural processes; accumulate	Lead, mercury and other heavy metals, nuclear waste, plastics, glass

B. By existence in nature

Type	Definition	Examples
Quantitative	Already present in nature; become pollutants when their concentration exceeds a threshold	CO_2, nitrogen oxides, phosphates
Qualitative	Do not occur in nature; man-made	Fungicides, herbicides, DDT, plastics

C. By form: primary (emitted directly: SO_2, CO) and secondary (formed in air: ozone, acid rain, smog). **By medium:** air, water, soil, noise, light and thermal pollution. Typical pollutants: NO_x, SO_x,

CO_x; soot, smoke, tar, dust; metals Hg, Pb, Zn, Ni, Cd, Cr; acids; pesticides; garbage; radioactive waste; e-waste.

Major sources of air pollution (2021 7a)

Pollutant	Sources
Carbon compounds: CO_2 (complete burning), CO (incomplete burning)	Fossil fuel combustion in power plants, vehicles, industries, brick kilns, cooking
Sulphur compounds: SO_2, H_2S, H_2SO_4	Coal-fired power plants, oil refineries, smelters
Nitrogen oxides NO_x	Automobiles, power plants, high-temperature combustion
Metals: Pb, Ni, Sn, Be, Ti	Metallurgical industries, leaded fuel, batteries
SPM (suspended particulate matter) (suspended particulate matter), $PM_{2.5}/PM_{10}$	Stone crushing, fly ash, construction, road dust, brick kilns
Hydrocarbons, VOCs	Vehicles, refineries, solvents, paints

Control: cyclones (mechanical collectors), wet scrubbers, ESP (electrostatic precipitators), bag filters, flue-gas desulphurisation, cleaner fuels and catalytic converters.

Major sources of water pollution and their control (2023 7a)

Source	Pollutants	Control
Industrial effluents (textile, tannery, chemical, power plant)	Chemicals, dyes, heavy metals, solvents, hot water	Effluent treatment plant (ETP) before discharge; reuse and recycle; zero liquid discharge; cooling towers for hot water
Domestic sewage	Organic matter, pathogens, detergents	Sewage treatment plants, sanitary latrines
Agricultural runoff	Fertilisers (nitrates, phosphates → eutrophication), pesticides	Controlled and organic fertiliser use, integrated pest management, buffer strips
Urban runoff and solid waste	Oil, garbage, organic matter, plastics	Proper solid waste management, drainage, landfill liners
Oil spills and petroleum leaks	Hydrocarbons	Double-hull tankers, booms and skimmers, leak detection
Suspended matter (silt), mining	Sediment, acid mine drainage	Settling ponds, erosion control, reforestation

> **Memory trick:** Disposal classes: **"Bio-quick, Persistent-slow, Non-bio-never"**. Existence: **Quantitative = too much of natural; Qualitative = man-made**.

B7 Dose-response relationship; lethal, sub-lethal, acute Must study

Possible · 47% Easy

PAST-PAPER QUESTIONS

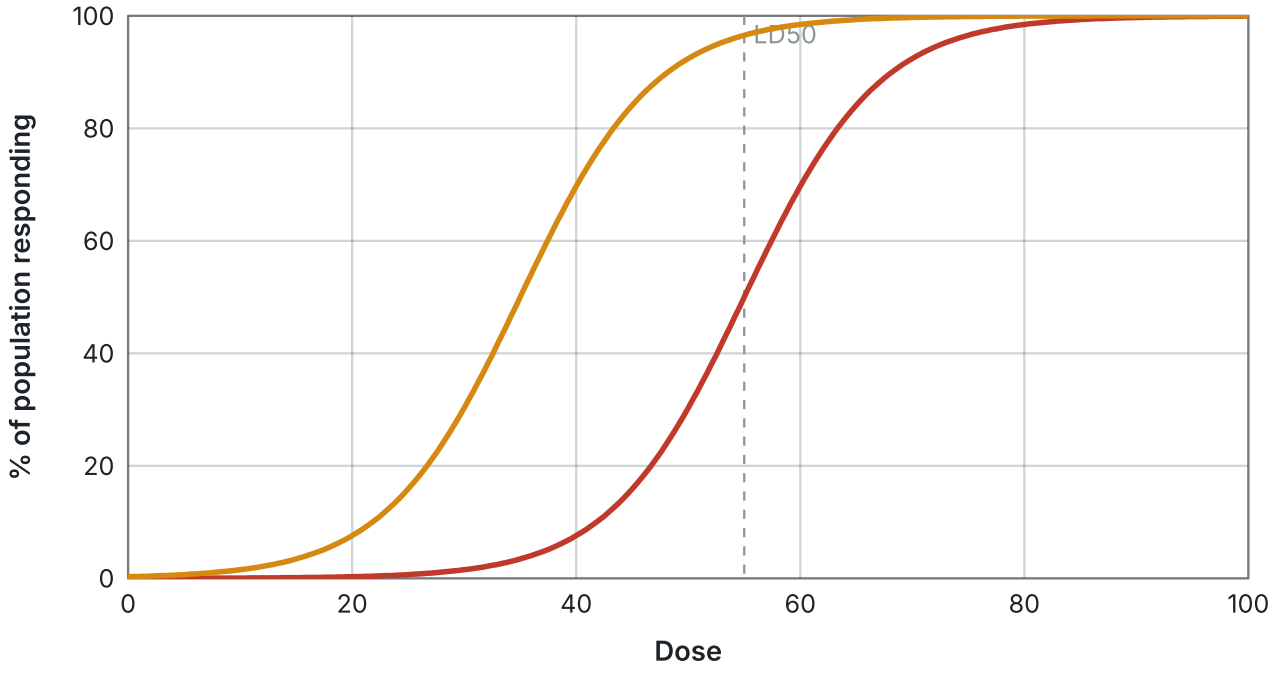
- **2023 Q6(a)** [8] What is meant by hazardous chemicals? Explain the dose-response relationship. Show (i) lethal, (ii) sub-lethal, (iii) acute in it.
- **2021 Q6(b)** [8] Explain the hazardous chemical dose-response relationship. Point out (i) lethal, (ii) sub-lethal and (iii) acute.
- **2020 Q5(a)** [6] Explain the dose-response relationship.

UNDERSTAND **Read this to understand; you do not need to write it**

Give more and more of a chemical to a group and count how many respond. The curve is an S: nothing at very low dose (below the threshold), then a steep rise, then everyone responds. **LD_{50}** = dose that kills half. Sub-lethal = harms but does not kill; lethal = kills; acute = effect from one big dose.

INTERACTIVE **Dose-response curve**

Move the sliders and watch the curve change.



Left of both curves: no observed effect (threshold). Between them: sub-lethal effects. Right: lethal.

Before you start: the more you get, the worse it is

Paracelsus (16th century): "the dose makes the poison". Plot the **dose** (amount received) on the x-axis and the **response** (effect on the body) on the y-axis. At very small doses the body copes and nothing happens; above a **threshold** effects start, first reversible, then permanent, and finally death.

Hazardous chemicals (2023 6a)

Hazardous chemicals are chemicals found to cause intrinsic health deterioration and/or physical catastrophe (fire, explosion, corrosion): toxic, corrosive, flammable, reactive, oxidising and radioactive substances (see B6).

Dose-response relationship (2023 6a, 2021 6b, 2020 5a)

All materials are not equally toxic and the effect depends on a numbers of factors such as

a) Concentration of the toxic materials
b) Characteristics of the toxic materials
c) The duration of exposure
d) Route of entry into the body
e) Physical condition of the victim

Dose-response curve: no effect below A, reversible effects A-B, permanent damage B-C, death C-D

Dose = the amount of a substance a person is exposed to (concentration × time, mg/kg body weight). **Response** = the biological effect produced. The dose-response relationship describes how the effect increases as the dose increases:

Region of curve	Dose	Response
Below **A**	Below the threshold	**No observable effect** (the body detoxifies/repairs); point A = threshold of response
A to B	Moderate	**Reversible effects**: irritation, headache, nausea; recover when exposure stops
B to C	High	**Permanent (irreversible) damage**: organ damage, cancer, nerve damage
C to D	Very high	**Death**; the lethal range (LD_{50} = dose that kills 50 % of a test population)

The response also depends on concentration, characteristics of the chemical, duration and frequency of exposure, route of entry, and the person's physical condition (age, health, sex).

Lethal, sub-lethal, acute on the curve (2023 6a, 2021 6b)

- **Lethal**: the dose/concentration that causes death by direct poisoning - the C-D part of the curve (LD_{50}, LC_{50} marked in it).
- **Sub-lethal**: a dose below the lethal level, which does not kill but affects growth, reproduction, behaviour or organs - the A-C region (reversible A-B and permanent B-C damage).
- **Acute**: an effect (usually severe, often death) produced within a short period after a single exposure to a high dose - the steep upper part of the curve reached in one exposure. (Chronic = from repeated exposure over a long time; cumulative = successive doses build up.)
- **IDLH** **(immediately dangerous to life or health)** (immediately dangerous to life or health): the maximum concentration from which a worker could escape within 30 minutes, if the respirator fails, without death or irreversible effects.

> **Memory trick:** Curve = **"Nothing, Reversible, Permanent, Dead"** (A-B-C-D). Sub-lethal = **sub** (below) lethal.

B5 Numerical: time-weighted average (TWA) exposure and comparison with TLV

Good to have · Possible · 33% · Easy

PAST-PAPER QUESTIONS

- **2024 Q8(c)** [13] TLV-TWA of a substance is 200 ppm. Shift 8:00 AM - 5:00 PM (9 h) with 1 h lunch (12:00-1:00 PM) with no exposure. Hourly concentrations (ppm) at 8:00, 9:00 ... 16:00: 137, 130, 129, 157, 136, 138, 160, 140, 147. (i) Calculate the TWA (time-weighted average), (ii) has the worker exceeded the TLV (threshold limit value)? (iii) Comment on the effect of exposure time and concentration on TLV-TWA.

UNDERSTAND **Read this to understand; you do not need to write it**

Multiply each concentration by how long it lasted, add, and divide by 8 hours. The lunch hour away from the area counts as zero.

CALCULATOR **Time-weighted average (8 one-hour readings)**

Change any number; every step is recalculated and shown with the numbers put in.

1. Sum of C × t: **137** + **130** + **129** + **157** + **138** + **160** + **140** + **147** = **1138** ppm·h
2. TWA: TWA = $\Sigma C_i t_i$ / 8 = **1138** / 8 = **142.25** ppm
3. Verdict: **Within the limit**

Before you start: averaging exposure over a day

The concentration changes from hour to hour. To compare with the 8-hour limit (TLV-TWA), we take a **weighted average**: multiply each concentration by the time it lasted, add them, and divide by 8 hours. An hour with no exposure (lunch away from the area) adds zero.

If the TWA is less than the TLV-TWA, the worker is within the safe limit.

Numerical (2024 8c)

Step (write this)	Why / what it means
TLV-TWA = 200 ppm; shift 8 AM-5 PM; lunch 12-1 PM with no exposure	Write the data: 9 hours at work, 8 hours exposed.
Exposed hours (each 1 h): 8-9: 137, 9-10: 130, 10-11: 129, 11-12: 157, 1-2: 138, 2-3: 160, 3-4: 140, 4-5: 147 ppm	Take the 8 readings of the exposed hours; the 12:00 reading (136) falls in the lunch hour, when exposure is zero.
$\Sigma C_i t_i$ = (137 + 130 + 129 + 157 + 138 + 160 + 140 + 147) × 1 h = 1138 ppm·h	Each concentration times its time (1 h each), then added.
TWA = 1138 / 8 = **142.25 ppm**	TLV-TWA is always referred to an 8-hour day.
142.25 ppm < 200 ppm ⇒ **the TLV is NOT exceeded**	Compare with the limit.

Side note: If the examiner wants all 9 readings with the lunch hour counted as 0 over the 9-hour day: TWA = 1138/9 = 126.4 ppm - still below 200 ppm, same conclusion. State your assumption clearly. Also check no

single reading crosses the STEL (short-term exposure limit)/ceiling: the highest, 160 ppm, is well below 200.

Effect of exposure time and concentration on TLV-TWA (iii)

- TWA is the product of **concentration × time** averaged over 8 h: doubling either the concentration or the exposure time doubles the contribution to the TWA.
- A short period of high concentration can be balanced by low-exposure periods, as long as the 8-h average stays below the TLV-TWA and short peaks stay below the STEL (15 min) and ceiling.
- Longer shifts (e.g. 10-12 h) or overtime increase the total dose and reduce recovery time, so the TLV must be reduced (adjusted) for extended shifts.
- Reducing the time (job rotation, breaks away from the area) or the concentration (ventilation, enclosure) both lower the TWA.

Memory trick: "Multiply, Add, Divide by 8, Compare". Lunch = zero.

WRITE IN EXAM **Write this in the exam (this much is enough for full marks)**

2024 8(c) TLV-TWA = 200 ppm; shift 8 AM–5 PM, lunch 12–1 PM (no exposure), so 8 exposed hours

$\Sigma C_i t_i$: (137 + 130 + 129 + 157 + 138 + 160 + 140 + 147) × 1 h

= 1138

= **1138 ppm·h**

TWA: $\Sigma C_i t_i$ / 8

= 1138 / 8

= **142.25 ppm**

142.25 ppm < 200 ppm, so the TLV-TWA is **not exceeded**; the highest hourly value (160 ppm) is also below 200 ppm.